CFA / LEVEL I

组合管理

Portfolio Management

知识点先行 · 原题跟练 · 中英文对照

Learn the concept. Work the original question. Explain the choice.

8 学习单元 Learning units	100 原题逐题对应 Mapped source items	48 重点例题 Priority examples
按考点重新归类 Grouped by tested concept	99 可正常计分 + 1 错配校勘 99 scorable + 1 corrupted source	含公式、步骤与错项解释 Formulas, steps and distractors

从你提供的两份材料出发，把分散的题目变成可复用的解题方法。每单元先讲知识点，再按子考点讲解原题；全部题目保留英文、中文、原题号和来源页码。

Built from your two supplied files, this guide turns separate questions into reusable solution methods. Each unit teaches the concepts before explaining the original questions by subtopic, retaining English, Chinese, original numbers and source pages.

使用范围 / SCOPE

题库《1000题(4).pdf》第 184-202 页的组合管理 Q1-Q100；答案《组合管理(1).pdf》第 1-100 页。两份材料的官方/模拟题出处无法核实，因此统一标为用户提供题库。

Portfolio Management Q1-Q100 from pages 184-202 of the supplied question PDF, matched to pages 1-100 of the answer PDF. Official-versus-mock provenance is unverified; all items are labeled as user-supplied questions.

整理日期 / Prepared: 2026-11-02

本文件是对上传题库的完整学习整理，不声称覆盖所有 2026 考纲内容。
This is a complete guide to the uploaded question set, not a claim of exhaustive 2026 curriculum coverage.

怎么最快学完这一套题

A practical route through this question set

第一遍建议用约 180 分钟读知识点并做 48 道重点例题。先独立选答案，再看推导和错项。第二遍做其余题，最后只重做做错、猜对和超时的题。时间是学习预算，可按薄弱点调整。

Budget about 180 minutes for a first pass through the concepts and 48 priority examples. Choose an answer before reading the reasoning and distractors. Complete the remaining questions on the second pass, then repeat only missed, guessed and slow items. Adjust the time budget to your weak areas.

学习单元 / Unit	题数 Items	分钟 Min	优先完成 / Priority
01 收益、风险偏好与分散化 Return, risk preference and diversification	15	30	Q24, Q53, Q66, Q90, Q98
02 有效前沿、CAL 与 CML Efficient frontier, CAL and CML	8	15	Q6, Q48, Q68, Q100
03 CAPM、Beta 与收益生成模型 CAPM, beta and return-generating models	18	30	Q40, Q52, Q79, Q81, Q87, Q93, Q94
04 风险调整绩效评价 Risk-adjusted performance evaluation	7	20	Q9, Q27, Q70, Q78, Q95
05 IPS、风险承受能力与资产配置 IPS, risk tolerance and asset allocation	24	30	Q14, Q37, Q38, Q56, Q62, Q77, Q83, Q86, Q91, Q97
06 投资者、投资工具与组合流程 Investors, investment vehicles and the portfolio process	10	15	Q1, Q23, Q34, Q57, Q63
07 行为金融：识别偏差的关键词 Behavioral finance: recognize the bias	8	20	Q10, Q11, Q17, Q21, Q28, Q33
08 风险管理、VaR 与风险度量 Risk management, VaR and risk measures	10	20	Q2, Q3, Q35, Q72, Q80, Q88

读题固定三步 / A THREE-STEP ROUTINE

① 圈出问什么：variance 还是 standard deviation；总收益还是超额收益。② 选公式或分类：总风险 σ、系统性风险 β、IPS 章节或行为偏差。③ 写出一个反证：另外两个选项错在什么条件或单位。

1. Identify the requested output: variance or standard deviation, total or excess return. 2. Choose the relevant formula or classification: σ, β, IPS section or bias. 3. State why each rejected choice fails in definition, assumption or units.

频次说明：上表题数只反映这套 100 题的构成，不代表正式考试权重。

Counts describe this source set and are not official exam weights.

目录与来源导航

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03 CAPM、Beta 与收益生成模型 CAPM, beta and return-generating models	23	18 题 / items
04 风险调整绩效评价 Risk-adjusted performance evaluation	39	7 题 / items
05 IPS、风险承受能力与资产配置 IPS, risk tolerance and asset allocation	46	24 题 / items
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本讲义的 8 个单元是为学习顺序而重组，不是原文件的章节编号。每个原题只在一个单元完整讲解；概念页会引用同题的计算结果，帮助建立联系。PDF 书签可直接跳转单元，答案表的题号可跳转原题解析。

The eight units are a teaching sequence, not the source file’s chapter numbering. Each original question has one complete solution location; concept pages may cite the same example to build connections. PDF bookmarks jump to units, and answer-key question numbers link to the solutions.

来源编号：题库 PDF 页码按整个 202 页文件计算；章节页码为 1-19。答案 PDF 第 n 页对应 Qn。
Question-source pages refer to the full 202-page file; chapter pages run from 1-19. Answer-source page n corresponds to Qn.

01 收益、风险偏好与分散化

Return, risk preference and diversification

本单元 15 题 | 首轮预算 30 分钟 | 重点：Q24, Q53, Q66, Q90, Q98

对应阅读主题 / Reading theme: Portfolio Risk and Return: Part I; Portfolio Management: An Overview

本单元先解决两个问题：组合赚多少，以及组合波动多大。收益可以直接按权重平均；风险必须考虑资产之间如何一起变动。

This unit answers two questions: what return does the portfolio earn, and how volatile is it? Returns are weighted averages; risk also depends on how the assets move together.

1.1 收益平均，风险不能直接平均

Average returns; account for dependence when measuring risk

$$E(R_p) = w_1 E(R_1) + w_2 E(R_2); \; w_1 + w_2 = 1$$

w 是资产占组合价值的权重。看到组合目标收益，先用这条式子反推权重。

w is each asset's portfolio weight. If the target portfolio return is given, solve this equation for the weights first.

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$$

σ 是标准差，σ² 是方差，ρ 是相关系数。交叉项有 2；权重也要平方。若问标准差，最后再开平方。

σ is standard deviation, σ² is variance, and ρ is correlation. Include the factor of 2 in the cross term and square the weights. Take a square root only if standard deviation is requested.

$$Cov(R_1,R_2) = \rho_{12}\sigma_1\sigma_2$$

若题目给方差，先对各资产方差开平方，才能放入交叉项。若给协方差，交叉项直接写成 2w1w2Cov。 Q53 的 5.5 %² = 0.00055，故 σ=√(0.00055/0.75)=2.708%。

If the inputs are variances, take their square roots before forming the correlation cross term. If covariance is given, use 2w1w2Cov directly. In Q53, 5.5 %² = 0.00055, so σ=√(0.00055/0.75)=2.708%.

看到的题目 / Cue	最短解题路径 / Shortest solution path
Q24：给方差，问组合方差 Variances in; portfolio variance out	0.25×0.05 + 0.25×0.06 + 0.5×0.75×√0.05×√0.06 = 0.04804 → B
Q66：给组合收益，求风险 Target return given; find risk	w1=(12.6%-6%)/(17%-6%)=0.6；ρ=0，σp=√(0.6²×0.24²+0.4²×0.12²)=15.18% → B
Q90：直接套双资产公式 Direct two-asset calculation	σp=√(0.4²×0.15²+0.6²×0.18²+2×0.4×0.6×0.2×0.15×0.18)=13.36% → C

1.2 为什么分散化有效

Why diversification works

在权重和各资产波动率不变、权重为正时，相关系数越低，组合方差越低。ρ=0 只表示没有线性共同变动，不代表组合没有风险；ρ=1 时才有 σp=w1σ1+w2σ2。ρ=-1 时还需要适当配重，才可能把两资产风险完全抵消。

Holding positive weights and asset volatilities fixed, lower correlation means lower portfolio variance. Zero correlation does not mean zero portfolio risk. At ρ=1, σp=w1σ1+w2σ2. At ρ=-1, complete risk cancellation still requires appropriate weights.

σ_p^2 = average variance / n + [(n-1)/n] × average covariance

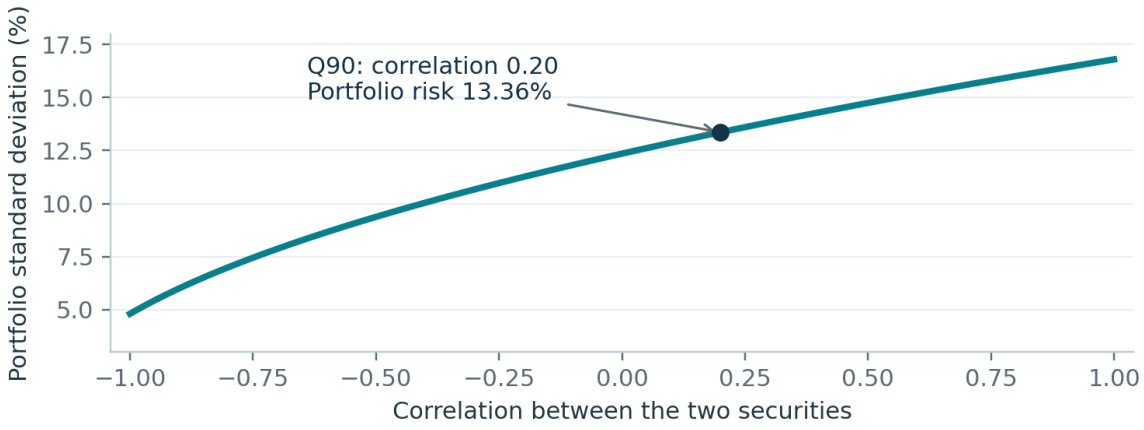
对等权组合，n 很大时第一项趋近 0，组合方差趋近平均协方差。因此，增加资产数量不能保证把市场共同风险消除。对应 Q98。

For an equally weighted portfolio, the first term vanishes as n grows, leaving average covariance. More holdings do not guarantee elimination of common market risk. See Q98.

辨析 / PRECISION NOTE

Q15 按原答案采用：分散化比率 = 等权组合标准差 ÷ 个别证券标准差的平均值；数值越低，分散化效果越强。其他资料可能把倒数也称 diversification ratio，必须先看定义。

Q15 uses portfolio standard deviation divided by the average standalone standard deviation. A lower value indicates stronger diversification. Some other sources use the reciprocal under the same name; check the definition first.



图示使用 Q90 的权重与波动率，只改变相关系数；标点为原题 ρ=0.20。

This chart keeps Q90's weights and volatilities and varies only correlation; the marked point uses the original ρ=0.20.

1.3 风险偏好与效用

Risk preference and utility

$U = E(R) - 0.5 A\sigma^2$

本讲义公式统一用小数收益率，例如 10%=0.10。A>0 厌恶风险，A=0 风险中性，A<0 偏好风险。

The formulas use decimal returns, such as 10%=0.10. A>0 denotes risk aversion, A=0 risk neutrality, and A<0 risk seeking.

风险厌恶者：同风险选高收益，同收益选低风险。风险中性者：只比较期望收益。无风险资产 $\sigma^2=0$，因此效用等于期望收益，与 A 无关。Q7 的答案页考这一点，但原题题干损坏，不能作为正常练习计分。

A risk-averse investor prefers higher return for the same risk or lower risk for the same return. A risk-neutral investor compares expected returns alone. With $\sigma^2=0$, utility equals expected return regardless of A. Q7's solution tests this point, but its corrupted stem must not be scored as a valid exercise.

收益分布的三个词要分开：variance 衡量波动；skewness 衡量不对称；kurtosis 衡量尾部厚度这一类分布特征。Q31 问极端收益的相对可能性，在给定选项中选 kurtosis。峰度不是某个损失阈值的完整概率模型。

Keep the distribution terms separate: variance measures dispersion, skewness measures asymmetry, and kurtosis characterizes tail heaviness. For Q31's choices, kurtosis best addresses extreme returns. Kurtosis alone does not determine the exact probability beyond a specified loss threshold.

01 原题与逐题解析

Original questions and explanations

先作答，再看答案；标为 KEY EXAMPLE 的题目优先完成。
Attempt the question before reading the solution; prioritize items labeled KEY EXAMPLE.

风险偏好、效用与收益分布

Risk preference, utility and return distributions

Q008 原题 / ORIGINAL QUESTION

题库 PDF 第 185 页（章节第 2 页）| 答案 PDF 第 8 页

A risk-neutral investor most likely seeks to maximize:

风险中性投资者最可能寻求最大化什么？

A	both risk and return.	同时最大化风险和收益。
B	return irrespective of risk.	最大化收益，不考虑风险。
C	return for a given level of risk.	在给定风险水平下最大化收益。

答案 / ANSWER: B

风险中性对应风险厌恶系数 A=0。效用函数 $U=E(r)-0.5A\sigma^2$ 因此简化为 U=E(r)，投资者只按预期收益排序，不因风险本身获得奖励或惩罚。

Risk neutrality means A=0 in $U=E(r)-0.5A\sigma^2$. Utility therefore equals expected return, so the investor ranks choices by expected return without either rewarding or penalizing risk itself.

排除干扰项 / ELIMINATE DISTRACTORS

A：风险中性并非主动追求风险。C：给定风险下追求更高收益是一般理性投资原则，不能最准确刻画风险中性。

A: Risk neutrality does not mean seeking risk. C: Maximizing return at a given risk is a general rational principle, not the most distinctive definition of risk neutrality.

速记 / RECALL: A=0：只看收益。

A=0: focus only on expected return.

Q041 原题 / ORIGINAL QUESTION

题库 PDF 第 191 页（章节第 8 页）| 答案 PDF 第 41 页

Risk-averse investors make investment decisions that maximize:

风险厌恶型投资者作出投资决策时，会最大化什么？

A	both return and risk.	风险与收益两者。
B	return irrespective of risk.	收益，而不考虑风险。
C	return for the same amount of risk.	在相同风险水平下的收益。

答案 / ANSWER: C

风险厌恶并不意味着完全拒绝风险，而是要求承担更高风险时得到相应的预期收益补偿。在风险相同的组合中，风险厌恶型投资者偏好更高收益；收益相同则偏好更低风险。

Risk aversion does not mean refusing all risk. It means requiring compensation for bearing more risk. Among portfolios with the same risk, a risk-averse investor prefers the higher expected return; at the same return, lower risk is preferred.

排除干扰项 / ELIMINATE DISTRACTORS

A 把风险本身当作需要增加的目标；B 描述风险中性而非风险厌恶。

A treats risk itself as a benefit. B describes risk neutrality rather than risk aversion.

速记 / RECALL: 同风险选高收益；同收益选低风险。

At equal risk choose higher return; at equal return choose lower risk.

Q055 原题 / ORIGINAL QUESTION

题库 PDF 第 193 页（章节第 10 页）| 答案 PDF 第 55 页

Long-term historical data on the risk-return trade-off of securities show that investors are most likely:

证券风险与收益权衡的长期历史数据表明，投资者最可能属于哪一类？

A	risk averse.	风险厌恶型。
B	risk neutral.	风险中性型。
C	risk seeking	风险偏好型。

答案 / ANSWER: A

本题考查风险溢价的经济含义：投资者通常要求较高预期收益来补偿承担的较高风险，这符合风险厌恶。这里讨论的是总体长期关系，不代表每一项高风险投资都一定获得较高实现收益。

The question tests the economic meaning of a risk premium: investors generally require higher expected returns as compensation for bearing more risk, consistent with risk aversion. This is a broad long-term relationship, not a guarantee that every risky investment earns a higher realized return.

排除干扰项 / ELIMINATE DISTRACTORS

B 不要求额外收益来补偿风险；C 可能愿意为承担更多风险而接受更低收益，均不符合本题逻辑。

B does not require compensation for risk. C may accept lower return to obtain more risk; neither fits the intended relationship.

速记 / RECALL: 要求风险补偿，说明风险厌恶。

Requiring compensation for risk indicates risk aversion.

Q031 原题 / ORIGINAL QUESTION

题库 PDF 第 189 页（章节第 6 页）| 答案 PDF 第 31 页

When evaluating the return distribution of an asset class, the probability of extreme returns is best assessed by the distribution's:

评估某资产类别收益分布时，极端收益的概率最适合通过哪项分布特征判断？

A	kurtosis.	峰度。
B	variance.	方差。
C	skewness.	偏度。

答案 / ANSWER: A

峰度用于刻画尾部厚度及极端观察值的倾向；较高峰度通常意味着相对正态分布更厚的尾部。方差描述总体离散程度，偏度描述左右不对称。

Kurtosis characterizes tail heaviness and the tendency for extreme observations; higher kurtosis generally indicates heavier tails relative to a normal distribution. Variance measures overall dispersion, while skewness measures asymmetry.

排除干扰项 / ELIMINATE DISTRACTORS

B：不能单凭总体离散度识别尾部形状。C：衡量不对称方向，而非尾部厚度本身。

B: Overall dispersion alone does not identify tail shape. C measures asymmetry rather than tail heaviness itself.

速记 / RECALL: 极端尾部看峰度；左右不对称看偏度。

Extreme tails: kurtosis. Left-right asymmetry: skewness.

概念辨析 / NOTE: 峰度是尾部风险的概括指标，不能单凭一个峰度数值计算任意阈值的精确极端收益概率。

Kurtosis summarizes tail behavior; it does not by itself determine the exact probability of exceeding an arbitrary return threshold.

Q007 原文错配 / CORRUPTED SOURCE

题库 PDF 第 185 页（章节第 2 页）| 答案 PDF 第 7 页

原题完整保留以供核对；不可按正常选择题计分。 / Preserved for audit; exclude from scoring.

An analyst gathers the following information about a company. If all purchases and sales were made on credit, the cash conversion cycle (based on a 360-day year) is:

某分析师收集了如下公司信息。若所有采购和销售均采用赊账，以一年 360 天计算，现金转换周期为：

指标 / Metric	数值 / Value
应付账款周转率 / Payables turnover	8
存货周转率 / Inventory turnover	2
应收账款周转率 / Receivables turnover	10

A	less than the utility generated for a risk-averse investor.	低于厌恶风险投资者所获得的效用。
B	equal to the utility generated for a risk-averse investor.	等于厌恶风险投资者所获得的效用。
C	greater than the utility generated for a risk-averse investor.	高于厌恶风险投资者所获得的效用。

答案文档标示：**B* / SOURCE KEY ONLY: B***

本题题干和选项错配，不能作为有效选择题作答。答案第 7 页标示 B，但它讨论的是无风险资产效用：$U=E(r)-0.5A\sigma^2$；由于 $\sigma^2=0$，各类投资者均有 $U=E(r)$，所以效用相同。这里只保留该经答案支持的知识点。

The stem and options are mismatched, so the printed question is not a valid multiple-choice problem. Answer page 7 marks B but discusses utility from a risk-free asset: $U=E(r)-0.5A\sigma^2$. Since $\sigma^2=0$, every investor has $U=E(r)$, regardless of risk aversion.

排除干扰项 / ELIMINATE DISTRACTORS

原选项比较的是效用，无法与现金转换周期对应；不得依据 B 倒推出或补造原题。此题不计入正常正确率。

The choices compare utility and cannot answer a cash-conversion-cycle question. Do not reconstruct an original stem from B. Exclude this item from ordinary accuracy scoring.

速记 / RECALL: 无风险资产：$\sigma^2=0$，因此 $U=E(r)$。

For a risk-free asset, $\sigma^2=0$, so $U=E(r)$.

原文校勘 / NOTE: 严重原文错配：题库第 185 页的公司周转率题干与效用选项不相容。B 仅为答案文档所标答案，不是损坏题干的可验证答案；原本完整题干无法从所给文件恢复。

Severe source mismatch: the turnover-ratio stem on question PDF page 185 is incompatible with the utility choices. B is only the supplied answer-key label, not a verifiable answer to the corrupted stem. The complete intended original stem cannot be recovered from these files.

相关性与分散化的方向判断

Correlation and the direction of diversification benefits

Q012 原题 / ORIGINAL QUESTION

题库 PDF 第 186 页（章节第 3 页）| 答案 PDF 第 12 页

When creating a long-only portfolio, which of the following correlation coefficients between assets would be most effective at reducing portfolio risk?

构建只做多组合时，下列哪一个资产间相关系数最有效地降低组合风险？

A	-0.5.	-0.5。
B	0.	0。
C	0.5.	0.5。

答案 / ANSWER: A

在权重和单项资产风险相同的条件下，组合方差中的协方差项为 $2w_1w_2\rho_{12}\sigma_1\sigma_2$。只做多时 $w_1w_2>0$，因此相关系数越低，组合方差越低；三个选项中 -0.5 最低。

Holding weights and individual risks fixed, the covariance term in portfolio variance is $2w_1w_2\rho_{12}\sigma_1\sigma_2$. With positive long-only weights, a lower correlation reduces variance. Of the choices, -0.5 is the lowest.

排除干扰项 / ELIMINATE DISTRACTORS

B、C 的相关系数都高于 -0.5，因此在其他条件相同时分散化效果较弱。

B and C have higher correlations than -0.5 and therefore provide less risk reduction, all else equal.

速记 / RECALL: 只做多、其他条件相同：相关性越低越好。

For long-only portfolios, lower correlation improves diversification, all else equal.

Q067 原题 / ORIGINAL QUESTION

题库 PDF 第 196 页（章节第 13 页）| 答案 PDF 第 67 页

An equally weighted portfolio is composed of two risky assets. If the correlation of asset returns is equal to zero, the portfolio standard deviation is:

一个等权组合由两项风险资产构成。若两者收益率的相关系数为零，组合标准差如何？

A	equal to zero.	等于零。
B	equal to the weighted average of the assets' standard deviations.	等于两项资产标准差的加权平均。
C	less than the weighted average of the assets' standard deviations.	小于两项资产标准差的加权平均。

答案 / ANSWER: C

当权重各为 0.5 且相关系数为 0，sigma_p=√(0.25×sigma_1²+0.25×sigma_2²)。对于两项标准差均为正的风险资产，这小于 0.5×sigma_1+0.5×sigma_2，但仍大于零。零相关带来分散收益，不代表风险完全抵消。

With 0.5 weights and zero correlation, sigma_p=√(0.25×sigma_1²+0.25×sigma_2²). For two risky assets with positive volatilities, this is below 0.5×sigma_1+0.5×sigma_2 but remains positive. Zero correlation creates diversification benefits without eliminating risk.

排除干扰项 / ELIMINATE DISTRACTORS

A 把零相关误解为零风险；B 是相关系数等于 +1 时的结果。

A mistakes zero correlation for zero risk. B is the result when correlation equals +1.

速记 / RECALL: 相关为 0：风险降低但不归零。

At zero correlation, risk falls but does not disappear.

Q096 原题 / ORIGINAL QUESTION

题库 PDF 第 202 页（章节第 19 页）| 答案 PDF 第 96 页

For an equally weighted portfolio, an increase in the correlations between asset returns most likely decreases the:

对于等权重组合，资产收益之间的相关性上升，最可能使下列哪项下降？

A	portfolio's expected return.	组合预期收益。
B	portfolio's standard deviation of returns.	组合收益标准差。
C	level of risk reduction provided by the portfolio.	组合带来的风险降低程度。

答案 / ANSWER: C

在其他条件不变时，相关性上升会提高协方差项，使组合方差上升、分散化效果下降。预期收益是各资产预期收益的加权平均，公式中没有相关性。

Holding other inputs constant, higher correlations increase covariance terms and portfolio variance, weakening diversification. Expected return is the weighted average of asset expected returns and does not depend on correlation.

排除干扰项 / ELIMINATE DISTRACTORS

A 没有随相关性机械下降的依据；B 方向相反，正权重组合的标准差一般上升。

A has no mechanical reason to fall when correlations rise. B has the direction reversed: volatility generally rises for a positive-weight portfolio.

速记 / RECALL: 相关性越高，分散化降风险越少。

Higher correlation means less risk reduction from diversification.

Q015 原题 / ORIGINAL QUESTION

题库 PDF 第 186 页（章节第 3 页）| 答案 PDF 第 15 页

The diversification ratio of a portfolio is best described as the ratio of the:

组合的分散化比率最适合描述为以下哪一个比值？

A	standard deviation of the equally weighted portfolio's returns to the average standard deviation of the individual securities' returns.	等权组合收益标准差，除以个别证券收益标准差的平均值。
B	standard deviation of the market-capitalization-weighted portfolio's returns to the standard deviation of the equally weighted portfolio's returns.	市值加权组合收益标准差，除以等权组合收益标准差。
C	average standard deviation of the individual securities' returns to the standard deviation of the market-capitalization- weighted portfolio's returns.	个别证券收益标准差的平均值，除以市值加权组合收益标准差。

答案 / ANSWER: A

按本题答案采用的定义，分散化比率=等权组合的标准差÷个别证券标准差的平均值。分子体现分散化后的风险，分母体现平均单项风险；该比率越低，分散化带来的风险降低越大。

Under the definition used in the supplied solution, the diversification ratio equals the standard deviation of an equally weighted portfolio divided by the average individual-security standard deviation. A lower ratio means greater risk reduction under this convention.

排除干扰项 / ELIMINATE DISTRACTORS

B：比较的是两种加权方案，不是本题定义。C：既倒置分子分母，也改为市值加权组合。

B compares weighting schemes rather than the specified definition. C reverses the ratio and uses a market-capitalization-weighted portfolio.

速记 / RECALL: 先辨定义：本题是“组合风险 ÷ 平均单项风险”。

Check the definition: here it is portfolio risk / average individual risk.

口径提醒 / NOTE: “Diversification ratio”在其他资料中也常用相反方向：加权平均单项波动率÷组合波动率。本题必须按所附答案的定义作答，不可混用“越大越好”和“越小越好”。

Other sources also use the reciprocal convention: weighted average individual volatility divided by portfolio volatility. Follow the supplied definition here; do not mix up whether a larger or smaller ratio is better.

Q020 原题 / ORIGINAL QUESTION

题库 PDF 第 187 页（章节第 4 页）| 答案 PDF 第 20 页

The risk-return trade-off of a portfolio of only risky assets most likely improves when a risk-free asset is added to the portfolio because:

在仅含风险资产的组合中加入无风险资产后，风险收益权衡最可能改善，其原因是：

A	the risk-free asset is uncorrelated with the other assets in the portfolio.	无风险资产与组合中其他资产不相关。
B	the lower return on the risk-free asset provides a diversification effect.	无风险资产较低的收益带来分散化效果。
C	the correlations among the risky assets decrease, providing a diversification effect.	风险资产之间的相关性下降，产生分散化效果。

答案 / ANSWER: A

按所附答案，选 A。实质是无风险资产方差与协方差均为零，使可行组合包含无风险资产与最优风险组合的线性组合；借贷条件允许时，可改善原有可行机会集。加入它并不会改变原有风险资产彼此的相关系数。

The supplied answer is A. The relevant property is zero variance and covariance for the risk-free asset, permitting linear combinations with an optimal risky portfolio and potentially improving the opportunity set under the borrowing/lending assumptions. Existing risky-asset correlations do not change.

排除干扰项 / ELIMINATE DISTRACTORS

B：低收益本身不会带来分散化收益。C：增加一种资产不等于改变其他资产之间的相关性。

B: A lower return by itself does not create diversification benefits. C: Adding an asset does not change correlations among existing assets.

速记 / RECALL: 关键是无风险资产的零方差、零协方差。

The key properties are zero variance and zero covariance.

口径提醒 / NOTE: 与第 5 题相同：“不相关”是题库惯用说法；恒定无风险收益的 Pearson 相关系数严格而言未定义，协方差才是零。改善并非对每一个已有组合都严格成立。

As in Q5, “uncorrelated” is source shorthand: Pearson correlation with a constant return is undefined, whereas covariance is zero. Strict improvement need not occur for every pre-existing portfolio.

Q098 重点例题 / KEY EXAMPLE

题库 PDF 第 202 页（章节第 19 页）| 答案 PDF 第 98 页

As the number of assets in an equally weighted portfolio becomes large, the portfolio's variance of returns most likely approaches:

当等权重组合中的资产数量变得很大时，组合收益方差最可能趋近于什么？

A	Zero.	零。
B	the average variance of the individual assets' returns.	单个资产收益方差的平均值。
C	the average covariance between the individual assets' returns.	资产收益之间协方差的平均值。

答案 / ANSWER: C

等权重组合方差可写为 σ_p^2 = 平均方差 / N + (1 - 1/N) × 平均协方差。随着 N 增大，在平均方差有界的通常条件下，第一项趋近零，第二项趋近平均协方差。

For an equally weighted portfolio, σ_p^2 = average variance / N + (1 - 1/N) × average covariance. As N grows, with bounded average variance, the first term vanishes and the portfolio variance approaches the average covariance.

排除干扰项 / ELIMINATE DISTRACTORS

A 只有在平均协方差也趋近零等特殊情况下才成立；B 忽略了个别资产方差被分散化稀释。

A requires special conditions, such as average covariance also approaching zero. B ignores the dilution of individual-asset variance through diversification.

速记 / RECALL: 资产再多，仍留下共同变动的协方差。

Even many assets leave the covariance from common movements.

组合风险的计算

Calculating portfolio risk

Q024 重点例题 / KEY EXAMPLE

题库 PDF 第 188 页（章节第 5 页）| 答案 PDF 第 24 页

Two assets have the following characteristics. The variance of returns for an equally weighted portfolio of the two assets is closest to:

两项资产具有下表特征。由这两项资产组成的等权组合，其收益方差最接近多少？

指标 / Metric	数值 / Value
资产 1 收益方差 / Asset 1 variance	0.05
资产 2 收益方差 / Asset 2 variance	0.06
两资产收益相关系数 / Correlation	0.75

A	0.038	0.038。
B	0.048.	0.048。
C	0.055.	0.055。

答案 / ANSWER: B

先由方差求协方差：$Cov_{12}=\rho\sqrt{(Var_1 Var_2)}=0.75\sqrt{(0.05\times0.06)}=0.041079$。再算组合方差：$\sigma^2_p=0.5^2\times0.05+0.5^2\times0.06+2\times0.5\times0.5\times0.041079=0.048040$，选 B。题目问方差，不需开平方。

First obtain covariance: $Cov_{12}=\rho\sqrt{(Var_1 Var_2)}=0.75\sqrt{(0.05\times0.06)}=0.041079$. Then $\sigma^2_p=0.5^2\times0.05+0.5^2\times0.06+2\times0.5\times0.5\times0.041079=0.048040$. Choose B; no final square root is needed because variance is requested.

排除干扰项 / ELIMINATE DISTRACTORS

A：协方差交叉项漏乘 2，会误得约 0.038。C：0.055 是两个单项方差的简单平均，忽略了权重平方及协方差结构。

A omits the factor of 2 in the covariance cross-term and gives about 0.038. C is the simple average of individual variances and ignores squared weights and covariance.

速记 / RECALL: 给方差先开方得到 σ；问方差最后别再开方。

Convert variances to volatilities for covariance; do not square-root the final variance.

Q053 重点例题 / KEY EXAMPLE

题库 PDF 第 193 页（章节第 10 页）| 答案 PDF 第 53 页

The correlation of returns between two securities with equal standard deviation of returns is 0.75. If the covariance of returns is 5.5%², the standard deviation of returns for each security is closest to:

两只证券的收益率标准差相同，收益率相关系数为 0.75。若协方差为 5.5 %²，每只证券的收益率标准差最接近多少？

A	2.7%.	2.7%。
B	3.7%.	3.7%。
C	7.3%.	7.3%。

答案 / ANSWER: A

两只证券的标准差相同，故 Cov=相关系数×sigma²。因此 sigma=√(5.5/0.75)%=(√7.3333)%=2.708%，约为 2.7%。若全部用小数收益率计算，5.5 %²=5.5×0.01²=0.00055，同样得到 √(0.00055/0.75)=0.02708。

Equal standard deviations imply Cov=correlation×sigma². Thus sigma=√(5.5/0.75)%=2.708%, or about 2.7%. In decimal-return units, 5.5 %²=5.5×0.01²=0.00055, giving √(0.00055/0.75)=0.02708.

排除干扰项 / ELIMINATE DISTRACTORS

B 把 sigma×sigma 错当成 2×sigma，错误地用 7.3333/2=3.6667；C 在得到 7.3333 的方差数值后忘记开平方。

B treats sigma×sigma as 2×sigma and incorrectly divides 7.3333 by 2. C stops at the variance value of 7.3333 and fails to take its square root.

速记 / RECALL: 协方差单位要平方：1 %² = 0.0001。

Covariance has squared units: 1 %² = 0.0001.

概念辨析 / NOTE: 原题的 5.5%² 表示 5.5 个“百分比平方”单位，不是 (5.5%)²，也不是 0.055。按此单位，原答案 2.7% 一致。

The source’s 5.5%² means 5.5 units of percentage squared, not (5.5%)² and not 0.055. This interpretation is consistent with its 2.7% answer.

Q066 重点例题 / KEY EXAMPLE

题库 PDF 第 196 页（章节第 13 页）| 答案 PDF 第 66 页

A portfolio consists of two securities with the following characteristics. If the portfolio has an expected return of 12.6% and the returns of the two securities are uncorrelated, the portfolio's standard deviation is closest to:

某组合由下表中的两只证券构成。若组合预期收益率为 12.6%，且两只证券的收益率不相关，组合收益率的标准差最接近多少？

证券 / Security	预期收益率 / Expected return	标准差 / Standard deviation
证券 1 / Security 1	17%	24%
证券 2 / Security 2	6%	12%

A	13.4%.	13.4%。
B	15.2%.	15.2%。
C	19.2%.	19.2%。

答案 / ANSWER: B

先用收益率反推权重：12.6%=w×17%+(1-w)×6%，所以 w=(12.6%-6%)/(17%-6%)=0.6，另一权重为 0.4。因为相关系数为 0，组合方差=$0.6^2\times0.24^2+0.4^2\times0.12^2$=0.02304。开平方得 sigma_p=0.15179，约 15.2%。

First infer the weight from return: 12.6%=w×17%+(1-w)×6%, so w=(12.6%-6%)/(17%-6%)=0.6, leaving 0.4 for the other security. With zero correlation, variance=$0.6^2\times0.24^2+0.4^2\times0.12^2$=0.02304. Taking the square root gives sigma_p=0.15179, or about 15.2%.

排除干扰项 / ELIMINATE DISTRACTORS

A 相当于错误采用各 50% 权重；C 的 19.2% 是标准差加权平均 0.6×24%+0.4×12%，只有相关系数为 +1 时才适用。

A incorrectly assumes 50/50 weights. C is the weighted-average volatility, 0.6×24%+0.4×12%, which applies here only if correlation is +1.

速记 / RECALL: 收益率线性加权；风险先算方差再开平方。

Returns are weighted linearly; risk requires variance first, then a square root.

原文校勘 / NOTE: 题库第 196 页的表头错位。本讲义依据答案第 66 页的明确计算，将三列恢复为“证券、预期收益率、标准差”；数值保持 17%/24% 与 6%/12%。

The question table on PDF page 196 has misaligned headers. The explicit calculations on answer page 66 establish the intended columns as security, expected return, and standard deviation; values remain 17%/24% and 6%/12%.

Q090 重点例题 / KEY EXAMPLE

题库 PDF 第 200 页（章节第 17 页）| 答案 PDF 第 90 页

A portfolio consisting of two securities has the following characteristics. If the correlation of returns between the two securities is 0.20, the portfolio's standard deviation of returns is closest to:

一个组合由两只证券组成，数据如下。若两只证券收益的相关系数为 0.20，组合收益标准差最接近多少？

证券 / Security	组合权重 / Weight	收益标准差 / Standard deviation
1	40%	15%
2	60%	18%

A	1.8%.	1.8%。
B	9.1%.	9.1%。
C	13.4%.	13.4%。

答案 / ANSWER: C

$\sigma_p^2 = w1^2\sigma1^2 + w2^2\sigma2^2 + 2w1w2\rho12\sigma1\sigma2 = 0.4^2 \times 0.15^2 + 0.6^2 \times 0.18^2 + 2 \times 0.4 \times 0.6 \times 0.20 \times 0.15 \times 0.18 = 0.017856$。再开平方：$\sigma_p = \sqrt{0.017856} = 0.133626 \approx 13.4\%$。

$\sigma_p^2 = w1^2\sigma1^2 + w2^2\sigma2^2 + 2w1w2\rho12\sigma1\sigma2 = 0.4^2 \times 0.15^2 + 0.6^2 \times 0.18^2 + 2 \times 0.4 \times 0.6 \times 0.20 \times 0.15 \times 0.18 = 0.017856$. Then take the square root: $\sigma_p = \sqrt{0.017856} = 0.133626 \approx 13.4\%$.

排除干扰项 / ELIMINATE DISTRACTORS

A 把方差 0.017856 直接当标准差；B 在方差公式中误用标准差且没有正确开平方。

A treats the variance 0.017856 as a standard deviation. B incorrectly substitutes standard deviations for variances and fails to take the correct square root.

速记 / RECALL: 算组合风险：先算方差，再开平方。

Portfolio volatility: calculate variance first, then take its square root.

02 有效前沿、CAL 与 CML

Efficient frontier, CAL and CML

本单元 8 题 | 首轮预算 15 分钟 | 重点：Q6, Q48, Q68, Q100

对应阅读主题 / Reading theme: Portfolio Risk and Return: Parts I and II

2.1 先分清三个“最优”

Distinguish three different portfolio choices

概念 / Concept	含义 / Meaning
全局最小方差组合 / GMV portfolio	所有风险资产组合中方差最低；位于最小方差边界最左端。 The lowest-variance risky portfolio; the leftmost point of the minimum-variance frontier.
最优风险组合 / Optimal risky portfolio	从无风险利率出发，找到与有效前沿相切、Sharpe 最高的风险资产组合。 The tangency risky portfolio with the highest Sharpe ratio relative to the risk-free rate.
投资者最优完整组合 / Optimal complete portfolio	用无风险资产与最优风险组合配重，选出最适合该投资者风险厌恶程度的位置。 The investor-specific mix of the risk-free asset and the optimal risky portfolio.

最小方差边界：每个给定期望收益下，方差最小的风险资产组合。有效前沿是该边界从 GMV 向上的那一段；下半段会被同风险、更高收益的组合支配。它不是另一条位于边界左上的曲线。

The minimum-variance frontier contains the lowest-variance risky portfolio for each expected return. Its upper branch from the GMV portfolio is the efficient frontier. Lower-branch portfolios are dominated by higher-return portfolios with the same risk; the efficient frontier is not a separate curve to the upper left.

2.2 有无风险资产后，用一条直线配重

With a risk-free asset, allocation follows a straight line

$$E(Rc) = Rf + y[E(Rp)-Rf];\ \sigma c = |y|\sigma p$$

y 是风险组合权重，1-y 是无风险资产权重。在常见 y≥0 情形，σc=yσp。0<y<1 为借出/持有无风险资产；y=1 全部投风险组合；y>1 借入资金加杠杆。

y is the risky-portfolio weight and 1-y is the risk-free weight. For the usual y≥0 case, σc=yσp. With 0<y<1 the investor lends; y=1 is fully invested in the risky portfolio; y>1 represents borrowing and leverage.

$$\text{CAL slope} = [E(Rp)-Rf] / \sigma p = \text{Sharpe ratio}$$

CAL 可由任一风险组合与无风险资产组成；CML 是以市场组合为风险组合的特殊 CAL。横轴都是总风险 σ，纵轴都是期望收益。

A CAL combines a risk-free asset with a chosen risky portfolio. The CML is the special CAL using the market portfolio. Both plot expected return against total risk σ.

原题 Q6：18%=3%+y(15%-3%)，所以 y=1.25；市场组合 125%，无风险资产 -25%，即借款 25% 加杠杆，选 B。

Original Q6: 18%=3%+y(15%-3%), so y=1.25. The weights are 125% in the market and -25% in the risk-free asset: 25% borrowing, answer B.

同质预期等资本市场理论假设下，投资者共用同一个最优风险组合；个人风险偏好决定的是 y，而不是该风险组合的构成。风险越厌恶，在同一 CAL 上选择越靠左。无风险利率改变时，切点可能改变，所以最优风险组合依赖 Rf。对应 Q68、Q92、Q100。

Under capital-market-theory assumptions including homogeneous expectations, investors share the same optimal risky portfolio. Personal risk aversion determines y rather than the composition of that risky portfolio. Greater aversion means a position further left on the same CAL. A change in Rf can change the tangency point, so the optimal risky portfolio depends on Rf. See Q68, Q92 and Q100.

辨析 / PRECISION NOTE

Q5 与 Q20：原答案把无风险资产与风险资产称为“零相关”。严格地说，无风险资产标准差为 0，Pearson 相关系数分母为 0，因而未定义；真正可用于组合公式的是协方差为 0。做本题按原答案 B/A，理解时记零协方差。

Q5 and Q20 use “zero correlation” with a risk-free asset. Strictly, a constant return has zero standard deviation, making Pearson correlation undefined. Its covariance with risky returns is zero. The supplied answers are B/A; the mathematically precise portfolio statement is zero covariance.

辨析 / PRECISION NOTE

加入无风险资产会扩展可选组合，但把现金加入某个固定风险组合时，该 CAL 的 Sharpe 斜率不会自动上升；风险资产之间的相关性也不会因此改变。

Access to a risk-free asset expands the investment opportunity set. Mixing cash with a fixed risky portfolio does not automatically increase that CAL’s Sharpe slope, and it does not change correlations among the risky assets.

02 原题与逐题解析

Original questions and explanations

先作答，再看答案；标为 KEY EXAMPLE 的题目优先完成。
Attempt the question before reading the solution; prioritize items labeled KEY EXAMPLE.

有效前沿与最小方差

Efficient frontier and minimum variance

Q048 重点例题 / KEY EXAMPLE

题库 PDF 第 192 页（章节第 9 页）| 答案 PDF 第 48 页

The Markowitz efficient frontier is best described as a curve that:

马科维茨有效前沿最恰当的描述是哪一条曲线？

A	lies above and to the left of the minimum-variance frontier.	位于最小方差前沿的左上方。
B	connects the minimum-variance portfolios for all possible returns.	连接所有可能收益水平所对应的最小方差组合。
C	contains all portfolios of risky assets that rational, risk-averse investors will choose.	包含理性、风险厌恶投资者会选择的所有风险资产组合。

答案 / ANSWER: C

在只考虑风险资产的均值-方差框架内，有效前沿是最小方差前沿从全球最小方差组合向上的那一段。它提供同等风险下最高的预期收益，因此理性、风险厌恶投资者只会从这一段选择风险资产组合。

In the risky-assets-only mean-variance framework, the efficient frontier is the upper portion of the minimum-variance frontier beginning at the global minimum-variance portfolio. It offers the highest expected return for each risk level, so rational risk-averse investors choose risky portfolios on this portion.

排除干扰项 / ELIMINATE DISTRACTORS

A 错在把有效前沿当成另一条独立曲线；B 描述完整最小方差前沿，包含收益较低的无效下半支。

A incorrectly treats the efficient frontier as a separate curve. B describes the entire minimum-variance frontier, including its inefficient lower branch.

速记 / RECALL: 有效前沿 = 最小方差前沿的上半支。

The efficient frontier is the upper branch of the minimum-variance frontier.

概念辨析 / NOTE: “所有会选择的组合”限于本题风险资产组合的均值-方差框架；加入无风险资产后，最优完整组合可以位于 CAL/CML 上。

The statement concerns risky portfolios in the mean-variance setting; with a risk-free asset, the optimal complete portfolio may instead lie on the CAL/CML.

Q060 原题 / ORIGINAL QUESTION

题库 PDF 第 194 页（章节第 11 页）| 答案 PDF 第 60 页

The global minimum-variance portfolio is a portfolio that lies:

全球最小方差组合位于什么位置？

A	anywhere along the minimum-variance frontier.	最小方差前沿上的任意位置。
B	at the left-most point of the minimum-variance frontier.	最小方差前沿最左侧的点。
C	at the upper right-most point of the minimum-variance frontier.	最小方差前沿最右上方的点。

答案 / ANSWER: B

当横轴为标准差、纵轴为预期收益率时，全球最小方差组合在所有可行风险资产组合中具有最低方差，因此位于最左侧。它也是有效上半支与无效下半支的连接点。

With standard deviation on the horizontal axis and expected return on the vertical axis, the global minimum-variance portfolio has the lowest variance of all feasible risky portfolios and lies farthest left. It joins the efficient upper branch to the inefficient lower branch.

排除干扰项 / ELIMINATE DISTRACTORS

A 仅保证在某个目标收益下方差最小，不保证全局最小；C 通常对应较高风险。

A may minimize variance for a particular target return but not globally. C generally corresponds to higher risk.

速记 / RECALL: GMV 只问“全局最小风险”，找最左点。

For GMV, look for the globally lowest risk: the leftmost point.

Q047 原题 / ORIGINAL QUESTION

题库 PDF 第 192 页（章节第 9 页）| 答案 PDF 第 47 页

The capital market line most likely consists of portfolios that:

资本市场线（CML）最可能由哪些组合构成？

A	are fully diversified.	完全分散化的组合。
B	have zero systematic risk.	系统性风险为零的组合。
C	have nonsystematic risk equal to beta.	非系统性风险等于 beta 的组合。

答案 / ANSWER: A

CML 上的组合由无风险资产和市场组合构成。市场组合已经充分分散，因此这些组合不承担可分散的非系统性风险；持有市场组合的比例决定它们的市场风险大小。

CML portfolios combine the risk-free asset with the market portfolio. Because the market portfolio is fully diversified, these combinations contain no diversifiable nonsystematic risk; their allocation to the market determines their market-risk exposure.

排除干扰项 / ELIMINATE DISTRACTORS

B 仅适用于纯无风险资产这一特殊点；C 错把 beta 当成非系统性风险，beta 实际衡量系统性风险。

B holds only at the pure risk-free point. C incorrectly treats beta as nonsystematic risk; beta measures systematic risk.

速记 / RECALL: CML：无风险资产 + 完全分散的市场组合。

CML: the risk-free asset plus the fully diversified market portfolio.

无风险资产与最优组合

Risk-free asset and the optimal portfolio

Q005 原题 / ORIGINAL QUESTION

题库 PDF 第 184 页（章节第 1 页）| 答案 PDF 第 5 页

The correlation between the risk-free asset and the optimal risky portfolio is expected to be:

无风险资产与最优风险资产组合之间的相关系数预期为：

A	negative	负值。
B	zero.	零。
C	positive.	正值。

答案 / ANSWER: B

按本题及所附答案的考试表述，选 B。用于组合方差计算的关键性质是无风险收益的方差为零，且与风险资产收益的协方差为零，因此它不增加协方差项。

Under the wording used by this question and its supplied solution, choose B. The operative property is that a constant risk-free return has zero variance and zero covariance with risky returns, so it contributes no covariance term to portfolio variance.

排除干扰项 / ELIMINATE DISTRACTORS

A、C 都不符合本题所采用的“零相关”惯用表述。计算时应依赖零方差、零协方差，而非给常数收益估计相关系数。

A and C do not match the convention adopted by the question. For calculations, use zero variance and zero covariance rather than estimating correlation with a constant.

速记 / RECALL: 考试选 B；严格数学上记“协方差为零”。

Exam answer B; the rigorous property is zero covariance.

口径提醒 / NOTE: 严格而言，若无风险收益恒定，则 σ_f=0，Pearson 相关系数 Cov/(σ_fσ_p) 的分母为零，因此未定义；题库将其简称为“零相关”。

Strictly, with constant risk-free returns, σ_f=0 and Pearson correlation Cov/(σ_fσ_p) is undefined because its denominator is zero. The source uses “zero correlation” as shorthand.

Q006 重点例题 / KEY EXAMPLE

题库 PDF 第 185 页（章节第 2 页）| 答案 PDF 第 6 页

An investor who can lend and borrow at the risk-free rate builds a portfolio using the risk-free asset and the market portfolio. The risk-free rate is 3% and the expected market return is 15%. If the expected portfolio return is 18%, the investor's portfolio is:

某投资者可以按无风险利率借入和贷出资金，并以无风险资产和市场组合构建投资组合。无风险利率为 3%，市场预期收益率为 15%，组合预期收益率为 18%。该投资组合属于：

A	a lending portfolio.	贷出组合。
B	a leveraged portfolio.	杠杆组合。
C	the optimal risky portfolio.	最优风险资产组合。

答案 / ANSWER: B

设市场组合权重为 w：18%=3%+w(15%-3%)，所以 w=1.25。投资者把自身资本的 125% 投入市场组合，同时借入 25%（无风险资产权重为 -25%），因此属于杠杆组合。

Let w be the market-portfolio weight: 18%=3%+w(15%-3%), so w=1.25. The investor places 125% of equity in the market portfolio and borrows 25%, giving the risk-free asset a weight of -25%. This is a leveraged portfolio.

排除干扰项 / ELIMINATE DISTRACTORS

A：贷出组合的市场权重小于 100%。C：完全持有市场组合时，预期收益仅为 15%。

A: A lending portfolio invests less than 100% in the market. C: Holding only the market portfolio would produce a 15% expected return.

速记 / RECALL: 目标收益高于市场收益：借钱加杠杆。

Target return above the market return implies borrowing and leverage.

Q068 重点例题 / KEY EXAMPLE

题库 PDF 第 196 页（章节第 13 页）| 答案 PDF 第 68 页

If all investors have homogeneous expectations, the total risk and expected return of portfolios consisting of the risk-free asset and the optimal risky portfolio are plotted on the:

如果所有投资者具有同质预期，由无风险资产与最优风险组合构成的组合，其总风险和预期收益应绘制在哪条线上？

A	capital market line.	资本市场线（CML）。
B	security market line.	证券市场线（SML）。
C	security characteristic line.	证券特征线（SCL）。

答案 / ANSWER: A

在资本市场理论的同质预期等假设下，所有投资者选择相同的最优风险组合，即市场组合。无风险资产与市场组合的不同权重构成 CML：横轴是总风险 σ，纵轴是预期收益 E(R)。

Under the capital market assumptions, including homogeneous expectations, investors share the same optimal risky portfolio: the market portfolio. Combinations of this portfolio and the risk-free asset lie on the CML, with total risk σ on the horizontal axis and expected return E(R) on the vertical axis.

排除干扰项 / ELIMINATE DISTRACTORS

B 的横轴是 β，衡量系统性风险；C 的两轴分别为市场超额收益与证券超额收益。

B uses beta, a measure of systematic risk, on its horizontal axis. C plots security excess returns against market excess returns.

速记 / RECALL: 市场组合 + 无风险资产 + 横轴 σ = CML。

Market portfolio + risk-free asset + σ on the x-axis = CML.

Q092 原题 / ORIGINAL QUESTION

题库 PDF 第 201 页（章节第 18 页）| 答案 PDF 第 92 页

With respect to capital market theory, which of the following statements is most accurate?

关于资本市场理论，下列哪项表述最准确？

A	The optimal risky portfolio is dependent on the risk-free rate.	最优风险组合取决于无风险利率。
B	The optimal risky portfolio is dependent on the investor's risk profile.	最优风险组合取决于投资者的风险偏好。
C	The investor's optimal portfolio must lie on the Markowitz efficient frontier.	投资者的最优组合必须位于马科维茨有效前沿上。

答案 / ANSWER: A

在给定风险资产有效前沿时，无风险利率决定 CAL 在纵轴上的起点；起点变化通常会改变切点，即最优风险组合。投资者风险厌恶程度决定的是在该风险组合与无风险资产之间如何分配资金。

For a given efficient frontier of risky assets, the risk-free rate sets the CAL's vertical intercept. Changing this intercept generally changes the tangency point, the optimal risky portfolio. Investor risk aversion determines how much to allocate between that risky portfolio and the risk-free asset.

排除干扰项 / ELIMINATE DISTRACTORS

B 混淆共同的最优风险组合与个人最终组合；C 忽视无风险资产，最终组合通常在最优 CAL 上，仅恰好全投切点风险组合时也位于风险资产有效前沿。

B confuses the common optimal risky portfolio with an individual's final portfolio. C ignores the risk-free asset: the final portfolio generally lies on the optimal CAL and is also on the risky-asset frontier only at the tangency portfolio.

速记 / RECALL: Rf 影响切点；风险厌恶影响切点组合的持有权重。

Rf affects the tangency point; risk aversion affects the weight invested in it.

Q100 重点例题 / KEY EXAMPLE

题库 PDF 第 202 页（章节第 19 页）| 答案 PDF 第 100 页

Two investors have indifference curves that are tangent to the same capital allocation line (CAL). If Investor 1 is more risk averse than Investor 2, Investor 1's optimal portfolio is:

两位投资者的无差异曲线都与同一条资本配置线（CAL）相切。若投资者 1 比投资者 2 更厌恶风险，投资者 1 的最优组合位于何处？

A	to the left of Investor 2's optimal portfolio on the CAL.	在 CAL 上位于投资者 2 最优组合的左侧。
B	at the same point on the CAL as Investor 2's optimal portfolio.	与投资者 2 位于 CAL 上的同一点。
C	to the right of Investor 2's optimal portfolio on the CAL.	在 CAL 上位于投资者 2 最优组合的右侧。

答案 / **ANSWER: A**

CAL 的横轴是总风险。面对相同投资机会，更厌恶风险的投资者选择较低风险、较低预期收益的组合，即沿 CAL 向左移动；通常表现为较低的风险组合权重。

Total risk is on the CAL’s horizontal axis. Facing the same investment opportunities, the more risk-averse investor chooses lower risk and lower expected return, placing the optimal portfolio farther left on the CAL, generally with a smaller allocation to the risky portfolio.

排除干扰项 / ELIMINATE DISTRACTORS

B 忽略风险偏好的差异；C 对应较低风险厌恶、较高风险承担。

B ignores the difference in preferences. C corresponds to lower risk aversion and greater risk-taking.

速记 / RECALL: 风险厌恶更强，沿同一 CAL 向左选。

Greater risk aversion moves the optimum left along the same CAL.

03 CAPM、Beta 与收益生成模型

CAPM, beta and return-generating models

本单元 18 题 | 首轮预算 30 分钟 | 重点：Q40, Q52, Q79, Q81, Q87, Q93, Q94

对应阅读主题 / Reading theme: Portfolio Risk and Return: Part II

3.1 Beta 是“跟市场一起动”的程度

Beta measures exposure to market movements

$$\beta i = Cov(Ri,Rm)/\sigma m^2 = \rho im\ \sigma i/\sigma m$$

总风险大，不一定 Beta 大。Q40 中 β=0.6×25%/15%=1，股票总波动比市场大，但系统性风险与市场相同。

Higher total volatility does not necessarily imply higher beta. In Q40, β=0.6×25%/15%=1: the stock has greater total volatility but the same systematic risk as the market.

$$\rho im = \beta i\ \sigma m/\sigma i;\ Cov(Ri,Rm) = \beta i\ \sigma m^2$$

看到 Beta、两个标准差求相关，用左式；看到 Beta 和市场标准差求协方差，用右式。相关系数无单位，必须在 -1 到 1 之间。

Use the left equation when beta and both volatilities are given. Use the right equation to recover covariance from beta and market volatility. Correlation is unitless and must lie between -1 and 1.

Q87：ρ=0.35×18%/12%=0.525 → B。Q93：Cov=1.8×0.35²=0.2205 → A。Q93 的资产波动率 70% 不是这个最短解法所需的数据。

Q87: ρ=0.35×18%/12%=0.525 → B. Q93: Cov=1.8×0.35²=0.2205 → A. The asset volatility of 70% in Q93 is not needed for this shortest solution.

3.2 市场给什么风险付报酬

Which risk earns compensation

$$\sigma i^2 = \beta i^2 \sigma m^2 + \sigma e^2$$

在单因子市场模型、残差与市场不相关的假设下，总方差 = 系统性方差 + 非系统性方差。比较个股特有风险，要减去 $\beta^2\sigma m^2$，而不是比较 σ 或 β。

In the single-factor market model with residuals uncorrelated with the market, total variance equals systematic variance plus residual variance. Compare stock-specific risk after subtracting $\beta^2\sigma m^2$; do not compare σ or β alone.

原题 Q79：三只股票的残差方差分别为 0.0208、0.0203、0.0112，所以 Stock 1 特有风险最高，选 A。资本市场理论只为系统性风险提供风险溢价；可分散的特有风险不应获得额外均衡报酬。

Original Q79 gives residual variances of 0.0208, 0.0203 and 0.0112. Stock 1 therefore has the highest stock-specific risk, answer A. Capital market theory prices systematic risk; diversifiable idiosyncratic risk does not earn an additional equilibrium premium.

3.3 CAPM：先认清市场收益与市场风险溢价

CAPM: distinguish market return from market risk premium

$$E(Ri) = Rf + \beta i[E(Rm)-Rf] = Rf + \beta i \times MRP$$

MRP 是 market risk premium。若题目已给 MRP，不能再减一次 Rf。SML 的截距是 Rf，斜率是 MRP，横轴是 β。

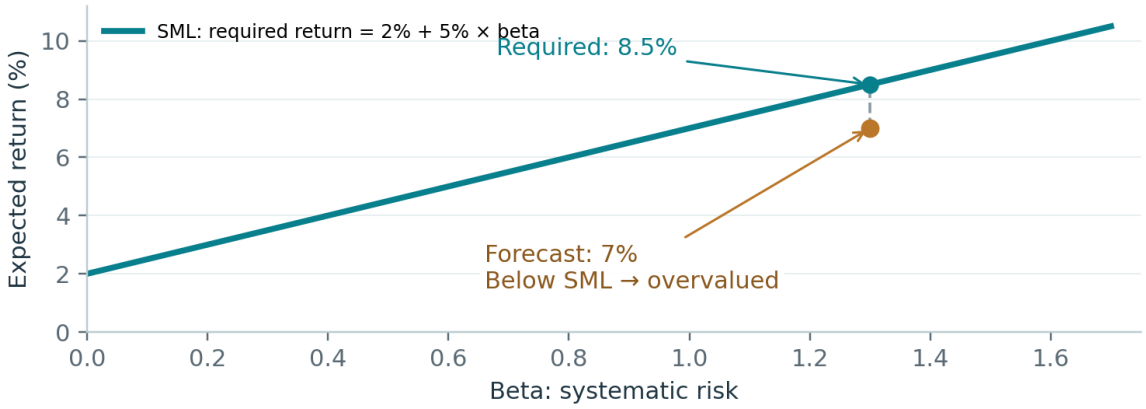
MRP means market risk premium. If MRP is already provided, do not subtract Rf again. The SML has intercept Rf, slope MRP and beta on the horizontal axis.

Q52：先算 β=0.8×35%/20%=1.4，再算 2%+1.4×(10%-2%)=13.2%，选 B。Q54：β=(5%-1%)/5%=0.8。Q94：MRP=(11%-2%)/1.5=6%。

Q52: first β=0.8×35%/20%=1.4, then 2%+1.4×(10%-2%)=13.2%, answer B. Q54: β=(5%-1%)/5%=0.8. Q94: MRP=(11%-2%)/1.5=6%.

判断估值时，把分析师预测收益与 CAPM 必要收益比较。预测收益高于必要收益：在 SML 上方，低估；预测收益低于必要收益：在 SML 下方，高估。Q81 的必要收益 2%+1.3×5%=8.5%，预测只有 7%，选高估 C。

For valuation, compare the analyst's forecast return with the CAPM required return. A forecast above the required return plots above the SML and indicates undervaluation; below it indicates overvaluation. Q81 requires 2%+1.3×5%=8.5%, versus a 7% forecast, so the stock is overvalued: C.



原题 Q81：SML 给出要求收益 8.5%；分析师预测 7%，位于线下。

Original Q81: the SML requires 8.5%; the analyst's 7% forecast lies below the line.

3.4 三条线与两个回归表达式

Three lines and two regression specifications

线 / Line	横轴 / x-axis	纵轴 / y-axis	斜率；截距 / Slope; intercept
CAL / CML	总风险 σ Total risk	期望收益 Expected return	Sharpe；Rf
SML	系统性风险 β Systematic risk	期望/必要收益 Expected/required return	MRP；Rf
SCL	市场超额收益 Rm-Rf	证券超额收益 Ri-Rf	β；Jensen α

CML 只描述无风险资产与市场组合形成的有效组合，其斜率是市场 Sharpe；SML 给出任一证券或组合的 CAPM 必要收益；SCL 则描述证券与市场超额收益的回归关系。

The CML describes efficient combinations of the risk-free asset and the market, with the market Sharpe ratio as its slope. The SML gives the CAPM required return for any security or portfolio. The SCL describes the regression relationship between security and market excess returns.

SCL: Ri - Rf = αi + βi(Rm - Rf) + εi

SCL 是超额收益回归线，其截距是 Jensen alpha。原始收益市场模型 Ri=ai+βiRm+εi 的截距 ai 通常不等于这个 alpha。

The SCL is a regression of excess returns, with Jensen's alpha as its intercept. The raw-return market model Ri=ai+βiRm+εi generally has a different intercept ai.

模型 / Model	看到什么就选它 / Recognition cue
统计因子 / Statistical factors	从观测收益的协方差结构提取因子，未必有清晰经济含义。 Extract factors from return covariance patterns; an economic interpretation is not required.
Fama-French 三因子 / Three-factor	市场、规模、价值。 Market, size and value.
Carhart 四因子 / Four-factor	三因子加 momentum 动量；Q29 选 B。 Adds momentum to the three factors; Q29 is B.
Market model 市场模型	用市场收益解释或预测某证券/公司的收益。 Use market returns to explain or forecast a security's returns.

辨析 / PRECISION NOTE

Q50 的 β<1 推导使用通常的正市场风险溢价假设。不要把“收益等于市场风险溢价”误读成“收益等于市场总收益”。

Q50's β<1 conclusion uses the usual positive market-risk-premium assumption. Do not confuse a return equal to the market risk premium with a return equal to the market's total return.

03 原题与逐题解析

Original questions and explanations

先作答，再看答案；标为 KEY EXAMPLE 的题目优先完成。
Attempt the question before reading the solution; prioritize items labeled KEY EXAMPLE.

Beta、协方差与系统性风险

Beta, covariance and systematic risk

Q040 重点例题 / KEY EXAMPLE

题库 PDF 第 191 页（章节第 8 页）| 答案 PDF 第 40 页

An analyst estimates the standard deviation of returns for the market portfolio to be 15% and the standard deviation of returns for a stock to be 25%. If the correlation of returns between the stock and the market portfolio is 0.6, the stock has:

市场组合收益率的标准差为 15%，某股票为 25%，两者的收益率相关系数为 0.6。该股票的系统性风险与市场组合相比如何？

A	less systematic risk than the market portfolio.	低于市场组合。
B	the same systematic risk as the market portfolio.	等于市场组合。
C	more systematic risk than the market portfolio.	高于市场组合。

答案 / **ANSWER: B**

先计算 beta：beta=相关系数×股票标准差÷市场标准差=0.6×25%÷15%=1。市场组合的 beta 也为 1，因此两者系统性风险相同。股票总波动率较高，额外部分可以来自非系统性风险。

Compute beta=correlation×stock volatility÷market volatility=0.6×25%÷15%=1. The market also has beta equal to 1, so systematic risk is the same. The stock can have higher total volatility because it also carries nonsystematic risk.

排除干扰项 / ELIMINATE DISTRACTORS

A、C 都不能仅凭 25% 与 15% 的标准差比较得出；必须同时考虑与市场的相关性。

Neither A nor C follows from comparing 25% with 15% alone; correlation with the market must also be included.

速记 / RECALL: 判断系统性风险看 beta，不单看标准差。

Compare systematic risk using beta, not standard deviation alone.

Q044 原题 / ORIGINAL QUESTION

题库 PDF 第 191 页（章节第 8 页）| 答案 PDF 第 44 页

According to capital market theory, the only type of risk that is priced is:

按照资本市场理论，唯一获得定价补偿的风险类型是什么？

A	systematic risk.	系统性风险。
B	diversifiable risk.	可分散风险。
C	idiosyncratic risk.	特有风险。

答案 / **ANSWER: A**

在资本市场理论的均衡框架中，投资者可以通过持有分散化组合消除个别公司的特有风险，因此不会仅因承担这部分风险而获得额外预期收益。无法通过分散化消除的系统性风险才获得风险溢价。

In capital market equilibrium, investors can diversify away company-specific risk and therefore do not earn an additional expected return merely for bearing it. Systematic risk cannot be diversified away and is the risk that earns a premium.

排除干扰项 / ELIMINATE DISTRACTORS

B、C 都属于可分散的非系统性风险。

B and C are both forms of diversifiable nonsystematic risk.

速记 / RECALL: CAPM 只补偿市场风险，不补偿持仓不分散。

CAPM rewards market risk, not a lack of diversification.

Q084 原题 / ORIGINAL QUESTION

题库 PDF 第 199 页（章节第 16 页）| 答案 PDF 第 84 页

According to capital market theory, an efficient market does not reward investors for taking on:

根据资本市场理论，有效市场不会因投资者承担下列哪种风险而给予额外预期回报？

A	market risk.	市场风险。
B	systematic risk.	系统性风险。
C	idiosyncratic risk.	特质风险。

答案 / **ANSWER: C**

特质风险又称非系统性风险、公司特有风险或可分散风险。投资者可以通过分散化降低它，因此资本市场理论不为承担这类风险提供额外预期风险溢价。

Idiosyncratic risk is also called nonsystematic, company-specific or diversifiable risk. Because diversification can reduce it, capital market theory provides no additional expected risk premium for bearing it.

排除干扰项 / ELIMINATE DISTRACTORS

A 市场风险和 B 系统性风险在此框架下指同一类不可分散风险，是风险溢价补偿的对象。

A, market risk, and B, systematic risk, refer to the non-diversifiable risk that is compensated in this framework.

速记 / RECALL: 可分散的风险，没有额外预期溢价。

Diversifiable risk earns no additional expected premium.

Q087 重点例题 / KEY EXAMPLE

题库 PDF 第 200 页（章节第 17 页）| 答案 PDF 第 87 页

An investor gathers the following information about a security and the market. The correlation between the security's returns and the market's returns is closest to:

投资者获得以下证券与市场的数据。证券收益与市场收益的相关系数最接近多少？

项目 / Item	数值 / Value
证券 Beta / Security beta	0.35
证券收益标准差 / Security return standard deviation	12%
市场收益标准差 / Market return standard deviation	18%

A	0.2.	0.2。
B	0.5.	0.5。
C	0.8	0.8。

答案 / ANSWER: B

βi = ρim × σi / σm，所以 ρim = βi × σm / σi = 0.35 × 18% / 12% = 0.525，最接近 0.5。

βi = ρim × σi / σm, so ρim = βi × σm / σi = 0.35 × 18% / 12% = 0.525, closest to 0.5.

排除干扰项 / ELIMINATE DISTRACTORS

A 常见错误是颠倒市场与证券的标准差；C 常见错误是把标准差比值误写成方差比值。

A can result from reversing the market and security standard deviations. C can result from using variances instead of standard deviations.

速记 / RECALL: 反求相关性：β × 市场波动 / 证券波动。

Recover correlation: β × market volatility / security volatility.

Q093 重点例题 / KEY EXAMPLE

题库 PDF 第 201 页（章节第 18 页）| 答案 PDF 第 93 页

An analyst gathers the following information. The covariance between the returns of the asset and the market is closest to:

分析师获得以下数据。该资产收益与市场收益之间的协方差最接近多少？

资产 / Asset	收益标准差 / Standard deviation	Beta / β
该资产 / Asset	70%	1.8
市场组合 / Market portfolio	35%	1.0

A	0.22.	0.22。
B	0.40.	0.40。
C	0.90.	0.90。

答案 / ANSWER: A

βi = Cov(Ri, Rm) / σm²，所以 Cov(Ri, Rm) = βiσm² = 1.8 × 0.35² = 0.2205，最接近 0.22。这里收益率以小数输入，协方差单位为小数收益率的平方。

βi = Cov(Ri, Rm) / σm², so Cov(Ri, Rm) = βiσm² = 1.8 × 0.35² = 0.2205, closest to 0.22. Returns are entered as decimals, so covariance is in squared decimal-return units.

排除干扰项 / ELIMINATE DISTRACTORS

B = β²σm²，是系统性方差；C = βσm / σi = 0.90，是相关系数，不是协方差。

B equals β²σm², systematic variance. C equals βσm / σi = 0.90, the correlation rather than covariance.

速记 / RECALL: 协方差 = β × 市场方差；β 不平方。

Covariance = β × market variance; do not square β.

Q079 重点例题 / KEY EXAMPLE

题库 PDF 第 198 页（章节第 15 页）| 答案 PDF 第 79 页

An analyst gathers the following information about three stocks. If the standard deviation of market returns is equal to 10%, the stock with the highest nonsystematic risk is:

分析师获得以下三只股票的数据。若市场收益标准差为 10%，哪只股票的非系统性风险最高？

股票 / Stock	Beta / β	收益标准差 / Return standard deviation
股票 1 / Stock 1	0.9	17%
股票 2 / Stock 2	1.1	18%
股票 3 / Stock 3	1.2	16%

A	Stock 1.	股票 1。
B	Stock 2.	股票 2。
C	Stock 3.	股票 3。

答案 / **ANSWER: A**

市场模型下，非系统性方差 = $\sigma i^2 - \beta i^2 \sigma m^2$。股票 1：$0.17^2 - 0.9^2 \times 0.10^2 = 0.0208$；股票 2：$0.18^2 - 1.1^2 \times 0.10^2 = 0.0203$；股票 3：$0.16^2 - 1.2^2 \times 0.10^2 = 0.0112$。股票 1 最大；比较方差即可判断标准差的大小。

Under the market model, nonsystematic variance = $\sigma i^2 - \beta i^2 \sigma m^2$. Stock 1: $0.17^2 - 0.9^2 \times 0.10^2 = 0.0208$. Stock 2: $0.18^2 - 1.1^2 \times 0.10^2 = 0.0203$. Stock 3: $0.16^2 - 1.2^2 \times 0.10^2 = 0.0112$. Stock 1 is highest; ranking variances also ranks their square roots.

排除干扰项 / ELIMINATE DISTRACTORS

B 的总风险最高，但并非非系统性风险最高；C 的 β 最大，反映的是系统性风险最高。

B has the highest total risk, not the highest nonsystematic risk. C has the highest beta and thus the highest systematic risk.

速记 / RECALL: 先用总方差减去系统性方差，不能只比 σ 或 β。

Subtract systematic variance from total variance; do not compare σ or β alone.

CAPM 与证券市场线

CAPM and the security market line

Q039 原题 / ORIGINAL QUESTION

题库 PDF 第 190 页（章节第 7 页）| 答案 PDF 第 39 页

The security market line plots the expected return of a portfolio against a measure of the portfolio's:

证券市场线（SML）把组合的预期收益率与组合的哪一种风险度量联系起来？

A	total risk	总风险。
B	systematic risk.	系统性风险。
C	unsystematic risk.	非系统性风险。

答案 / ANSWER: B

SML 的横轴是 beta，衡量资产或组合对市场风险的敏感度，即系统性风险。纵轴是预期收益率，方程为 E(R)=Rf+beta×[E(Rm)-Rf]。

The SML plots beta, the sensitivity to market risk, on the horizontal axis and expected return on the vertical axis. Its equation is E(R)=Rf+beta×[E(Rm)-Rf], so it prices systematic risk.

排除干扰项 / ELIMINATE DISTRACTORS

A 对应以标准差为横轴的 CAL/CML；C 是可分散风险，CAPM 不为它单独定价。

A is associated with CAL/CML diagrams that use standard deviation. C is diversifiable risk and receives no separate CAPM risk premium.

速记 / RECALL: SML 横轴 beta；CML 横轴标准差。

SML uses beta; CML uses standard deviation.

Q075 原题 / ORIGINAL QUESTION

题库 PDF 第 197 页（章节第 14 页）| 答案 PDF 第 75 页

The slope of the security market line is most likely the:

证券市场线（SML）的斜率最可能是什么？

A	security's beta.	证券的 Beta。
B	market risk premium.	市场风险溢价。
C	market risk premium divided by the market standard deviation.	市场风险溢价除以市场收益标准差。

答案 / ANSWER: B

SML 方程为 E(Ri) = Rf + βi[E(Rm) - Rf]。横轴是 β，所以每增加一单位 β，要求收益增加一个市场风险溢价；斜率就是 E(Rm) - Rf。

The SML is E(Ri) = Rf + βi[E(Rm) - Rf]. Because beta is on the horizontal axis, each additional unit of beta increases required return by the market risk premium. The slope is E(Rm) - Rf.

排除干扰项 / ELIMINATE DISTRACTORS

A 的 β 是 SCL 的斜率；C 的“市场风险溢价 / 市场标准差”是 CML 的斜率。

A, beta, is the SCL slope. C, the market risk premium divided by market standard deviation, is the CML slope.

速记 / RECALL: 先看横轴：SML 横轴 β，斜率市场溢价。

Check the x-axis: SML uses β, so its slope is the market premium.

Q052 重点例题 / KEY EXAMPLE

题库 PDF 第 193 页（章节第 10 页）| 答案 PDF 第 52 页

An analyst gathers the following information. According to the CAPM, the expected return of the security is closest to:

分析师收集了下表中的信息。根据 CAPM，该证券的预期收益率最接近多少？

指标 / Measure	数值 / Value
无风险利率 / Risk-free rate	2%
市场组合预期收益率 / Expected market return	10%
市场组合标准差 / Market standard deviation	20%
证券标准差 / Security standard deviation	35%
证券与市场的相关系数 / Security-market correlation	0.8

A	5.7%.	5.7%。
B	13.2%.	13.2%。
C	16.0%	16.0%。

答案 / ANSWER: B

第一步，beta=0.8×35%÷20%=1.4。第二步，市场风险溢价=10%-2%=8%。第三步，E(Ri)=2%+1.4×8%=13.2%，选 B。

First, beta=0.8×35%÷20%=1.4. Second, the market risk premium is 10%-2%=8%. Finally, E(Ri)=2%+1.4×8%=13.2%, so B is correct.

排除干扰项 / ELIMINATE DISTRACTORS

A 把证券与市场标准差的比值颠倒：beta=0.8×20%/35%=0.457，才得到约 5.7%；C 误把 10% 市场总收益当作溢价：2%+1.4×10%=16.0%。

A reverses the volatility ratio, giving beta=0.8×20%/35%=0.457 and a return near 5.7%. C wrongly uses the 10% market return as the premium: 2%+1.4×10%=16.0%.

速记 / RECALL: 先用相关系数求 beta，再把市场总收益减 Rf。

Compute beta from correlation, then subtract Rf from the market return.

Q054 原题 / ORIGINAL QUESTION

题库 PDF 第 193 页（章节第 10 页）| 答案 PDF 第 54 页

An analyst gathers the following information about an asset and the market. Based on the CAPM, the asset's beta is closest to:

分析师收集了下表中关于某资产与市场的信息。根据 CAPM，该资产的 beta 最接近多少？

指标 / Measure	数值 / Value
无风险利率 / Risk-free rate	1%
市场风险溢价 / Market risk premium	5%
资产预期收益率 / Expected asset return	5%

A	0.80.	0.80。
B	1.00.	1.00。
C	1.25.	1.25。

答案 / ANSWER: A

由 CAPM 移项：beta=[E(Ri)-Rf]÷MRP=(5%-1%)÷5%=0.80。题目已经直接给出市场风险溢价 5%，不需要再从它减去无风险利率。

Rearrange CAPM: beta=[E(Ri)-Rf]÷MRP=(5%-1%)÷5%=0.80. The 5% given is already the market risk premium, so do not subtract the risk-free rate from it again.

排除干扰项 / ELIMINATE DISTRACTORS

B 来自忽略证券收益中的无风险部分，或把已给出的溢价再次减 Rf；C 相当于把应有的比值颠倒。

B can result from ignoring the risk-free component of the asset return or subtracting Rf from the premium again. C reverses the required ratio.

速记 / RECALL: 看到 market risk premium，直接当 CAPM 分母。

When market risk premium is given, use it directly as the CAPM denominator.

Q094 重点例题 / KEY EXAMPLE

题库 PDF 第 201 页（章节第 18 页）| 答案 PDF 第 94 页

A security with a beta of 1.5 has an expected return of 11% according to the CAPM. If the risk-free rate is 2%, the market risk premium is closest to:

某证券 Beta 为 1.5，按 CAPM 计算的预期收益为 11%。若无风险利率为 2%，市场风险溢价最接近多少？

A	4.0%.	4.0%。
B	6.0%.	6.0%。
C	7.3%.	7.3%。

答案 / ANSWER: B

CAPM：E(Ri) = Rf + βi × MRP。移项得 MRP = [E(Ri) - Rf] / βi = (11% - 2%) / 1.5 = 6%。若问市场预期收益，则还要加回 Rf，得到 8%。

CAPM: E(Ri) = Rf + βi × MRP. Rearranging gives MRP = [E(Ri) - Rf] / βi = (11% - 2%) / 1.5 = 6%. The expected market return would be 8% after adding back Rf.

排除干扰项 / ELIMINATE DISTRACTORS

A 把已求出的风险溢价再次减去 Rf；C = 11% / 1.5，漏掉先减 Rf。

A subtracts Rf again from the already-calculated premium. C divides 11% by 1.5 without first subtracting Rf.

速记 / RECALL: 求市场溢价：先减 Rf，再除 β。

To find the market premium, subtract Rf and then divide by β.

Q050 原题 / ORIGINAL QUESTION

题库 PDF 第 192 页（章节第 9 页）| 答案 PDF 第 50 页

The expected return for a security is equal to the market's risk premium. If the risk-free rate is positive and the CAPM holds, the beta of the security is:

某证券的预期收益率等于市场风险溢价。若无风险利率为正且 CAPM 成立，该证券的 beta 是多少？

A	less than 1.	小于 1。
B	equal to 1.	等于 1。
C	greater than 1.	大于 1。

答案 / **ANSWER: A**

令市场风险溢价为 MRP。CAPM 给出 E(Ri)=Rf+beta×MRP，题目又给 E(Ri)=MRP，因此 beta=1-Rf/MRP。在通常假设 MRP>0 且 Rf>0 时，beta<1。

Let the market risk premium be MRP. CAPM gives E(Ri)=Rf+beta×MRP, while the question states E(Ri)=MRP. Thus beta=1-Rf/MRP. Under the usual positive-MRP assumption and Rf>0, beta is less than 1.

排除干扰项 / ELIMINATE DISTRACTORS

B 会得到证券收益等于整个市场预期收益 Rf+MRP；C 在正风险溢价下会要求比市场更高的收益。

B would make the security’s expected return equal to the full market return Rf+MRP. C would require a return above the market under a positive risk premium.

速记 / RECALL: 市场收益 E(Rm) 与市场风险溢价 E(Rm)-Rf 不能混用。

Do not confuse the market return E(Rm) with its risk premium E(Rm)-Rf.

口径提醒 / NOTE: 原题默认通常资本市场设定：市场风险溢价为正；若允许负溢价，单凭题面不足以得出相同结论。

The source assumes the usual positive market risk premium. If a negative premium were allowed, the same conclusion would not follow from the stated information alone.

Q081 重点例题 / KEY EXAMPLE

题库 PDF 第 199 页（章节第 16 页）| 答案 PDF 第 81 页

An analyst gathers the following information about the security market line (SML) and a stock. According to capital market theory, if the analyst believes the stock will have a return of 7%, the stock is:

分析师获得以下 SML 与某股票的数据。若分析师预计该股票收益率为 7%，按资本市场理论，该股票的估值状况如何？

项目 / Item	数值 / Value
SML 截距 / SML intercept	2%
SML 斜率 / SML slope	5%
股票 Beta / Stock beta	1.3

A	undervalued.	低估。
B	properly valued.	合理估值。
C	overvalued.	高估。

答案 / ANSWER: C

SML 截距是 Rf = 2%，斜率是市场风险溢价 = 5%。要求收益率 = 2% + 1.3 × 5% = 8.5%。预测收益 7% 低于要求收益 8.5%，对应点在 SML 下方，说明当前价格过高。

The SML intercept is Rf = 2%, and its slope is the market risk premium of 5%. Required return = 2% + 1.3 × 5% = 8.5%. The forecast return of 7% is below the required 8.5%, placing the stock below the SML and indicating overvaluation.

排除干扰项 / ELIMINATE DISTRACTORS

A 常见误算是把已经给出的 5% 风险溢价再次减去 Rf；B 常见误算是漏乘 β，只算 2% + 5%。

A can result from subtracting Rf again from the already-given market premium. B can result from omitting beta and calculating only 2% + 5%.

速记 / RECALL: 预测收益低于要求收益 → SML 下方 → 高估。

Forecast return below required return → below the SML → overvalued.

回归线与因子模型

Regression lines and factor models

Q043 原题 / ORIGINAL QUESTION

题库 PDF 第 191 页（章节第 8 页）| 答案 PDF 第 43 页

Which of the following lines is plotted on a graph with the excess return of a security on the y-axis and the excess return of the market on the x-axis?

在纵轴为某证券超额收益、横轴为市场超额收益的图中，所绘制的是哪条线？

A	Capital market line	资本市场线。
B	Security market line	证券市场线。
C	Security characteristic line	证券特征线。

答案 / ANSWER: C

证券特征线（SCL）描述证券超额收益与市场超额收益的回归关系：(Ri-Rf)=alpha+beta×(Rm-Rf)+误差。其斜率为 beta，截距为 alpha。

The security characteristic line (SCL) is the regression of a security's excess return on the market's excess return: (Ri-Rf)=alpha+beta×(Rm-Rf)+error. Its slope is beta and its intercept is alpha.

排除干扰项 / ELIMINATE DISTRACTORS

A 的横轴为总风险标准差；B 的横轴为 beta，而不是市场超额收益。

A uses total-risk standard deviation on the horizontal axis. B uses beta rather than market excess return.

速记 / RECALL: SCL 看收益对收益；SML 看收益对 beta。

SCL relates return to return; SML relates return to beta.

Q073 原题 / ORIGINAL QUESTION

题库 PDF 第 197 页（章节第 14 页）| 答案 PDF 第 73 页

The intercept on the y-axis of the security characteristic line is:

证券特征线（SCL）在纵轴上的截距是什么？

A	beta	Beta。
B	Jensen's alpha	詹森 Alpha。
C	the risk-free rate of return.	无风险收益率。

答案 / ANSWER: B

超额收益形式的 SCL 为 Ri - Rf = αi + βi(Rm - Rf) + εi。纵轴截距为 Jensen's alpha；斜率为 β。

In excess-return form, the SCL is Ri - Rf = αi + βi(Rm - Rf) + εi. Its vertical intercept is Jensen's alpha and its slope is beta.

排除干扰项 / ELIMINATE DISTRACTORS

A 把斜率误当截距；C 的 Rf 是 SML 的截距，并非超额收益 SCL 的截距。

A confuses the slope with the intercept. C is the intercept of the SML, not of the excess-return SCL.

速记 / RECALL: SCL：截距 α，斜率 β。

SCL: intercept α; slope β.

Q064 原题 / ORIGINAL QUESTION

题库 PDF 第 195 页（章节第 12 页）| 答案 PDF 第 64 页

The market model is most likely used to predict

市场模型最可能用于预测什么？

A	market returns in a future period.	未来某期的市场收益率。
B	economic growth in a future period.	未来某期的经济增长。
C	company-specific returns in a future period.	未来某期某公司的股票收益率。

答案 / ANSWER: C

市场模型将某证券收益率写作 Ri=alpha_i+beta_i×Rm+误差。给定或预测市场收益 Rm 后，可用于估计该公司股票的收益；它本身不是市场收益或 GDP 增长的预测模型。

The market model expresses a security's return as Ri=alpha_i+beta_i×Rm+error. Given an assumed or forecast market return, it can estimate that company's stock return. It does not itself forecast the market return or GDP growth.

排除干扰项 / ELIMINATE DISTRACTORS

A 把解释变量 Rm 当成被预测对象；B 不属于这一证券收益模型的直接用途。

A treats the explanatory market return as the forecast target. B is outside the model's direct purpose.

速记 / RECALL: Market model：用市场收益解释单只证券收益。

The market model uses market return to explain an individual security's return.

概念辨析 / NOTE: 此处 company-specific returns 指"该公司证券的收益"，不等于可以预测模型中的公司特有随机误差。

Here company-specific returns means the return on that company's security, not a prediction of its idiosyncratic random residual.

Q025 原题 / ORIGINAL QUESTION

题库 PDF 第 188 页（章节第 5 页）| 答案 PDF 第 25 页

With respect to return-generating models, statistical factor models:

关于收益生成模型，统计因子模型：

A	only include factors that have economic meaning.	只包含有经济含义的因子。
B	identify factors that explain the covariance in observed returns.	识别能够解释已观察收益协方差的因子。
C	only include factors that have a fundamental connection to returns.	只包含与收益存在基本面联系的因子。

答案 / ANSWER: B

统计因子模型从收益数据的协方差结构提取共同因子，例如主成分或因子分析。因子可以有效解释数据，但未必具有清楚的经济含义或基本面解释。

Statistical factor models extract common factors from the covariance structure of observed returns, for example through principal-components or factor analysis. These factors may explain the data without having a clear economic or fundamental interpretation.

排除干扰项 / ELIMINATE DISTRACTORS

A：统计提取并不要求每个因子都有经济解释。C：这是把统计模型与基本面因子模型混淆。

A: Statistical extraction need not yield an economic interpretation. C confuses statistical factors with fundamental factors.

速记 / RECALL: 统计模型：先从数据找共同变动，再谈解释。

Statistical models identify common variation in data; interpretation is secondary.

Q029 原题 / ORIGINAL QUESTION

题库 PDF 第 189 页（章节第 6 页）| 答案 PDF 第 29 页

Momentum, defined as relative past stock returns, is most likely a factor in:

动量，即股票过去的相对收益，最可能是哪一种模型的因子？

A	fundamental factor models.	基本面因子模型。
B	the Carhart four-factor model.	Carhart 四因子模型。
C	the Fama-French three-factor model.	Fama-French 三因子模型。

答案 / ANSWER: B

Carhart 四因子模型在 Fama-French 的市场、规模和价值三个因子之外，再加入动量因子。因此见到“过去相对收益”或 momentum，优先联想到第四个因子。

The Carhart model adds momentum to the Fama-French market, size, and value factors. Thus, relative past stock returns identify the fourth Carhart factor.

排除干扰项 / ELIMINATE DISTRACTORS

A：动量不以公司基本面特征定义。C：经典 Fama-French 三因子模型不含动量因子。

A: Momentum is not defined by company fundamentals. C: The classic Fama-French three-factor model does not include momentum.

速记 / RECALL: FF3：市场、规模、价值；Carhart 再加动量。

FF3: market, size, value; Carhart adds momentum.

04 风险调整绩效评价

Risk-adjusted performance evaluation

本单元 7 题 | 首轮预算 20 分钟 | 重点：Q9, Q27, Q70, Q78, Q95

对应阅读主题 / Reading theme: Portfolio Risk and Return: Part II

这一单元先问：投资者承受的是组合全部风险，还是把组合视为已充分分散后的一个组成部分？未充分分散时，不能忽略特有风险。

First ask whether the investor bears the portfolio's entire risk or views it within a fully diversified investment setting. An undiversified portfolio's idiosyncratic risk cannot be ignored.

指标 / Measure	公式 / Formula	口径与用途 / Risk basis and use
Sharpe ratio 夏普比率	(Rp-Rf)/σp	总风险；CAL 斜率；未充分分散的组合优先。 Total risk; CAL slope; appropriate for an undiversified portfolio.
Treynor ratio 特雷诺比率	(Rp-Rf)/βp	系统性风险；通常用于已充分分散的组合。 Systematic risk; usually for well-diversified portfolios.
Jensen alpha 詹森阿尔法	Rp-[Rf+βp(Rm-Rf)]	实际收益减 CAPM 要求收益；系统性风险调整。 Actual minus CAPM-required return; adjusts for systematic risk.
M² 风险调整收益	Rf+Sharpe_p×σm	把组合风险调到市场标准差后的收益水平。 Return after scaling portfolio volatility to market volatility.
M² alpha	M²-Rm	再减市场收益，才得到超额绩效。 Subtract market return to obtain relative risk-adjusted performance.

4.1 最常考的四种代入

Four recurring calculation patterns

Q70：Sharpe=(15.2%-3.1%)/11.7%=1.034，选 A。它是比率，不能把 1.034 再写成 1.034% 当作同一含义。

Q70: Sharpe=(15.2%-3.1%)/11.7%=1.034, answer A. This is a ratio; 1.034 and 1.034% do not mean the same thing.

Q27：Jensen α=7%-[1%+1.2×(5%-1%)]=1.2%，选 B。中括号是 CAPM 要求的 5.8%，不是市场收益 5%。

Q27: Jensen α=7%-[1%+1.2×(5%-1%)]=1.2%, answer B. The bracketed 5.8% is the CAPM required return, not the market return of 5%.

Q9：M²=2%+0.8×12%=11.6%；题目问 M² alpha，还要减 8%，得到 3.6%，选 A。

Q9: M²=2%+0.8×12%=11.6%. The question asks for M² alpha, so subtract 8% to obtain 3.6%, answer A.

Q95：先用 Sharpe×σp=0.8×19%=15.2% 得到组合超额收益；再算 β=0.7×19%/10%=1.33；Treynor=0.152/1.33=0.1143，选 B。不需要单独知道 Rf。

Q95: recover portfolio excess return as Sharpe×σp=0.8×19%=15.2%. Then β=0.7×19%/10%=1.33, so Treynor=0.152/1.33=0.1143, answer B. Rf is not separately needed.

4.2 一眼选指标

Choose the metric from the wording

出现 not fully diversified → Sharpe。出现 only systematic risk → 在给定选项中找 Beta 类指标，如 Jensen alpha 或 Treynor。出现 slope of CAL → Sharpe。出现 risk-adjusted return minus market → M^2 alpha。

Not fully diversified points to Sharpe. Only systematic risk points to a beta-based measure such as Jensen's alpha or Treynor. Slope of CAL means Sharpe. Risk-adjusted return minus the market means M^2 alpha.

辨析 / PRECISION NOTE

本讲义把 M^2 收益水平与 M^2 alpha 分开，与 Q9 原答案一致。不同资料可能直接把相对市场差值称 M^2；读题时以定义为准。Q76 原解析对 Treynor 与 SML 斜率的说法过宽：一般组合的 Treynor 不一定等于 MRP；按预期收益计算、满足 CAPM 公允定价且 $\beta \neq 0$ 时才相等。

This guide distinguishes the M^2 return level from M^2 alpha, following Q9. Other sources may use M^2 for the difference from the market. Q76's original explanation overstates the Treynor/SML relationship: a general portfolio's Treynor need not equal MRP; equality holds when computed from expected returns under CAPM fair pricing with $\beta \neq 0$.

04 原题与逐题解析

Original questions and explanations

先作答，再看答案；标为 KEY EXAMPLE 的题目优先完成。
Attempt the question before reading the solution; prioritize items labeled KEY EXAMPLE.

先选对风险口径

Choose the risk measure first

Q030 原题 / ORIGINAL QUESTION

题库 PDF 第 189 页（章节第 6 页）| 答案 PDF 第 30 页

Which of the following measures uses only systematic risk to evaluate portfolio performance?

下列哪一种指标仅使用系统性风险评价组合绩效？

A	M^2	M^2。
B	Sharpe ratio	夏普比率。
C	Jensen's alpha	Jensen alpha。

答案 / ANSWER: C

Jensen alpha 以 CAPM 为基准，用 beta 代表系统性风险，比较实际收益与相应要求收益。Sharpe 和 M^2 则依赖标准差，衡量总风险。

Jensen's alpha benchmarks return against CAPM, using beta to measure systematic risk. Sharpe and M^2 depend on standard deviation and therefore use total risk.

排除干扰项 / ELIMINATE DISTRACTORS

A、B 都使用总波动率 σ，而不是只使用 beta。

A and B use total volatility σ rather than beta alone.

速记 / RECALL: σ：Sharpe、M^2；β：Jensen、Treynor。

σ: Sharpe and M^2; β: Jensen and Treynor.

Q078 重点例题 / KEY EXAMPLE

题库 PDF 第 198 页（章节第 15 页）| 答案 PDF 第 78 页

Which of the following measures is most appropriate to evaluate the performance of a portfolio that is not fully diversified?

评价一个尚未充分分散化的组合，最适合使用哪个绩效指标？

A	Sharpe ratio	夏普比率。
B	Treynor ratio	特雷诺比率。
C	Jensen's alpha	詹森 Alpha。

答案 / ANSWER: A

未充分分散化意味着组合仍承担显著的非系统性风险，此时应评价总风险。Sharpe 使用标准差，包含系统性与非系统性风险，因此最合适。

An incompletely diversified portfolio still bears material nonsystematic risk, so total risk is relevant. Sharpe uses standard deviation, covering both systematic and nonsystematic risk, and is therefore the most appropriate choice.

排除干扰项 / ELIMINATE DISTRACTORS

B 和 C 以 β 所反映的系统性风险为基础，更适合用于已充分分散化、非系统性风险很小的组合。

B and C evaluate performance relative to systematic risk measured by beta and are more suitable when nonsystematic risk is negligible.

速记 / RECALL: 没分散好，用总风险 σ；优先 Sharpe。

Not well diversified: evaluate total risk σ; prefer Sharpe.

Q076 原题 / ORIGINAL QUESTION

题库 PDF 第 197 页（章节第 14 页）| 答案 PDF 第 76 页

Which of the following performance measures is equal to the slope of the capital allocation line?

下列哪个绩效指标等于资本配置线（CAL）的斜率？

A	M^2	M^2 指标。
B	Sharpe ratio	夏普比率。
C	Treynor ratio	特雷诺比率。

答案 / ANSWER: B

CAL 连接无风险资产与某一风险组合，纵轴是预期收益，横轴是总风险 σ。其斜率为 [E(Rp) - Rf] / σp，正是该风险组合的 Sharpe ratio。

The CAL connects the risk-free asset and a specified risky portfolio, with expected return on the vertical axis and total risk σ on the horizontal axis. Its slope is [E(Rp) - Rf] / σp, the risky portfolio's Sharpe ratio.

排除干扰项 / ELIMINATE DISTRACTORS

A 的 M^2 以收益率单位表示风险调整结果；C 的 Treynor 以 β 为风险分母，不能作为以 σ 为横轴的 CAL 斜率。

A, M^2, expresses a risk-adjusted result in return units. C, Treynor, uses beta rather than σ as its risk denominator and therefore is not the slope of the CAL.

速记 / RECALL: CAL 斜率 = 超额收益 / 总风险 = Sharpe。

CAL slope = excess return / total risk = Sharpe.

概念辨析 / NOTE: 答案页把 Treynor ratio 概括为 SML 斜率，表述过宽。SML 的斜率恒为市场风险溢价；只有满足 CAPM 公允定价且 β 非零时，组合的预期 Treynor 比率才等于该斜率。

The answer page loosely identifies the Treynor ratio with the SML slope. The SML slope is the market risk premium; a portfolio's expected Treynor ratio equals it only under CAPM fair pricing with nonzero beta.

四种绩效指标的计算

Calculate the four performance measures

Q070 重点例题 / KEY EXAMPLE

题库 PDF 第 196 页（章节第 13 页）| 答案 PDF 第 70 页

A portfolio has an annual return of 15.2% and a standard deviation of returns of 11.7%. If the risk-free rate is 3.1%, the portfolio's Sharpe ratio is closest to:

某组合年收益率为 15.2%，收益标准差为 11.7%，无风险利率为 3.1%。其夏普比率最接近多少？

A	1.03.	1.03。
B	1.30.	1.30。
C	1.56.	1.56。

答案 / ANSWER: A

Sharpe = (Rp - Rf) / σp = (15.2% - 3.1%) / 11.7% = 12.1% / 11.7% = 1.0342，约为 1.03。它表示每承担一单位总风险所获得的超额收益。

Sharpe = (Rp - Rf) / σp = (15.2% - 3.1%) / 11.7% = 12.1% / 11.7% = 1.0342, approximately 1.03. It measures excess return per unit of total risk.

排除干扰项 / ELIMINATE DISTRACTORS

B = 15.2% / 11.7%，漏减无风险利率；C = (15.2% + 3.1%) / 11.7%，错误地加上无风险利率。

B omits the subtraction of the risk-free rate. C adds the risk-free rate instead of subtracting it.

速记 / RECALL: Sharpe 分子先减 Rf，分母用标准差。

For Sharpe, subtract Rf first and divide by standard deviation.

Q027 重点例题 / KEY EXAMPLE

题库 PDF 第 188 页（章节第 5 页）| 答案 PDF 第 27 页

An analyst gathers the following information about a portfolio and the market. Jensen's alpha for the portfolio is:

某分析师收集了如下组合与市场数据。该组合的 Jensen alpha 为：

指标 / Metric	数值 / Value
组合收益 / Portfolio return	7.0%
市场收益 / Market return	5.0%
无风险利率 / Risk-free rate	1.0%
组合 beta / Portfolio beta	1.2

A	0.0%.	0.0%。
B	1.2%.	1.2%。
C	2.2%.	2.2%。

答案 / ANSWER: B

CAPM 要求收益=R_f+β_p(R_m-R_f)=1%+1.2×(5%-1%)=5.8%。Jensen alpha=实际组合收益-CAPM 要求收益=7%-5.8%=1.2%。

The CAPM required return is R_f+β_p(R_m-R_f)=1%+1.2×(5%-1%)=5.8%. Jensen's alpha is the portfolio return minus this required return: 7%-5.8%=1.2%.

排除干扰项 / ELIMINATE DISTRACTORS

A：把市场收益 5% 错当市场风险溢价，会误算 7%-(1%+1.2×5%)=0%。C：若漏减 CAPM 中的无风险收益 1%，就会误得 2.2%。

A uses the 5% market return instead of the 4% market risk premium, giving 7%-(1%+1.2×5%)=0%. C omits the risk-free component from required return and incorrectly gives 2.2%.

速记 / RECALL: Alpha = 实际收益 - 与 beta 匹配的要求收益。

Alpha = actual return - the required return for the portfolio's beta.

Q009 重点例题 / KEY EXAMPLE

题库 PDF 第 185 页（章节第 2 页）| 答案 PDF 第 9 页

The market has a return of 8% and a standard deviation of returns of 12%. The risk-free rate is 2%. If a portfolio has a Sharpe ratio of 0.8, the portfolio's M2 alpha is closest to:

市场收益率为 8%，收益标准差为 12%，无风险利率为 2%。某组合的夏普比率为 0.8，该组合的 M² alpha 最接近：

A	3.6%.	3.6%。
B	5.6%.	5.6%。
C	11.6%.	11.6%。

答案 / ANSWER: A

先把组合调整到与市场相同的总风险：调整后收益=R_f+Sharpe_p×σ_m=2%+0.8×12%=11.6%。M² alpha 再减市场收益：11.6%-8%=3.6%。

First scale the portfolio to market volatility: adjusted return=R_f+Sharpe_p×σ_m=2%+0.8×12%=11.6%. M² alpha is the adjusted return minus the market return: 11.6%-8%=3.6%.

排除干扰项 / ELIMINATE DISTRACTORS

B：错误地减去市场风险溢价 6%，而不是市场收益 8%，得到 5.6%。C：11.6% 是调整后收益，还没有减去市场收益。

B subtracts the 6% market risk premium instead of the 8% market return, incorrectly giving 5.6%. C is the adjusted portfolio return before subtracting the market return.

速记 / RECALL: M² alpha = R_f + Sharpe_p×σ_m - R_m。

M² alpha = R_f + Sharpe_p×σ_m - R_m.

概念辨析 / NOTE: 文献中 M² 既可能指调整后的收益，也可能指相对市场的差额。本题明确问 M² alpha，因此必须减去市场收益。

M² terminology sometimes denotes adjusted return and sometimes its difference from the market. This item explicitly asks for M² alpha, so subtract the market return.

Q095 重点例题 / KEY EXAMPLE

题库 PDF 第 201 页（章节第 18 页）| 答案 PDF 第 95 页

An analyst gathers the following information about a portfolio and the market. The portfolio's Treynor ratio is closest to:

分析师获得以下组合与市场数据。该组合的特雷诺比率最接近多少？

项目 / Item	数值 / Value
组合 Sharpe / Portfolio Sharpe ratio	0.8
组合波动率 / Portfolio volatility	19%
市场波动率 / Market volatility	10%
组合与市场相关性 / Portfolio-market correlation	0.7

A	0.060.	0.060。
B	0.114.	0.114。
C	0.413.	0.413。

答案 / **ANSWER: B**

第一步，超额收益 = Sharpe × σp = 0.8 × 0.19 = 0.152。第二步，βp = ρpm × σp / σm = 0.7 × 0.19 / 0.10 = 1.33。第三步，Treynor = (Rp - Rf) / βp = 0.152 / 1.33 = 0.1143，约为 0.114。

Step 1: excess return = Sharpe × σp = 0.8 × 0.19 = 0.152. Step 2: βp = ρpm × σp / σm = 0.7 × 0.19 / 0.10 = 1.33. Step 3: Treynor = (Rp - Rf) / βp = 0.152 / 1.33 = 0.1143, approximately 0.114.

排除干扰项 / ELIMINATE DISTRACTORS

A 在 β 公式里误用方差比值；C 把市场与组合的标准差比值颠倒。

A uses variances instead of standard deviations in beta. C reverses the portfolio and market standard-deviation ratio.

速记 / RECALL: Sharpe 先还原超额收益；相关性与波动算 β；最后除 β。

Recover excess return from Sharpe, calculate β from correlation and volatilities, then divide by β.

05 IPS、风险承受能力与资产配置

IPS, risk tolerance and asset allocation

本单元 24 题 | 首轮预算 30 分钟 | 重点：Q14, Q37, Q38, Q56, Q62, Q77, Q83, Q86, Q91, Q97

对应阅读主题 / Reading theme: Basics of Portfolio Planning and Construction

5.1 投资政策说明书是客户需求的落地文件

The IPS turns the client’s needs into portfolio instructions

IPS（investment policy statement）先写目标，再写约束。目标是风险与收益；约束可记为流动性、期限、税务、法律监管、独特情况。收益要求不能脱离风险能力，所选相对业绩基准应具有可投资性。

The IPS sets objectives and constraints. Objectives cover risk and return. Constraints cover liquidity, time horizon, taxes, legal/regulatory restrictions and unique circumstances. Return requirements must be consistent with risk capacity, and a relative-return benchmark should be investable.

区分 / Distinction	判断依据 / What to look for
承担风险的能力 / Ability or capacity	客观财务条件：投资期限、资产相对负债、未来现金需求。期限更长、财务缓冲更厚，其他条件不变时能力通常更高。 Objective finances: horizon, assets relative to liabilities and cash needs. Longer horizons and stronger financial cushions generally support more capacity.
承担风险的意愿 / Willingness	主观心理态度：害怕损失的程度、风险厌恶、问卷反应。 Subjective attitude: fear of losses, risk aversion and questionnaire responses.
整体风险承受度 / Overall tolerance	同时考虑能力与意愿；高意愿不能弥补支付必要支出所需资金不足。 Consider both. Strong willingness cannot compensate for insufficient capacity to meet essential spending.

Q37、Q46、Q51 都用时间期限考 ability；Q49 用 wealth relative to liabilities 考 ability。不要看到客户“很自信”就判断其客观风险能力高。

Q37, Q46 and Q51 use horizon to test ability; Q49 uses wealth relative to liabilities. A client’s self-confidence does not establish objective risk capacity.

约束可以由客户设定，也可以由法律强制。养老金高风险资产的法定上限属于法律监管约束；为节税避开利息收入属于税务约束；客户自行限制房地产投资不自动属于监管约束。对应 Q45、Q59、Q85。

Constraints can be client-imposed or legally required. A statutory cap on pension holdings of high-risk assets is regulatory; avoiding interest income for tax reasons is a tax constraint. A voluntary real-estate limit is not automatically a regulatory constraint. See Q45, Q59 and Q85.

5.2 IPS 章节定位题：按动词找位置

IPS section questions: use the action verb

章节 / Section	负责什么 / What it covers	原题 / Questions
Objectives and constraints 目标与约束	客户的具体需求；风险、收益及限制。 Client-specific needs, objectives and limitations.	Q74
Statement of duties and responsibilities 职责与责任	谁决策、谁执行、经理有多大自由裁量权。 Who does what and the manager's discretion.	Q77
Procedures 程序	IPS 如何以及何时更新、复核。 How and when the IPS itself is reviewed.	Q62
Investment guidelines 投资指引	允许的资产、策略、衍生品使用及组合操作边界。 Permitted assets, strategies, derivatives and portfolio operating limits.	Q56
Evaluation and review 评价与复核	怎样评价投资结果、反馈业绩。 How investment results are evaluated and fed back.	Q38

辨析 / PRECISION NOTE

Q38 的 review 是评价投资结果；Q62 的 review 是复核 IPS 文件。只背英文 review 一个词容易选错。

In Q38, review concerns investment results. In Q62, it concerns the IPS document itself. Memorizing the word review without its object leads to mistakes.

5.3 长期配置、短期偏离与再平衡

Long-term allocation, tactical deviations and rebalancing

战略资产配置 SAA：把 IPS 的目标与约束，加上长期资本市场预期，转换为资产类别政策权重。战术资产配置 TAA：在允许范围内短期偏离政策权重。再平衡 rebalancing：纠正市场涨跌造成的权重漂移，恢复既定风险敞口；不是以创造 alpha 为其首要定义。

Strategic asset allocation combines IPS objectives/constraints with long-term capital market expectations to set policy weights. Tactical allocation allows short-term deviations within limits. Rebalancing corrects weight drift and restores policy risk exposures; generating alpha is not its defining purpose.

定义资产类别时，要"类内相似、类间不同"：同一类别内资产应具有共同特征与正相关；不同类别间偏好低相关，从而获得分散化。Q86 问 within an asset class，选 positive；Q83 问 between asset classes，在 0、0.5、1 中选 0。

Define asset classes using similarity within classes and differentiation across classes. Assets within a class should share characteristics and positive correlation; low correlations across classes support diversification. Q86 asks within a class: positive. Q83 asks between classes: choose 0 among 0, 0.5 and 1.

Top-down 自上而下：宏观经济 → 资产/行业 → 公司证券。SAA 的风险收益由各类资产的预期收益、风险和相关性共同决定；证券选择或战术偏离属于后续主动决策。

Top-down analysis proceeds from the macroeconomy to assets/sectors and then securities. SAA risk and return depend jointly on asset-class returns, risks and correlations. Security selection and tactical deviations are additional active decisions.

5.4 两类高频应用：短期用钱与退休目标

Two recurring applications: near-term cash and retirement goals

Q91：总投资期限 15 年，不会消除一年后要付学费的流动性需求。应把这部分负债单独匹配；在给定选项中，商业票据最合适。真实选择仍需匹配到期日与信用质量。

Q91: a 15-year overall horizon does not remove the need to pay tuition in one year. Match that liability separately. Commercial paper is the best of the listed choices, with maturity and credit quality still relevant in practice.

FVnominal = FVtoday × (1+inflation)^n
Required r = (FVnominal/PV)^(1/n) - 1

先把以今天购买力表示的目标变为未来名义金额，再求投资回报率。无新增投入时，不能把缺口除以年数当复利回报。

Convert the target in today’s purchasing power into a future nominal amount, then solve for the compound investment return. With no new contributions, dividing the funding gap by years is not a compound return calculation.

Q97：名义目标=500,000×1.02^15=672,934；r=(672,934/150,000)^(1/15)-1=10.52%，选 C。等价于先求实际回报 8.36%，再用 (1+rreal)×1.02-1 转为名义回报。

Q97: nominal target=500,000×1.02^15=672,934. Required r=(672,934/150,000)^(1/15)-1=10.52%, answer C. Equivalently, find the 8.36% real return and convert using (1+rreal)×1.02-1.

05 原题与逐题解析

Original questions and explanations

先作答，再看答案；标为 KEY EXAMPLE 的题目优先完成。
Attempt the question before reading the solution; prioritize items labeled KEY EXAMPLE.

风险能力、意愿与约束

Risk capacity, willingness and constraints

Q037 重点例题 / KEY EXAMPLE

题库 PDF 第 190 页（章节第 7 页）| 答案 PDF 第 37 页

A client's time horizon is most appropriately used by an investment adviser to determine the client's:

投资顾问最适合用客户的投资期限来判断客户的哪一项？

A	risk attitude.	风险态度。
B	ability to take risk.	承担风险的能力。
C	willingness to take risk.	承担风险的意愿。

答案 / ANSWER: B

投资期限是客观财务条件，影响客户承受短期亏损、等待恢复和调整储蓄的空间，因此主要衡量风险承担能力。风险态度和意愿则更接近心理偏好。

The investment horizon is an objective financial circumstance. It affects the capacity to absorb short-term losses, wait for recovery, and adjust saving, so it primarily measures ability to take risk. Attitude and willingness concern psychological preferences.

排除干扰项 / ELIMINATE DISTRACTORS

A、C 主要反映主观心理取向，不能仅凭投资期限判断。

A and C describe subjective preferences and cannot be inferred from the horizon alone.

速记 / RECALL: 期限、财富、负债看能力；问卷、情绪看意愿。

Horizon, wealth, and liabilities indicate ability; preferences indicate willingness.

Q046 原题 / ORIGINAL QUESTION

题库 PDF 第 192 页（章节第 9 页）| 答案 PDF 第 46 页

An investor's ability to take risk:

关于投资者承担风险的能力，以下哪项最准确？

A	is a function of risk tolerance.	它是风险承受程度的函数。
B	is typically assessed by a psychometric questionnaire.	通常通过心理测量问卷评估。
C	increases with the length of the investment horizon, all else being equal.	其他条件相同时，投资期限越长，风险承担能力越高。

答案 / ANSWER: C

较长投资期限通常提供更大空间来应对短期波动、恢复资产和调整现金流，因此提高客观风险承担能力。整体风险承受程度应结合能力与意愿来判断，不能把心理意愿当作财务能力。

A longer horizon generally provides more room to absorb temporary fluctuations, rebuild assets, and adjust cash flows, increasing objective ability to take risk. Overall risk tolerance reflects both ability and willingness; psychological willingness is not the same as financial capacity.

排除干扰项 / ELIMINATE DISTRACTORS

A 混淆整体风险承受程度与其组成因素；B 主要用于评估风险意愿或态度。

A confuses overall risk tolerance with one of its components. B primarily assesses willingness or attitude toward risk.

速记 / RECALL: 投资期限影响能力；心理问卷主要测意愿。

Investment horizon affects ability; psychological questionnaires mainly assess willingness.

Q049 原题 / ORIGINAL QUESTION

题库 PDF 第 192 页（章节第 9 页）| 答案 PDF 第 49 页

Which of the following most likely affects a client's ability to take risk? The client's:

以下客户特征中，哪项最可能影响其承担风险的能力？

A	utility function	效用函数。
B	degree of risk aversion	风险厌恶程度。
C	level of wealth relative to liabilities	相对于负债的财富水平。

答案 / ANSWER: C

财富相对于负债越充裕，通常越有能力吸收投资损失并满足未来支出，因此这是客观风险能力指标。必须把资产与应付款项、支出责任一起看。

Greater wealth relative to liabilities generally improves the capacity to absorb investment losses while meeting future spending needs. This is an objective capacity measure; assets should be considered alongside obligations.

排除干扰项 / ELIMINATE DISTRACTORS

A、B 更直接体现风险偏好或意愿，而非客观财务承受能力。

A and B primarily describe risk preferences or willingness rather than objective financial capacity.

速记 / RECALL: 看风险能力时，用“资产减责任”的思路。

Assess risk capacity by considering assets together with obligations.

Q051 原题 / ORIGINAL QUESTION

题库 PDF 第 192 页（章节第 9 页）| 答案 PDF 第 51 页

Which of the following characteristics is most likely used to determine an investor's ability to take risk? The investor's:

以下哪项最可能用于判断投资者承担风险的能力？

A	risk attitude.	风险态度。
B	self-confidence.	自信程度。
C	years until retirement.	距离退休的年数。

答案 / ANSWER: C

距离退休的年数反映投资期限、未来劳动收入及退休前调整储蓄的空间，属于客观风险能力的判断依据。风险态度与自信主要属于心理因素。

Years until retirement indicates the investment horizon, remaining employment income, and time to adjust saving before retirement. It is therefore relevant to objective risk-taking ability. Risk attitude and self-confidence are mainly psychological factors.

排除干扰项 / ELIMINATE DISTRACTORS

A、B 主要涉及风险意愿，不能替代对现金流和期限的客观分析。

A and B concern willingness and cannot replace objective analysis of cash flows and horizon.

速记 / RECALL: 退休还有多久，看 ability；敢不敢亏，看 willingness。

Time until retirement informs ability; comfort with losses informs willingness.

Q045 原题 / ORIGINAL QUESTION

题库 PDF 第 191 页（章节第 8 页）| 答案 PDF 第 45 页

Which of the following is most accurate regarding an investment policy statement (IPS)?

关于投资政策声明（IPS），以下哪项最准确？

A	Policies on sustainable investing require a separate IPS.	可持续投资政策需要另立一份 IPS。
B	Investment constraints can be determined by the client or by the law.	投资约束可以由客户或法律决定。
C	Clients can specify different spending goals, but each goal must have the same risk tolerance and return objective.	客户可以有不同的支出目标，但每个目标必须具有相同的风险承受水平和收益目标。

答案 / ANSWER: B

IPS 的投资约束既可以来自客户的具体需要，例如流动性或投资限制，也可以来自法律法规。它的作用是把投资目标及执行限制写清楚。

IPS constraints can originate from client-specific needs, such as liquidity or investment restrictions, or from laws and regulations. The IPS makes both the investment objectives and implementation limits explicit.

排除干扰项 / ELIMINATE DISTRACTORS

A：可持续投资政策可纳入同一 IPS；C：不同目标可以对应不同期限、风险承受水平和收益要求。

A: sustainable-investing policies can be incorporated in the same IPS. C: different goals can have different horizons, risk tolerances, and return objectives.

速记 / RECALL: IPS 同时容纳客户要求与法律限制。

The IPS incorporates both client requirements and legal constraints.

Q059 原题 / ORIGINAL QUESTION

题库 PDF 第 194 页（章节第 11 页）| 答案 PDF 第 59 页

In the investment policy statement of a pension fund, a countrywide limit on the proportion of high-risk assets that can be held in long-term pension portfolios is most likely a:

某国统一限制长期退休金组合中高风险资产的持有比例。在退休基金的 IPS 中，这最可能属于什么约束？

A	liquidity constraint.	流动性约束。
B	legal and regulatory constraint.	法律与监管约束。
C	time horizon constraint.	投资期限约束。

答案 / ANSWER: B

该上限由全国适用的规则设定，基金必须遵守，因此应归入法律与监管约束。判断约束类别要看限制的来源，而不只是看它限制了哪类资产。

A countrywide rule imposes an external limit that the pension fund must obey, making it a legal and regulatory constraint. Classify a constraint by its source, not merely by the type of asset it restricts.

排除干扰项 / ELIMINATE DISTRACTORS

A 涉及变现及现金支付需要；C 涉及资金使用时间，均不是题目所述限制的直接来源。

A concerns cash needs and convertibility to cash. C concerns when funds are needed; neither is the direct source of this limit.

速记 / RECALL: 法规规定的投资上限 = legal and regulatory constraint。

An investment cap imposed by law is a legal and regulatory constraint.

Q085 原题 / ORIGINAL QUESTION

题库 PDF 第 199 页（章节第 16 页）| 答案 PDF 第 85 页

Which of the following is most likely a legal and regulatory constraint in an investment policy statement?

下列哪项最可能属于 IPS 中的法律与监管约束？

A	A pension fund's decision to limit investments in real estate	养老基金自行决定限制房地产投资。
B	A taxable investor's requirement to avoid investments in securities generating interest income	应税投资者要求避免投资产生利息收入的证券。
C	A public company director's restriction on trading the company's stock shortly before the publication of financial results	上市公司董事在财务业绩公布前不久受到公司股票交易限制。

答案 / ANSWER: C

上市公司董事的交易限制涉及内幕信息与相关法律监管要求，应记入 IPS 的法律与监管约束，避免投资经理代客户进行不合规交易。

A listed-company director’s trading restriction concerns inside information and related legal or regulatory requirements. It belongs under legal and regulatory constraints so the manager can avoid prohibited trading on the client’s behalf.

排除干扰项 / ELIMINATE DISTRACTORS

A 只说明基金自主决定，没有说明法律强制；B 是税务考虑，应归入税务约束。

A describes a voluntary decision without stating a legal requirement. B is a tax consideration and belongs under tax constraints.

速记 / RECALL: 客户自行偏好、税务需要、法律强制，分别归类。

Distinguish voluntary preferences, tax needs and legally imposed restrictions.

Q091 重点例题 / KEY EXAMPLE

题库 PDF 第 201 页（章节第 18 页）| 答案 PDF 第 91 页

An investor has a 15-year time horizon but needs to withdraw funds from her portfolio in one year's time to pay for tuition fees. Which of the following investments is most suitable to cover the investor's liquidity requirement due to the tuition fees?

某投资者的整体投资期限为 15 年，但一年后需要提款缴付学费。下列哪项最适合用于满足这笔学费的流动性需求？

A	Commercial paper	商业票据。
B	Private equity securities	私募股权证券。
C	Large-capitalization stocks	大盘股票。

答案 / ANSWER: A

应将一年后必须支付的学费单独配置到期限和流动性较匹配的低风险资产，而不能只依据整体 15 年期限选资产。三个选项中，短期商业票据最符合需要；实际仍须考虑发行人信用及到期日匹配。

Funds needed for tuition in one year should be allocated to relatively low-risk assets with suitable maturity and liquidity, rather than invested solely by reference to the overall 15-year horizon. Of the choices, short-term commercial paper is most suitable, subject to issuer credit quality and maturity matching.

排除干扰项 / ELIMINATE DISTRACTORS

B 私募股权流动性低；C 股票虽较易卖出，但一年后的价格存在明显波动风险。

B, private equity, is illiquid. C, equities, may be readily saleable but their value in one year is uncertain.

速记 / RECALL: 长期总目标不能覆盖短期必须付款的需求。

A long overall horizon does not remove a near-term funding need.

IPS 章节定位与收益目标

IPS section identification and return objectives

Q004 原题 / ORIGINAL QUESTION

题库 PDF 第 184 页（章节第 1 页）| 答案 PDF 第 4 页

With regard to an investment policy statement, which of the following statements about return objectives is most accurate?

就投资政策说明书而言，关于收益目标，下列哪项最准确？

A	A return objective cannot be a required rate of return.	收益目标不能是必要收益率。
B	Return objectives must be set independent of risk objectives.	收益目标必须独立于风险目标设定。
C	When setting a relative return objective, a good benchmark should be investable.	设定相对收益目标时，良好的基准应当可以投资。

答案 / ANSWER: C

相对收益目标是相对于基准衡量的目标，基准应可投资，才能反映可实现的比较对象。收益目标可以是满足支出或退休需要的必要收益率，也必须与承担风险的能力相协调。

A relative return objective is assessed against a benchmark. An investable benchmark represents an achievable alternative. A return objective may be a required rate needed to fund goals, and it must be consistent with the investor's risk objectives.

排除干扰项 / ELIMINATE DISTRACTORS

A：必要收益率可以成为收益目标。B：收益和风险必须一并考虑。

A: A required rate of return can be an objective. B: Return and risk objectives must be considered together.

速记 / RECALL: 相对收益看基准；基准应可投资。

Relative return requires an investable benchmark.

Q038 重点例题 / KEY EXAMPLE

题库 PDF 第 190 页（章节第 7 页）| 答案 PDF 第 38 页

Which of the following sections of an investment policy statement most likely provides guidance on obtaining feedback on investment results?

投资政策声明（IPS）的哪一部分最可能就获取投资结果的反馈提供指引？

A	Investment Guidelines	投资指引。
B	Evaluation and Review	评估与复核。
C	Statement of Duties and Responsibilities	职责与责任声明。

答案 / ANSWER: B

评估与复核部分规定如何衡量及评价投资结果，例如使用哪些基准、如何报告绩效以及如何进行结果反馈。它与投资结果的检查最直接相关。

The evaluation and review section explains how investment results are measured and assessed, including benchmarks, performance reporting, and feedback. It is the section most directly concerned with reviewing outcomes.

排除干扰项 / ELIMINATE DISTRACTORS

A 规定投资实施边界；C 分配参与者的责任和授权。

A sets implementation guidelines. C assigns responsibilities and authority.

速记 / RECALL: 查投资结果选 Evaluation and Review；查 IPS 如何更新看 Procedures。

Reviewing results points to Evaluation and Review; revising the IPS points to Procedures.

Q056 重点例题 / KEY EXAMPLE

题库 PDF 第 194 页（章节第 11 页）| 答案 PDF 第 56 页

Information regarding the permissible use of derivatives in a portfolio is most likely found in which of the following sections of an investment policy statement?

组合中衍生品的允许使用范围，最可能列在 IPS 的哪一部分？

A	Procedures	程序。
B	Investment Guidelines	投资指引。
C	Statement of Duties and Responsibilities	职责与责任声明。

答案 / ANSWER: B

投资指引描述投资实施的规则和边界，例如允许的资产、策略、杠杆或衍生品用途。是否允许用期货对冲、是否限制期权交易，都属于这一部分。

Investment guidelines define the rules and boundaries for implementation, including permitted assets, strategies, leverage, and uses of derivatives. Permission to hedge with futures or restrictions on options trading belong here.

排除干扰项 / ELIMINATE DISTRACTORS

A 主要规定 IPS 更新或特殊情况的处理程序；C 规定谁负责、谁有权限。

A mainly describes procedures for updating the IPS or dealing with contingencies. C specifies who is responsible and authorized.

速记 / RECALL: “可以买什么、怎么用衍生品”看 Investment Guidelines。

Permitted instruments and derivative uses belong in Investment Guidelines.

Q062 重点例题 / KEY EXAMPLE

题库 PDF 第 195 页（章节第 12 页）| 答案 PDF 第 62 页

Which of the following sections of an investment policy statement (IPS) most likely explain how and when the IPS should be reviewed?

IPS 的哪一部分最可能说明应在何时以及如何复核 IPS 本身？

A	Procedures	程序。
B	Investment Guidelines	投资指引。
C	Statement of Duties and Responsibilities	职责与责任声明。

答案 / ANSWER: A

程序部分规定 IPS 如何保持更新，以及何时或因何种事件需要重新审视。这里复核的对象是政策文件本身，和第 38 题对投资结果的评估不同。

The procedures section explains how the IPS is kept current and when, or after which events, it should be reviewed. Here the object being reviewed is the policy document itself, unlike the investment-results review in Q38.

排除干扰项 / ELIMINATE DISTRACTORS

B 规定投资实施规则；C 分配责任和权限，不是本题问的复核流程。

B governs investment implementation. C allocates responsibilities and authority rather than specifying the review process asked about here.

速记 / RECALL: 复核 IPS 本身 → Procedures；复核投资表现 → Evaluation and Review。

Reviewing the IPS itself → Procedures; reviewing investment performance → Evaluation and Review.

Q074 原题 / ORIGINAL QUESTION

题库 PDF 第 197 页（章节第 14 页）| 答案 PDF 第 74 页

With respect to an investment policy statement, which of the following is most closely linked to the client's distinctive needs?

在投资政策声明（IPS）中，哪个部分与客户的个别需求联系最紧密？

A	The evaluation and review section	评估与审查部分。
B	The objectives and constraints sections	目标与约束部分。
C	The statement of duties and responsibilities	职责与责任声明。

答案 / ANSWER: B

目标部分记录客户的收益与风险目标；约束部分记录流动性、期限、税务、法律监管及特殊需求。两者共同把客户的个别情况转化为投资组合要求。

Objectives record the client’s return and risk goals. Constraints record liquidity, time horizon, taxes, legal and regulatory requirements, and unique needs. Together they translate the client’s individual circumstances into portfolio requirements.

排除干扰项 / ELIMINATE DISTRACTORS

A 说明如何评价投资结果；C 说明各参与方负责什么，都不是客户需求的核心记录位置。

A describes assessment of investment results. C assigns responsibilities to the parties; neither is the principal location for the client’s specific objectives and needs.

速记 / RECALL: 客户想要什么、受什么限制：目标 + 约束。

What the client needs and what limits the portfolio: objectives + constraints.

Q077 重点例题 / KEY EXAMPLE

题库 PDF 第 198 页（章节第 15 页）| 答案 PDF 第 77 页

Which of the following sections of an investment policy statement for a pension plan most likely specifies the discretion that portfolio managers have with respect to executing the investment strategy?

养老金计划的 IPS 中，哪个部分最可能规定组合经理执行投资策略时具有多大的自主决定权？

A	Procedures	程序。
B	Investment Constraints	投资约束。
C	Statement of Duties and Responsibilities	职责与责任声明。

答案 / ANSWER: C

职责与责任声明界定客户、托管人、投资经理等参与方的职责、权限与治理安排。经理有权自主作出哪些执行决定，属于授权与责任分配。

The statement of duties and responsibilities defines the roles, authority and governance arrangements of the client, custodian and investment managers. The manager’s discretion in implementing the strategy is a matter of delegated authority and responsibility.

排除干扰项 / ELIMINATE DISTRACTORS

A 程序关注 IPS 如何更新及如何处理特定情况；B 约束关注流动性、期限、税务、监管及特殊限制。

A concerns how to update the IPS and respond to contingencies. B records liquidity, horizon, tax, regulatory and unique constraints.

速记 / RECALL: “谁负责、谁有权决定”看职责与责任。

“Who is responsible and who has authority?” belongs under duties and responsibilities.

Q097 重点例题 / KEY EXAMPLE

题库 PDF 第 202 页（章节第 19 页）| 答案 PDF 第 97 页

An adviser gathers the following information about a client's retirement income needs:
The current balance of the client's retirement fund is $150,000.
The client plans to retire in 15 years' time.
$500,000 in today's money is needed to fund retirement.
No further retirement fund contributions will be made.
Inflation is expected to average 2% per year over the next 15 years. Based only on the above information and ignoring taxes, the minimum annual rate of return required to meet the client's retirement income objective is closest to:

顾问获得以下退休资金需求：
目前退休基金余额为 150,000 美元。
客户计划 15 年后退休。
按今天购买力计算，退休需要 500,000 美元。
未来不再追加退休基金供款。
未来 15 年预计年均通胀为 2%。仅依据上述信息且忽略税项，满足退休目标所需的最低年收益率最接近多少？

A	6.2%.	6.2%。
B	8.4%.	8.4%。
C	10.5%.	10.5%。

答案 / ANSWER: C

先把实际目标转为 15 年后的名义金额：FV = 500,000 × 1.02^15 ≈ 672,934。再令 150,000 × (1 + r)^15 = FV，得 r = [500,000 × 1.02^15 / 150,000]^(1/15) - 1 ≈ 10.525%。等价地，先求实际收益约 8.357%，再用 (1 + r名义) = (1 + r实际) × 1.02。

First inflate the real target into a nominal future amount: FV = 500,000 × 1.02^15 ≈ 672,934. Solve 150,000 × (1 + r)^15 = FV: r = [500,000 × 1.02^15 / 150,000]^(1/15) - 1 ≈ 10.525%. Equivalently, the required real return is approximately 8.357%, and (1 + nominal return) = (1 + real return) × 1.02.

排除干扰项 / ELIMINATE DISTRACTORS

B 的 8.4% 仅为实际收益要求，忽略通胀；A 把通胀方向处理反了。实际收益与通胀的精确结合是相乘，不是简单相加。

B, 8.4%, is only the real-return requirement and ignores inflation. A adjusts inflation in the wrong direction. The exact combination of real return and inflation is multiplicative, not simple addition.

速记 / RECALL: 今天购买力的未来目标，先通胀放大，再求复合收益。

For a future target stated in today’s money, inflate it first, then solve the compound return.

战略配置与再平衡

Strategic allocation and rebalancing

Q014 重点例题 / KEY EXAMPLE

题库 PDF 第 186 页（章节第 3 页）| 答案 PDF 第 14 页

Which of the following is most likely developed by combining a client's investment objectives and constraints with long-term capital market expectations?

将客户的投资目标和约束与长期资本市场预期相结合，最可能形成什么？

A	Risk budget	风险预算。
B	Strategic asset allocation	战略资产配置。
C	Investment policy statement	投资政策说明书。

答案 / ANSWER: B

IPS 界定客户的目标和约束；再加入各资产类别的长期预期收益、风险和相关性，才能确定长期政策权重，即战略资产配置（SAA）。

The IPS specifies client objectives and constraints. Combining these with long-run expected asset-class returns, risks, and correlations determines policy weights, which form the strategic asset allocation (SAA).

排除干扰项 / ELIMINATE DISTRACTORS

A：风险预算是分配或限制风险使用量。C：IPS 先明确客户要求，SAA 才是与市场预期结合后的配置结果。

A: A risk budget allocates or limits risk usage. C: The IPS states client requirements; SAA is the allocation derived from those requirements and market expectations.

速记 / RECALL: 客户目标与约束 + 长期市场预期 → SAA。

Client objectives and constraints + long-term market expectations → SAA.

Q016 原题 / ORIGINAL QUESTION

题库 PDF 第 186 页（章节第 3 页）| 答案 PDF 第 16 页

In regard to the asset allocation process, a top-down analysis most likely begins with an examination of :

在资产配置过程中，自上而下分析最可能从检查哪一项开始？

A	macroeconomic growth.	宏观经济增长。
B	a company's board of directors.	公司的董事会。
C	the expected growth of a company's competitors.	公司竞争对手的预期增长。

答案 / ANSWER: A

自上而下从宏观环境入手，例如经济增长、通胀和利率，再推进到资产类别、行业与个别证券。公司治理和竞争对手分析属于更细层面的分析。

Top-down analysis starts with the macroeconomic environment, such as growth, inflation, and interest rates, then proceeds to asset classes, industries, and securities. Governance and competitors are more granular considerations.

排除干扰项 / ELIMINATE DISTRACTORS

B、C 都是公司或行业层面的因素，不是自上而下分析的通常起点。

B and C are company- or industry-level issues rather than the usual starting point of top-down analysis.

速记 / RECALL: Top-down：宏观 → 行业 / 资产类别 → 个股。

Top-down: macro → industries / asset classes → securities.

Q019 原题 / ORIGINAL QUESTION

题库 PDF 第 187 页（章节第 4 页）| 答案 PDF 第 19 页

Portfolio rebalancing is best described as a process aimed at:

组合再平衡最适合描述为旨在实现哪项目标的过程？

A	generating alpha.	创造超额收益 alpha。
B	aligning portfolio weights with the tactical asset allocation decision.	使组合权重符合战术资产配置决策。
C	restoring the portfolio's original exposures to systematic risk factors.	恢复组合原先对系统性风险因子的敞口。

答案 / **ANSWER: C**

市场价格变化会使实际资产权重偏离政策权重。再平衡把权重拉回目标范围，恢复原定的系统性风险敞口和组合风险水平。

Market movements cause actual asset weights to drift from policy weights. Rebalancing restores target exposures to systematic risk factors and the intended portfolio risk profile.

排除干扰项 / ELIMINATE DISTRACTORS

A：再平衡不以创造 alpha 为定义目标。B：战术配置是主动偏离长期政策配置，概念不同。

A: Generating alpha is not the defining purpose. B: Tactical allocation deliberately deviates from long-term policy weights and is a separate decision.

速记 / RECALL: 再平衡 = 把漂移后的风险敞口拉回目标。

Rebalancing restores target exposures after portfolio drift.

Q026 原题 / ORIGINAL QUESTION

题库 PDF 第 188 页（章节第 5 页）| 答案 PDF 第 26 页

Which of the following statements about asset allocation is most accurate?

关于资产配置，下列哪项最准确？

A	Investors should diversify their wealth between asset classes in order to eliminate systematic risk.	投资者应跨资产类别分散财富，以消除系统性风险。
B	Investors with a below-average risk tolerance should have an above-average weight in alternative investments.	风险容忍度低于平均水平的投资者，应把高于平均水平的权重配置到另类投资。
C	Adding asset classes with a low correlation to existing asset classes improves an investor's risk-return trade-off.	加入与现有资产类别相关性较低的资产类别，有助于改善风险收益权衡。

答案 / **ANSWER: C**

资产配置不仅看各类资产的单独收益和风险，也看它们如何共同变动。加入相关性较低的资产类别能提供分散化机会，从而改善组合的风险收益关系，但不能保证消除整体市场风险。

Asset allocation considers not only each asset class's return and risk but also their co-movement. Lower-correlated asset classes can improve diversification and the portfolio risk-return trade-off, without guaranteeing elimination of overall market risk.

排除干扰项 / ELIMINATE DISTRACTORS

A：分散化主要降低非系统性风险，不能普遍消除系统性风险。B：另类投资的风险、杠杆与流动性差异很大，不能简单超配。

A: Diversification mainly reduces nonsystematic risk and cannot generally eliminate market risk. B: Alternatives differ widely in risk, leverage, and liquidity; low risk tolerance does not imply an overweight.

速记 / RECALL: 分散化看相关性，不能保证消除市场风险。

Diversification depends on correlation and cannot guarantee elimination of market risk.

Q058 原题 / ORIGINAL QUESTION

题库 PDF 第 194 页（章节第 11 页）| 答案 PDF 第 58 页

Which of the following investment principles best explains the use of strategic asset allocation in portfolio construction?

哪项投资原则最能解释组合构建为何采用战略资产配置？

A	Returns on similar assets reflect exposures to certain sets of systematic factors.	相似资产的收益反映对某些共同系统性因素的敞口。
B	Nonsystematic risk accounts for most of the change in portfolio value over the long term.	非系统性风险解释组合长期价值变化的大部分。
C	Deviating from policy exposures to systematic risk factors may add value to the portfolio.	偏离政策规定的系统性风险敞口可能为组合增值。

答案 / **ANSWER: A**

战略资产配置通过长期资产类别权重来确定组合对系统性风险因素的政策敞口。相似资产通常受相近的系统性因素驱动，因此按资产类别配置具有经济意义。

Strategic asset allocation uses long-term asset-class weights to establish policy exposures to systematic risk factors. Similar assets tend to respond to similar systematic factors, giving asset-class allocation an economic foundation.

排除干扰项 / ELIMINATE DISTRACTORS

B 误把非系统性风险当成分散组合长期变化的主要来源；C 描述战术性偏离政策权重的理由。

B incorrectly treats nonsystematic risk as the main driver of long-term changes in a diversified portfolio. C describes the rationale for tactical deviations from policy weights.

速记 / RECALL: SAA 决定长期系统性敞口；TAA 负责主动偏离。

SAA establishes long-term systematic exposures; TAA actively deviates from them.

Q061 原题 / ORIGINAL QUESTION

题库 PDF 第 195 页（章节第 12 页）| 答案 PDF 第 61 页

Which of the following principles best explains the focus on a strategic asset allocation when constructing a client's IPS?

构建客户 IPS 时重视战略资产配置，最能由哪项原则解释？

A	Adding assets with high correlation improves the risk-return trade-off.	加入高相关性的资产可以改善风险收益权衡。
B	Nonsystematic risk accounts for most of a portfolio's change in value over the long term.	非系统性风险解释组合长期价值变化的大部分。
C	The returns to groups of similar assets predictably reflect exposures to certain systematic factors.	相似资产组的收益有规律地反映对某些系统性因素的敞口。

答案 / **ANSWER: C**

战略资产配置把长期风险预算落实到资产类别上。其基础是不同类别具有不同的共同风险因素，而相似资产的收益通常反映相似的系统性风险敞口；所以选择长期权重会决定组合的主要风险收益特征。

Strategic asset allocation implements the long-term risk budget across asset classes. Its basis is that asset classes have common risk drivers and similar assets tend to reflect similar systematic exposures; long-term policy weights therefore determine the portfolio’s principal risk-return characteristics.

排除干扰项 / ELIMINATE DISTRACTORS

A 应强调较低相关性带来的分散收益；B 将可分散的非系统性风险误当成主要长期驱动。

A should emphasize lower correlation for diversification benefits. B incorrectly identifies diversifiable nonsystematic risk as the principal long-term driver.

速记 / RECALL: 同类资产共享因素，长期配置决定主要风险。

Similar assets share factors; long-term allocation determines principal risks.

Q069 原题 / ORIGINAL QUESTION

题库 PDF 第 196 页（章节第 13 页）| 答案 PDF 第 69 页

The risk-return profile of a portfolio's strategic asset allocation is most likely determined by the expected returns and risks of the individual asset classes and the:

一个组合的战略资产配置风险收益特征，最可能由各资产类别的预期收益、风险以及下列哪项共同决定？

A	correlations between those asset classes.	各资产类别之间的相关性。
B	use of security selection for each of those asset classes.	各资产类别内采用的证券选择策略。
C	allowable deviation of portfolio weights from policy weights for those asset classes.	组合权重相对于政策权重的允许偏离幅度。

答案 / **ANSWER: A**

在既定政策权重下，组合预期收益取决于各资产类别的预期收益；组合风险还取决于类别之间的相关性。战略资产配置的基础输入因此是预期收益、风险和相关性。

For specified policy weights, expected portfolio return depends on asset-class expected returns, while portfolio risk also depends on cross-asset-class correlations. Expected returns, risks and correlations are therefore the core inputs to strategic asset allocation.

排除干扰项 / ELIMINATE DISTRACTORS

B 属于证券选择带来的额外主动风险；C 属于权重偏离或再平衡管理，不能代替战略配置的相关性输入。

B introduces additional active risk from security selection. C concerns permitted weight deviations or rebalancing and does not replace correlations as an input to strategic allocation.

速记 / RECALL: 配置风险看三项输入：收益、波动、相关性。

Allocation inputs: expected return, volatility and correlation.

Q083 重点例题 / KEY EXAMPLE

题库 PDF 第 199 页（章节第 16 页）| 答案 PDF 第 83 页

When defining asset classes for a strategic asset allocation, which of the following pairwise correlations between asset class returns is most preferable?

为战略资产配置划分资产类别时，下列哪种资产类别之间的两两收益相关性最理想？

A	0.0	0.0。
B	0.5	0.5。
C	1.0	1.0。

答案 / **ANSWER: A**

不同资产类别应能够提供分散化收益，所以类别之间的相关性较低更理想。三个选项中 0.0 最低；负相关若存在也可能进一步增强分散化，但不在本题选项内。

Different asset classes should provide diversification, so lower correlations between classes are preferable. Of the choices, 0.0 is lowest. Negative correlation may provide further diversification, but it is not an option here.

排除干扰项 / ELIMINATE DISTRACTORS

B 比 0 更同步；C 完全正相关，几乎不能通过不同类别获得相关性方面的分散化好处。

B moves more closely together than zero correlation. C is perfectly positively correlated and provides no correlation-based diversification benefit.

速记 / RECALL: 资产类别之间：相关性低。

Between asset classes: prefer low correlation.

Q086 重点例题 / KEY EXAMPLE

题库 PDF 第 200 页（章节第 17 页）| 答案 PDF 第 86 页

When defining asset classes, the paired correlations of assets within an asset class should be:

划分资产类别时，同一资产类别内部各资产之间的两两相关性应当如何？

A	negative	负相关。
B	zero.	零相关。
C	positive.	正相关。

答案 / **ANSWER: C**

同一资产类别应由相对同质的资产组成，其收益通常受相似风险因子驱动，因此类内相关性应较高且为正。与第 83 题对照：类内相似，类间分散。

An asset class should contain relatively homogeneous assets whose returns are driven by similar risk factors. Within-class correlations should therefore be relatively high and positive. Contrast Q83: similarity within classes, diversification between classes.

排除干扰项 / ELIMINATE DISTRACTORS

A 和 B 更接近跨类别分散化所寻求的低相关特征，不能体现同一类别内部的同质性。

A and B resemble the low-correlation properties sought across classes and do not express within-class homogeneity.

速记 / RECALL: 类内高相关；类间低相关。

High correlation within a class; low correlation between classes.

06 投资者、投资工具与组合流程

Investors, investment vehicles and the portfolio process

本单元 10 题 | 首轮预算 15 分钟 | 重点：Q1, Q23, Q34, Q57, Q63

对应阅读主题 / Reading theme: Portfolio Management: An Overview

6.1 养老金：承诺的对象决定谁承担风险

Pensions: what is promised determines who bears the risk

计划 / Plan	承诺与风险 / Promise and risk bearer
Defined benefit, DB 设定受益计划	承诺退休福利；雇主/发起人承担满足该福利的主要资金与投资风险。风险容忍度取决于期限、成熟度、资助状态等，不能一律说低。 Promises retirement benefits. The sponsor bears the main funding and investment risk. Tolerance depends on maturity, horizon and funding rather than being uniformly low.
Defined contribution, DC 设定提存计划	承诺投入金额或规则，最终退休财富取决于投资表现；雇员承担投资与通胀风险。 Promises contributions or contribution rules. Retirement wealth depends on performance; the employee bears investment and inflation risk.

6.2 机构投资者看“负债什么时候来”

For institutions, ask when liabilities need to be paid

银行：流动性至关重要。捐赠基金 endowment：通常长期甚至永久存续，短期支出需要不能直接推出整体期限短。保险公司：不同账户负债功能不同，general account 与 surplus account 不应假定相同风险容忍度。主权财富基金 sovereign wealth fund：由政府拥有，并非只能投资本国。

Banks place paramount importance on liquidity. Endowments usually have long or perpetual horizons; current spending alone does not make their overall horizon short. Insurers’ general and surplus accounts serve different liability functions and need not share risk tolerance. Sovereign wealth funds are government-owned and are not restricted by definition to domestic investments.

6.3 工具比较：先看怎样交易和定价

Compare vehicles by trading and pricing mechanics

工具 / Vehicle	题库采用的典型特征 / Typical features used in these questions
ETF 交易所交易基金	交易所盘中交易；普通开放式 ETF 可创造/赎回份额，不能笼统归为封闭式基金；分红取决于产品政策。 Trades intraday. Ordinary open-end ETFs support creation/redemption and are not generally closed-end funds. Distributions depend on product policy.
Open-end mutual fund 开放式共同基金	通常按每日净值申购赎回，份额数量可变；原题比较下，最低投资额通常高于买一份 ETF。 Typically transacts at daily NAV with variable shares. In the source comparison, its investment minimum is typically higher than one ETF share.
Robo-adviser 智能投顾	自动化服务降低门槛，通常使用低成本指数工具，服务较低可投资资产客户。 Automation supports smaller accounts, often using low-cost index vehicles.
Smart beta 聪明贝塔	按规则暴露于价值、规模等因子；相较纯市值加权策略通常费用和换手更高。 Rules-based factor exposure, typically with higher fees and turnover than plain capitalization weighting.

辨析 / PRECISION NOTE

Q23 的“通常派息”和 Q65 的最低投资额都是题库中的典型判断，不是所有产品的绝对规则。现实中存在累积型 ETF、不同份额类别和低/零门槛共同基金。

Q23’s distributions and Q65’s minimum investment are typical source conventions, not universal product rules. Accumulating ETFs, different share classes and low/no-minimum mutual funds exist.

6.4 组合流程三阶段

Three stages of portfolio management

阶段 / Stage	内容 / Activities
Planning 规划	了解客户，制定 IPS。 Understand the client and formulate the IPS.
Execution 执行	确定资产配置、分析证券、构建与执行组合。 Set allocation, analyze securities, construct and implement the portfolio.
Feedback 反馈	监控、再平衡、衡量和评价业绩。 Monitor, rebalance, measure and evaluate performance.

组合思维关注各资产对整体组合的影响。它有助于降低单一公司灾难性损失的影响，但不会保证熊市不亏钱，也不会仅靠增加持仓数量消除市场系统性风险。

A portfolio perspective considers how each holding affects the whole portfolio. It can reduce damage from a single-company disaster, but it does not guarantee protection against market downturns or eliminate systematic market risk merely by adding holdings.

06 原题与逐题解析

Original questions and explanations

先作答，再看答案；标为 KEY EXAMPLE 的题目优先完成。
Attempt the question before reading the solution; prioritize items labeled KEY EXAMPLE.

投资者与养老金

Investors and pension arrangements

Q001 重点例题 / KEY EXAMPLE

题库 PDF 第 184 页（章节第 1 页）| 答案 PDF 第 1 页

Which of the following statements about pension plans is most accurate?

关于退休金计划，下列哪项最准确？

A	Defined benefit plans typically have a low risk tolerance.	设定受益计划通常具有较低的风险承受能力。
B	Defined contribution plans typically have a low risk tolerance.	设定提存计划通常具有较低的风险承受能力。
C	The sponsor of a defined benefit plan specifies the obligation owed to participants.	设定受益计划的发起人明确规定应向参加者承担的给付义务。

答案 / ANSWER: C

DB（设定受益）先承诺退休给付，因此发起人必须明确未来给付义务。DC（设定提存）先明确供款，最终退休资产取决于投资表现。两者的风险承受能力都不能一概判定为低。

A defined benefit (DB) plan promises retirement benefits, so its sponsor specifies the benefit obligation. A defined contribution (DC) plan specifies contributions; the eventual retirement wealth depends on investment performance. Neither type invariably has low risk tolerance.

排除干扰项 / ELIMINATE DISTRACTORS

A：DB 的风险承受能力取决于计划成熟度和资金状况等。B：DC 的风险承受能力随参加者的年龄、财务状况和目标而变。

A: A DB plan's risk tolerance depends on maturity and funding, among other factors. B: DC participants differ in age, financial circumstances, and goals.

速记 / RECALL: DB 承诺给付；DC 约定供款。

DB specifies benefits; DC specifies contributions.

Q063 重点例题 / KEY EXAMPLE

题库 PDF 第 195 页（章节第 12 页）| 答案 PDF 第 63 页

Which of the following best describes a characteristic of defined contribution pension plans?

以下哪项最能描述设定提存退休计划（defined contribution plan）的特点？

A	The employee accepts the investment and inflation risk.	雇员承担投资风险与通胀风险。
B	The employer is responsible for adequately funding the plan.	雇主负责确保计划资金充足。
C	Defined contribution plans typically have a higher cost to the company than defined benefit plans.	公司承担的成本通常高于设定受益计划。

答案 / **ANSWER: A**

DC 计划预先确定供款规则，最终退休财富取决于缴费、投资表现和通胀，因此雇员承担投资及购买力风险。它没有像 DB 计划那样向雇员承诺一个预先定义的退休福利金额。

A DC plan specifies contributions, while eventual retirement wealth depends on contributions, investment performance, and inflation. The employee therefore bears investment and purchasing-power risk, rather than receiving the predefined benefit promised by a DB plan.

排除干扰项 / ELIMINATE DISTRACTORS

B 更接近 DB 计划中雇主为承诺福利补足资金的责任；C 并非 DC 计划的一般特征。

B describes the employer's responsibility to fund promised DB benefits. C is not a general feature of DC plans.

速记 / RECALL: DC 定供款，员工承担结果；DB 定福利，雇主承担承诺。

DC fixes contributions and leaves outcomes with the employee; DB promises benefits.

Q034 重点例题 / KEY EXAMPLE

题库 PDF 第 190 页（章节第 7 页）| 答案 PDF 第 34 页

Which of the following statements about different types of investors is most accurate?

关于不同类型的投资者，下列哪项最准确？

A	For banks, the liquidity of their investments is a paramount concern.	对银行而言，投资的流动性是首要考虑之一。
B	For endowments, investment horizons are short due to their short-term spending needs.	对捐赠基金而言，由于存在短期支出需要，投资期限较短。
C	For insurance companies, the risk tolerance of their general and surplus accounts is typically the same.	对保险公司而言，一般账户和盈余账户通常具有相同的风险容忍度。

答案 / **ANSWER: A**

银行需要应对存款提取和支付等负债需求，因此高度关注投资流动性。捐赠基金虽有年度支出，整体通常具有长期甚至永久期限；保险公司支持保单负债的一般账户与盈余资本账户的风险目标也不同。

Banks must meet withdrawals and payment obligations, making investment liquidity paramount. Endowments can have annual spending needs while maintaining very long or perpetual horizons. Insurance general accounts supporting policy liabilities and surplus accounts have different risk objectives.

排除干扰项 / ELIMINATE DISTRACTORS

B：短期现金需求不等于整体投资期限短。C：一般账户与盈余账户的负债约束及风险承受能力不同。

B: Near-term cash needs do not imply a short overall horizon. C: General and surplus accounts face different liability constraints and risk tolerances.

速记 / RECALL: 银行重流动性；捐赠基金重长期；保险账户分开看。

Banks prioritize liquidity; endowments are long-term; insurance accounts differ.

Q042 原题 / ORIGINAL QUESTION

题库 PDF 第 191 页（章节第 8 页）| 答案 PDF 第 42 页

Sovereign wealth funds are best described as investment funds:

主权财富基金最恰当的定义是什么？

A	owned by governments.	由政府拥有的投资基金。
B	traded as closed-end country funds.	以封闭式国家基金形式交易的基金。
C	restricted from investing in foreign securities.	被禁止投资境外证券的基金。

答案 / **ANSWER: A**

主权财富基金的核心特征是政府所有权。其资金来源、目标和投资范围可以不同，且通常可持有多种国内或国际资产。

Government ownership is the defining feature of a sovereign wealth fund. Funding sources, objectives, and investment mandates differ, and funds may invest in domestic or international assets.

排除干扰项 / ELIMINATE DISTRACTORS

B 的交易形式并非主权财富基金的定义；C 的境外投资禁令并不普遍存在。

B describes a trading structure rather than the defining ownership feature. C is not a general restriction.

速记 / RECALL: Sovereign wealth fund = government-owned investment fund。

A sovereign wealth fund is a government-owned investment fund.

投资工具与管理模式

Investment vehicles and management approaches

Q023 重点例题 / KEY EXAMPLE

题库 PDF 第 188 页（章节第 5 页）| 答案 PDF 第 23 页

Exchange-traded funds (ETFs):

关于交易所交易基金（ETF），下列哪项正确？

A	are priced once a trading day.	每个交易日仅定价一次。
B	usually pay out dividends to shareholders.	通常向份额持有人派发股息。
C	are generally structured as closed-end funds.	一般采用封闭式基金结构。

答案 / **ANSWER: B**

本题答案 B 描述常见派息型 ETF：收到投资收益后可向持有人分配。ETF 在交易所盘中交易，并非只按日末价格交易；其申购赎回机制也不同于一般封闭式基金。

Answer B describes common distributing ETFs, which pass investment income to shareholders. ETFs trade intraday on an exchange rather than only at an end-of-day price, and their creation/redemption structure differs from that of typical closed-end funds.

排除干扰项 / ELIMINATE DISTRACTORS

A：更接近传统开放式共同基金按日末 NAV 交易的特征。C：ETF 不能一般等同为封闭式基金。

A resembles the end-of-day NAV dealing of traditional open-end mutual funds. C incorrectly equates ETFs with closed-end funds.

速记 / RECALL: ETF 盘中交易；是否派息要看份额类别。

ETFs trade intraday; distributions depend on the share class.

概念辨析 / NOTE: “通常派息”不是所有 ETF 的共同必备特征。累积型 ETF 可把收入留在基金内再投资；本题按给定选项与答案选择 B。

Paying distributions is not universal. Accumulating ETF share classes may reinvest income inside the fund. B is the source answer within the supplied choices.

Q065 原题 / ORIGINAL QUESTION

题库 PDF 第 195 页（章节第 12 页）| 答案 PDF 第 65 页

Open-end mutual funds typically:

开放式共同基金通常具有什么特点？

A	are priced intraday.	盘中定价。
B	have a fixed number of shares outstanding.	流通份额数量固定。
C	have a larger required minimum investment than ETFs.	最低投资金额高于 ETF。

答案 / ANSWER: C

按本题的一般产品比较，传统开放式共同基金通常设有最低申购金额，而 ETF 通常可按交易单位购买，因此 C 最合适。开放式基金通常按每日计算的资产净值办理申购赎回，并可增发或注销份额。

Under the source's general product comparison, traditional open-end mutual funds often require a minimum subscription, whereas ETFs can usually be bought in trading units, making C the best answer. Open-end funds normally transact at a daily NAV and create or redeem shares.

排除干扰项 / ELIMINATE DISTRACTORS

A 更符合 ETF 的盘中交易；B 更符合传统封闭式基金的结构。

A describes intraday ETF trading. B is more characteristic of a traditional closed-end fund structure.

速记 / RECALL: 共同基金：按 NAV 申赎；ETF：盘中交易。

Mutual funds transact at NAV; ETFs trade intraday.

概念辨析 / NOTE: "通常"不是普遍规则；不同基金及平台可能有低门槛或零门槛安排，不能把本题比较当作所有产品的硬性定义。

"Typically" is not universal: funds and platforms may have low or no minimums, so this comparison is not a defining rule for every product.

Q032 原题 / ORIGINAL QUESTION

题库 PDF 第 189 页（章节第 6 页）| 答案 PDF 第 32 页

Robo-advisers most likely.

智能投顾最可能具有以下哪项特点？

A	face high barriers to entry.	面临很高的进入门槛。
B	cater to the demand from investors with lower levels of investable assets.	满足可投资资产规模较小投资者的需求。
C	prefer actively managed funds to index funds when constructing client portfolios.	构建客户组合时偏好主动管理基金而非指数基金。

答案 / ANSWER: B

智能投顾通过自动化问卷、资产配置和再平衡降低提供服务的单位成本，因此能够服务资产规模较小的客户。常见模式使用成本较低的被动基金或 ETF。

Robo-advisers automate profiling, asset allocation, and rebalancing to lower service costs, allowing them to serve clients with smaller investable portfolios. Common models rely on relatively low-cost passive funds or ETFs.

排除干扰项 / ELIMINATE DISTRACTORS

A：本题所描述的技术模式不以高进入门槛为典型特征。C：常见模式更偏好低成本指数产品。

A: High barriers are not the typical feature emphasized by this model. C: Low-cost index products are more characteristic.

速记 / RECALL: 自动化降成本，使小额客户也能获得配置服务。

Automation lowers costs and makes portfolio services viable for smaller accounts.

Q082 原题 / ORIGINAL QUESTION

题库 PDF 第 199 页（章节第 16 页）| 答案 PDF 第 82 页

Relative to passive market-cap-weighted strategies, smart beta strategies typically have:

与被动市值加权策略相比，Smart Beta 策略通常具有怎样的管理费与组合换手率？

A	lower management fees and higher portfolio turnover.	管理费较低，组合换手率较高。
B	higher management fees and lower portfolio turnover.	管理费较高，组合换手率较低。
C	higher management fees and higher portfolio turnover.	管理费较高，组合换手率较高。

答案 / ANSWER: C

本题考查一般比较：Smart Beta 使用规则化的因子或非市值权重，需要调整权重来维持目标暴露，因此管理费与换手率通常高于简单市值加权被动策略。关键词是 typically，并非每一只具体基金都如此。

The question tests a typical comparison: smart beta uses rules-based factor or non-cap-weighted exposures and rebalances to maintain them. Management fees and turnover are generally higher than for simple passive market-cap-weighted strategies. “Typically” does not imply that every individual fund follows this pattern.

排除干扰项 / ELIMINATE DISTRACTORS

A 将管理费方向写反；B 将换手率方向写反。

A reverses the typical fee comparison. B reverses the typical turnover comparison.

速记 / RECALL: 相对简单市值加权：Smart Beta 通常费率更高、换手更多。

Versus simple cap weighting, smart beta typically has higher fees and turnover.

组合思维与管理步骤

Portfolio perspective and management stages

Q036 原题 / ORIGINAL QUESTION

题库 PDF 第 190 页（章节第 7 页）| 答案 PDF 第 36 页

The portfolio approach to investing most likely:

采用组合投资方法最可能带来什么效果？

A	prevents portfolio losses during market downturns.	防止组合在市场下跌期间出现亏损。
B	reduces the systematic risk of individual assets in a portfolio.	降低组合中各项资产本身的系统性风险。
C	helps avoid disastrous investment outcomes during normal market conditions.	有助于在正常市场条件下避免灾难性的投资结果。

答案 / ANSWER: C

组合方法关注资产之间的共同变动，而不仅是每项资产单独的表现。分散持仓可减少公司特有风险，从而降低单一投资失败对整体财富的冲击，但不能保证不亏损。

The portfolio approach considers how assets move together rather than evaluating each holding in isolation. Diversification reduces company-specific risk and the damage from a single failed investment, but it does not guarantee against losses.

排除干扰项 / ELIMINATE DISTRACTORS

A 错在保证市场下跌不亏损；B 混淆系统性风险与可分散的非系统性风险。

A incorrectly promises protection from market losses. B confuses systematic risk with diversifiable nonsystematic risk.

速记 / RECALL: 分散化主要处理个别资产风险，不能消除市场风险。

Diversification primarily addresses individual-asset risk, not market risk.

Q057 重点例题 / KEY EXAMPLE

题库 PDF 第 194 页（章节第 11 页）| 答案 PDF 第 57 页

In the portfolio management process, the feedback step most likely involves:

组合管理流程中的反馈阶段最可能包括哪项活动？

A	rebalancing the portfolio.	再平衡组合。
B	deciding on an asset allocation.	决定资产配置。
C	understanding the client's constraints.	了解客户约束。

答案 / **ANSWER: A**

反馈阶段检查绩效及客户条件变化，监控组合，并在权重偏离政策目标时进行再平衡。决定资产配置属于执行阶段的核心工作；了解客户约束属于规划阶段。

The feedback stage reviews performance and changes in client circumstances, monitors the portfolio, and rebalances when weights drift from policy targets. Asset allocation is central to execution, while understanding client constraints belongs to planning.

排除干扰项 / ELIMINATE DISTRACTORS

B 属于执行；C 属于规划。

B belongs to execution. C belongs to planning.

速记 / RECALL: Planning 了解客户；Execution 建组合；Feedback 评估与再平衡。

Planning understands the client; execution builds the portfolio; feedback reviews and rebalances.

07 行为金融：识别偏差的关键词

Behavioral finance: recognize the bias

本单元 8 题 | 首轮预算 20 分钟 | 重点：Q10, Q11, Q17, Q21, Q28, Q33

对应阅读主题 / Reading theme: The Behavioral Biases of Individuals

认知错误 cognitive errors：推理、记忆或信息处理出错，通常较能通过信息和教育纠正。情绪偏差 emotional biases：由情绪和冲动驱动，通常更难改变。分类是平均倾向，不意味着所有个体都容易或难以纠正。

Cognitive errors involve faulty reasoning, memory or information processing and are often more responsive to education and information. Emotional biases arise from feelings or impulses and are generally harder to change. These are tendencies rather than guarantees about every individual.

偏差 / Bias	关键词与后果 / Cue and consequence	题号 / Q
Overconfidence 过度自信；情绪 / Emotional	“我判断很准” → 低估风险、过度集中、可能过度交易。 Overrating one’s skill → underestimating risk, concentration, possible overtrading.	10,13
Illusion of control 控制幻觉；认知 / Cognitive	“熟悉或参与就能控制结果” → 集中持仓与过度交易。 Believing involvement confers control → concentration and overtrading.	17
Self-control 自我控制偏差；情绪 / Emotional	为了眼前享受牺牲长期储蓄目标。 Sacrificing long-term saving for immediate satisfaction.	11
Regret aversion 后悔厌恶；情绪 / Emotional	怕独自做错后悔 → 随大流、羊群行为。 Fear of regretting an independent decision → herding.	21
Loss aversion 损失厌恶；情绪 / Emotional	对损失更敏感，可能不愿实现亏损；比认知类偏差更难纠正。 Strong sensitivity to losses; reluctance to realize losses; often hard to correct.	28
Representativeness 代表性；认知 / Cognitive	把好公司形象推成好投资；halo effect 光环效应；忽略基准概率。 Treating a good-company image as evidence of a good investment; halo effect and base-rate neglect.	33
Hindsight 后见之明；认知 / Cognitive	事后觉得“我早知道会这样”。 Believing after the event that the outcome was predictable.	28 选项 / Option

7.1 两组最容易混淆的行为

Two commonly confused patterns

过度自信与控制幻觉都可能造成集中持仓。题目若强调预测能力被高估，想 overconfidence；若强调能控制本来不可控的结果，想 illusion of control。相同后果不表示相同成因。

Both overconfidence and illusion of control can cause concentrated holdings. Overstated forecasting skill points to overconfidence; believing one can control an uncontrollable outcome points to illusion of control. Similar outcomes do not imply identical causes.

Q13 问对盈利增长预测过度自信可能解释什么：对成长股过度乐观可能把价格推高，随后价值股相对表现更好，对应 value anomaly。Base-rate neglect 更直接属于代表性偏差；disposition effect（卖赢留亏）更常关联损失厌恶。 这些对应关系不是唯一因果；本题选三个选项中最合适的一项。

Q13 asks what overconfidence in earnings-growth forecasts may explain. Excessive optimism can overprice growth stocks and contribute to a value anomaly. Base-rate neglect is more directly associated with representativeness, while the disposition effect of selling winners and retaining losers is commonly linked to loss aversion. These are not exclusive causal links; choose the best of the three options.

07 原题与逐题解析

Original questions and explanations

先作答，再看答案；标为 KEY EXAMPLE 的题目优先完成。
Attempt the question before reading the solution; prioritize items labeled KEY EXAMPLE.

先分认知与情绪

Separate cognitive errors from emotional biases

Q022 原题 / ORIGINAL QUESTION

题库 PDF 第 187 页（章节第 4 页）| 答案 PDF 第 22 页

Cognitive errors:

关于认知错误，下列哪项正确？

A	stem from impulses and intuition.	源于冲动和直觉。
B	result in the same decision as assumed by traditional finance theory.	导致与传统金融理论假设相同的决策。
C	can often be corrected or eliminated through better information, education, and advice.	通常可借助更好的信息、教育和建议予以纠正或消除。

答案 / ANSWER: C

认知错误涉及信息处理、记忆、逻辑或概率判断失误，因此可用证据、教育和流程改善。情绪偏差则更深植于感受与冲动，通常更难仅靠说明道理纠正。

Cognitive errors involve faulty information processing, memory, logic, or probability judgments, so information, education, and better decision processes can often mitigate them. Emotional biases are rooted more deeply in feelings and impulses and are usually harder to correct through explanation alone.

排除干扰项 / ELIMINATE DISTRACTORS

A：更符合情绪偏差。B：偏差会使决策偏离传统理性模型，而非完全相同。

A is more characteristic of emotional biases. B is wrong because biases cause departures from traditional rational-choice assumptions.

速记 / RECALL: 认知错误可教育纠正；情绪偏差更难纠正。

Cognitive errors are often educable; emotional biases are harder to correct.

Q028 重点例题 / KEY EXAMPLE

题库 PDF 第 189 页（章节第 6 页）| 答案 PDF 第 28 页

Which of the following behavioral biases is most likely the hardest to correct?

下列哪一种行为偏差通常最难纠正？

A	Hindsight bias	后见之明偏差。
B	Loss-aversion bias	损失厌恶偏差。
C	Representativeness bias	代表性偏差。

答案 / ANSWER: B

损失厌恶属于情绪偏差，根植于对损失的强烈感受；即使投资者理解理性分析，感受仍可能持续。后见之明和代表性属于认知错误，通常更容易用记录、数据与教育改善。

Loss aversion is emotional and rooted in strong feelings about losses, which may persist despite rational explanation. Hindsight and representativeness are cognitive errors that can often be improved through records, evidence, and education.

排除干扰项 / ELIMINATE DISTRACTORS

A、C 属于认知偏差；与情绪偏差相比，通常较容易纠正。

A and C are cognitive biases and are generally more amenable to correction than emotional biases.

速记 / RECALL: 三选一找情绪偏差：损失厌恶。

Among these choices, the emotional bias is loss aversion.

用行为识别具体偏差

Identify the bias from the behavior

Q010 重点例题 / KEY EXAMPLE

题库 PDF 第 185 页（章节第 2 页）| 答案 PDF 第 10 页

Which of the following is most likely a consequence of overconfidence bias? Investors:

以下哪项最可能是过度自信偏差的后果？投资者：

A	holding poorly diversified portfolios.	持有分散化不足的组合。
B	continuing to hold classes of assets with which they are familiar.	继续持有自己熟悉的资产类别。
C	holding investments in a loss position longer than justified, in the hope that they will return to breakeven.	过久持有亏损投资，期望价格回到盈亏平衡点。

答案 / ANSWER: A

过度自信使投资者高估自己的预测和选股能力，低估出错概率，因此容易集中持仓并过度交易。持有分散化不足的组合是直接后果。

Overconfidence leads investors to overestimate forecasting and security-selection ability and underestimate the chance of error. Concentrated holdings, inadequate diversification, and excessive trading can result.

排除干扰项 / ELIMINATE DISTRACTORS

B：所附答案归为禀赋效应，熟悉并已持有资产会提高主观估值。C：更接近损失厌恶及处置效应。

B is classified as endowment bias in the supplied solution: familiarity and ownership increase perceived value. C is associated with loss aversion and the disposition effect.

速记 / RECALL: 自信过头：仓位集中、交易过多。

Overconfidence: concentrated positions and excessive trading.

Q011 重点例题 / KEY EXAMPLE

题库 PDF 第 186 页（章节第 3 页）| 答案 PDF 第 11 页

7%. If the risk-free rate is 3.1%, the portfolio's Sharpe ratio is closest to:

为了短期满足而未能采取行动追求长期目标，最能描述哪一种情绪偏差？

A	1.03.	自我控制偏差。
B	1.30.	禀赋效应。
C	1.56.	损失厌恶偏差。

答案 / ANSWER: A

自我控制偏差表现为难以延迟满足，例如当前消费过多而退休储蓄不足。核心冲突是短期诱惑压过长期目标。

Self-control bias is difficulty delaying gratification, such as consuming too much today and saving too little for retirement. Short-term temptation displaces long-term objectives.

排除干扰项 / ELIMINATE DISTRACTORS

B：禀赋效应是高估自己已经拥有的资产。C：损失厌恶是同等金额的损失带来更强烈的不适。

B: Endowment bias overvalues assets already owned. C: Loss aversion makes a loss more painful than an equal gain is pleasurable.

速记 / RECALL: 短期满足压过长期计划 = 自我控制偏差。

Short-term gratification defeats long-term plans = self-control bias.

Q017 重点例题 / KEY EXAMPLE

题库 PDF 第 187 页（章节第 4 页）| 答案 PDF 第 17 页

Which of the following is most likely a consequence of the illusion of control bias?

下列哪项最可能是控制幻觉偏差的后果？

A	An investor's portfolio turnover is too low.	投资者的组合换手率过低。
B	The investor's portfolio contains concentrated positions in companies.	投资者在某些公司上持仓过于集中。
C	An investor uses a simple forecasting model for portfolio construction.	投资者使用简单预测模型构建组合。

答案 / ANSWER: B

控制幻觉是高估自己能够影响实际上不能控制的投资结果。投资者可能对熟悉或自认能掌控的公司押注过大，造成集中持仓；也可能过度交易。

Illusion of control is overestimating one's influence over outcomes that are not controllable. Investors may place excessive bets on companies they believe they understand or can influence, producing concentrated positions and sometimes excessive trading.

排除干扰项 / ELIMINATE DISTRACTORS

A：通常更可能交易过多，而非换手过低。C：控制幻觉反而可能推动投资者构建过度复杂的模型，误以为细节能控制不确定性。

A: Excessive rather than insufficient trading is more characteristic. C: Illusion of control may instead encourage overly detailed models, mistaking complexity for control over uncertainty.

速记 / RECALL: 误以为能掌控结果 → 过度集中。

Believing outcomes are controllable can lead to excessive concentration.

Q021 重点例题 / KEY EXAMPLE

题库 PDF 第 187 页（章节第 4 页）| 答案 PDF 第 21 页

Which of the following best describes a potential consequence of the regret-aversion bias for financial market participants?

下列哪项最能描述后悔厌恶偏差对金融市场参与者的潜在影响？

A	Engaging in herding behaviour	从众行为。
B	Borrowing excessively to finance present consumption	过度借贷以满足当前消费。
C	Misidentifying risk tolerances because of how questions about risk tolerance were framed	由于风险承受能力问题的提问方式，错误识别自身的风险容忍度。

答案 / ANSWER: A

后悔厌恶使投资者避免独自做出事后可能显得错误的决定。跟随多数人能降低“只有我选错”的心理负担，因此容易产生从众行为。

Regret aversion discourages decisions that may look uniquely wrong in hindsight. Following the crowd reduces the psychological burden of being the only one mistaken, so it can produce herding.

排除干扰项 / ELIMINATE DISTRACTORS

B：属于自我控制问题。C：属于框架效应，问题表达改变了决策。

B is associated with self-control problems. C is framing bias, where the presentation of a question changes the response.

速记 / RECALL: 怕独自后悔 → 跟随大众。

Fear of individual regret → follow the crowd.

Q033 重点例题 / KEY EXAMPLE

题库 PDF 第 189 页（章节第 6 页）| 答案 PDF 第 33 页

Which of the following is most closely associated with representativeness bias?

下列哪项与代表性偏差关系最密切？

A	Momentum	动量。
B	The halo effect	光环效应。
C	Bubbles and crashes	泡沫与崩盘。

答案 / ANSWER: B

代表性偏差用表面特征或熟悉的类别代替完整概率判断。光环效应表现为因一家公司的某个优点（例如优质产品或管理者声望），就推断其股票也是好投资，忽略价格和其他证据。

Representativeness substitutes a salient feature or familiar category for a full assessment. Under the halo effect, one favorable attribute, such as strong products or a respected manager, leads investors to assume the stock is also attractive, overlooking valuation and other evidence.

排除干扰项 / ELIMINATE DISTRACTORS

A、C 可与多种行为机制相关，但光环效应与代表性偏差的联系最直接。

A and C can involve several behavioral mechanisms; the halo effect is the most direct association among these choices.

速记 / RECALL: 好公司不自动等于好股票：警惕光环效应。

A good company is not automatically a good investment: beware the halo effect.

Q013 原题 / ORIGINAL QUESTION

题库 PDF 第 186 页（章节第 3 页）| 答案 PDF 第 13 页

Which of the following can best be explained by overconfidence when predicting companies' earnings growth rates?

在预测公司盈利增长率时，过度自信最能解释以下哪一项？

A	Base-rate neglect	忽视基础概率。
B	The value anomaly	价值股异常。
C	The disposition effect	处置效应。

答案 / ANSWER: B

投资者可能过度相信热门成长股的盈利预测，把高增长外推得过远，使成长股价格偏高；对低预期的价值股则定价偏低。这可以帮助解释价值股相对成长股的超额表现，即价值股异常。

Investors may be too confident in forecasts for popular growth companies and extrapolate high growth too far, overpricing growth stocks relative to low-expectation value stocks. This can help explain the value anomaly.

排除干扰项 / ELIMINATE DISTRACTORS

A：忽视基础概率主要属于代表性偏差。C：卖赢留亏的处置效应更接近损失厌恶。

A: Base-rate neglect is mainly associated with representativeness. C: The disposition effect of selling winners and retaining losers is linked to loss aversion.

速记 / RECALL: 过度相信增长预测，可导致成长股贵、价值股便宜。

Overconfident growth forecasts can overprice growth stocks relative to value stocks.

08 风险管理、VaR 与风险度量

Risk management, VaR and risk measures

本单元 10 题 | 首轮预算 20 分钟 | 重点：Q2, Q3, Q35, Q72, Q80, Q88

对应阅读主题 / Reading theme: Introduction to Risk Management

8.1 风险管理不是把所有风险降到零

Risk management does not minimize every risk

目标是在可接受风险范围内提高企业价值：看预期收益与潜在损失的平衡，不是准确预测每次危机，也不是只看高概率事件。低概率、高损失事件同样需要考虑。

The goal is to enhance value within tolerable risk by balancing expected gains and potential losses. It is neither perfect crisis prediction nor attention only to high-probability events; low-probability severe losses also matter.

概念 / Concept	区别 / Distinction
Risk governance 风险治理	由董事会/高层自上而下确定框架，采用企业整体视角。 A top-down framework set by the board/senior leadership with an enterprise-wide view.
Risk tolerance 风险容忍度	愿意并能够接受多少风险；考虑财务资源与法律监管环境等。 How much risk is acceptable and bearable, considering resources and the regulatory environment.
Risk budgeting 风险预算	用标准差、Beta、VaR 或情景损失等度量，把风险额度量化并分配。 Quantify and allocate risk using measures such as volatility, beta, VaR or scenario losses.
Self-insurance 自保	自行承受某类风险，可能设置损失准备；不等于风险预算。 Retain a risk, perhaps with loss reserves; this is not risk budgeting.

8.2 风险分类只抓核心来源

Classify risk by its source

金融风险：市场、信用、流动性。非金融风险：会计、税务、法律监管、模型、操作等。Q18 在信用/流动性/会计中选会计；Q71 在税务/信用/会计中选信用。风险之间会相互放大，分类不表示互不关联。

Financial risks include market, credit and liquidity risk. Non-financial categories include accounting, tax, legal/regulatory, model and operational risks. Q18 selects accounting; Q71 selects credit. Risks can amplify each other, so categories are not independent compartments.

8.3 Greeks：先问谁对谁敏感

Greeks: identify which quantity responds to which driver

度量 / Measure	含义 / Meaning
Delta, Δ	衍生品价值对标的价格的一阶敏感度。 First-order sensitivity of derivative value to the underlying price.
Gamma, Γ	Delta 对标的价格变化的敏感度；Q72 选 C。 Sensitivity of delta to the underlying price; Q72 is C.
Vega	衍生品价值对波动率的敏感度。 Sensitivity of value to volatility.
Rho	衍生品价值对利率的敏感度。 Sensitivity of value to interest rates.

8.4 VaR 看门槛，尾部均值看更坏以后

VaR is a threshold; tail averages look beyond it

VaR 必须连同期限和概率一起读。Q88 的一天 5% VaR=500 万英镑，表示在模型假设下，一天损失达到或超过该门槛的概率约为 5%。长期平均频率对应 1/0.05=20 个交易日一次，选 C；不表示每第 20 天固定亏损。

Read VaR together with its horizon and tail probability. In Q88, one-day 5% VaR of GBP5 million means an approximately 5% chance of a daily loss at or beyond that threshold under the model. The long-run average frequency is 1/0.05=20 trading days, answer C; losses do not occur on a fixed 20-day schedule.

VaR 不是最大可能损失，也不说明越过门槛后平均亏多少。Expected shortfall/CVaR 通常衡量 VaR 尾部的条件平均损失。Loss given default（LGD）则以“已经违约”为条件，是不同的条件事件。

VaR is neither the maximum possible loss nor the average severity beyond its threshold. Expected shortfall/CVaR typically measures mean loss in the VaR tail. Loss given default conditions on default, which is a different event.

辨析 / PRECISION NOTE

Q80 按给定选项及原答案选 C：expected loss given default。应记“违约条件下的平均损失”，不要把 LGD、CVaR 和无条件预期信用损失当成同一个概念。若 LGD 为损失率，预期信用损失金额的常见分解是 PD×LGD×EAD。

For Q80, the supplied choices and solution select C: expected loss given default. Interpret it as average loss conditional on default. LGD, CVaR and unconditional expected credit loss are distinct. When LGD is a loss rate, a standard decomposition of expected credit loss in currency is PD×LGD×EAD.

08 原题与逐题解析

Original questions and explanations

先作答，再看答案；标为 KEY EXAMPLE 的题目优先完成。
Attempt the question before reading the solution; prioritize items labeled KEY EXAMPLE.

风险治理与预算

Risk governance and budgeting

Q002 重点例题 / KEY EXAMPLE

题库 PDF 第 184 页（章节第 1 页）| 答案 PDF 第 2 页

The process of risk management is best described as the set of decisions that maximizes a company's value while:

风险管理过程最适合描述为：在以下哪种条件下，使公司价值最大化的一系列决策？

A	minimizing the risk taken.	尽量减少所承担的风险。
B	bearing a tolerable level of risk.	承担可接受水平的风险。
C	predicting the potential risk correctly.	准确预测潜在风险。

答案 / ANSWER: B

风险管理的目标是在企业可承受的风险范围内创造价值。完全压低风险可能同时放弃有价值的机会；风险管理也不要求准确预测每一次风险事件。

Risk management seeks to maximize value while bearing a tolerable amount of risk. Minimizing risk can sacrifice valuable opportunities, and effective risk management does not require perfect prediction of every adverse event.

排除干扰项 / ELIMINATE DISTRACTORS

A：最小风险不等于最大企业价值。C：准确预测不是风险管理的必要条件或总体目标。

A: Minimum risk is not equivalent to maximum enterprise value. C: Accurate prediction is neither the overall objective nor a prerequisite.

速记 / RECALL: 目标是可承受风险下的价值最大化。

Maximize value subject to tolerable risk.

Q003 重点例题 / KEY EXAMPLE

题库 PDF 第 184 页（章节第 1 页）| 答案 PDF 第 3 页

Which of the following is least consistent with effective risk governance?

下列哪项与有效的风险治理最不一致？

A	Taking an enterprise-wide view	采取覆盖整个企业的视角。
B	Defining the enterprise's risk tolerance	界定企业的风险容忍度。
C	Following a bottom-up process to direct risk management activities	采用自下而上的流程指导风险管理活动。

答案 / ANSWER: C

风险治理由董事会和高层制定风险方向与容忍度，并从全企业层面统筹，即治理方向通常自上而下。部门可以自下而上报告风险信息，但这不等于由底层决定全企业治理方向。

Risk governance sets enterprise-wide direction and tolerance through the board and senior management, generally using a top-down approach. Bottom-up risk reporting can support that process, but it does not replace top-down governance.

排除干扰项 / ELIMINATE DISTRACTORS

A、B 都是有效风险治理的核心要素；题目要求选 least consistent。

A and B are core elements of effective governance. Notice the negative wording: least consistent.

速记 / RECALL: 治理定方向：全企业、自上而下。

Governance sets direction: enterprise-wide and top-down.

Q035 重点例题 / KEY EXAMPLE

题库 PDF 第 190 页（章节第 7 页）| 答案 PDF 第 35 页

Which of the following is most likely undertaken during the risk budgeting process?

以下哪项最可能在风险预算过程中进行？

A	Assessing risk appetite	评估风险偏好。
B	Setting a limit for value at risk (VaR)	设定风险价值（VaR）限额。
C	Establishing a reserve to cover potential future losses	建立准备金，以覆盖未来潜在损失。

答案 / ANSWER: B

风险预算把机构愿意承担的总风险分配给投资组合、策略或业务，并落实为可监控的限额。设定 VaR 限额，就是将总风险约束变成具体风险预算。

Risk budgeting allocates the total risk an institution is willing to bear across portfolios, strategies, or business units and translates it into measurable limits. A VaR limit is a concrete expression of that budget.

排除干扰项 / ELIMINATE DISTRACTORS

A 属于先确定整体风险承受程度的治理工作；C 是自我保险（self-insurance），以准备金承受损失，并不直接分配风险限额。

A determines overall risk tolerance before budgeting. C is self-insurance: accepting risk and setting aside a reserve to fund losses, rather than allocating risk limits.

速记 / RECALL: 风险偏好先定方向，风险预算再定额度。

Risk appetite sets the direction; risk budgeting sets the limits.

Q089 原题 / ORIGINAL QUESTION

题库 PDF 第 200 页（章节第 17 页）| 答案 PDF 第 89 页

A good risk management process should:

良好的风险管理流程应当做到什么？

A	predict when a crisis will occur.	预测危机何时发生。
B	consider the balance between expected returns and losses.	考虑预期收益与损失之间的权衡。
C	only consider losses occurring from events that have a high likelihood.	只考虑高概率事件造成的损失。

答案 / **ANSWER: B**

风险管理的核心是判断为争取收益所承担的损失风险是否值得、是否可承受。既要讨论常见结果，也要评估低概率但损失巨大的情景。

Risk management assesses whether the potential losses accepted in pursuit of returns are worthwhile and tolerable. It considers ordinary outcomes as well as low-probability scenarios with severe losses.

排除干扰项 / ELIMINATE DISTRACTORS

A 把准确预测危机当成必要能力；C 忽略尾部事件，可能低估重大损失。

A incorrectly makes precise crisis prediction the objective. C ignores tail events and can understate severe potential losses.

速记 / RECALL: 管理可承受的风险收益权衡，也要看小概率大损失。

Manage a tolerable risk-return trade-off, including rare but severe losses.

Q099 原题 / ORIGINAL QUESTION

题库 PDF 第 202 页（章节第 19 页）| 答案 PDF 第 99 页

In risk management, which of the following should be taken into account when determining an enterprise's risk tolerance?

风险管理中，确定企业风险承受度时应考虑下列哪项？

A	Management compensation	管理层薪酬。
B	The enterprise's value at risk (VaR)	企业的风险价值（VaR）。
C	The government and regulatory landscape	政府与监管环境。

答案 / **ANSWER: C**

风险承受度界定企业愿意且能够承担多大的风险，必须符合外部制度与监管要求。政府政策及监管环境可能限制企业可承担的风险，因此属于应考虑的因素。

Risk tolerance defines the level of risk an enterprise is willing and able to bear and must be consistent with external and regulatory constraints. Government policy and the regulatory environment can restrict acceptable risk and therefore must be considered.

排除干扰项 / ELIMINATE DISTRACTORS

A 不应以管理层个人报酬来设定企业整体风险承受度；B 的 VaR 是测量当前或潜在风险的工具，并非风险承受度本身。

A, managers’ compensation, should not determine the enterprise’s acceptable risk. B, VaR, measures risk exposure and is not itself the enterprise’s risk tolerance.

速记 / RECALL: 先定可承受风险，再用 VaR 等指标衡量和监控。

Define acceptable risk first, then measure and monitor it with tools such as VaR.

金融与非金融风险

Financial and non-financial risks

Q018 原题 / ORIGINAL QUESTION

题库 PDF 第 187 页（章节第 4 页）| 答案 PDF 第 18 页

Which of the following is best classified as a non-financial risk?

下列哪项最适合归类为非金融风险？

A	Credit risk	信用风险。
B	Liquidity risk	流动性风险。
C	Accounting risk	会计风险。

答案 / ANSWER: C

会计风险来自会计处理、报告或规则应用等问题，属于非金融风险。信用风险与交易对手不能履约有关；流动性风险与难以及时筹资或以合理价格成交有关，均属于金融风险。

Accounting risk arises from accounting treatment, reporting, or application of accounting rules and is a non-financial risk. Credit risk concerns failure to perform obligations, and liquidity risk concerns funding or transacting at reasonable prices; both are financial risks.

排除干扰项 / ELIMINATE DISTRACTORS

A、B 属于金融风险，不能因它们可能由运营事件触发就改变本题分类。

A and B are financial risks, even though operational events may sometimes trigger them.

速记 / RECALL: 市场、信用、流动性是金融；会计是非金融。

Market, credit, and liquidity are financial; accounting is non-financial.

Q071 原题 / ORIGINAL QUESTION

题库 PDF 第 197 页（章节第 14 页）| 答案 PDF 第 71 页

Which of the following is best classified as a financial risk?

下列哪项最适合归类为金融风险？

A	Tax risk	税务风险。
B	Credit risk	信用风险。
C	Accounting risk	会计风险。

答案 / ANSWER: B

信用风险是交易对手或债务人无法履行付款义务的风险，属于金融风险。考试中主要金融风险通常归为市场风险、信用风险与流动性风险。

Credit risk is the risk that a counterparty or debtor fails to meet payment obligations. It is classified as financial risk; the main financial-risk categories in this framework are market, credit and liquidity risk.

排除干扰项 / ELIMINATE DISTRACTORS

A 税务风险和 C 会计风险属于非金融风险；它们仍然可能造成财务损失。

A, tax risk, and C, accounting risk, are non-financial risks, although they can still cause financial losses.

速记 / RECALL: 金融风险记：市场、信用、流动性。

Core financial risks: market, credit and liquidity.

敏感度、尾部与信用损失

Sensitivity, tail risk and credit loss

Q072 重点例题 / KEY EXAMPLE

题库 PDF 第 197 页（章节第 14 页）| 答案 PDF 第 72 页

The risk management measure that captures the sensitivity of a derivative's delta to a change in the value of the underlying best describes:

衡量衍生品 Delta 对标的资产价格变化敏感度的风险指标是什么？

A	rho	Rho。
B	vega.	Vega。
C	gamma.	Gamma。

答案 / ANSWER: C

Delta = ∂V/∂S，衡量衍生品价格对标的价格的敏感度。Gamma = ∂Delta/∂S = ∂²V/∂S²，衡量 Delta 如何随标的价格变化，因此属于二阶敏感度。

Delta = ∂V/∂S measures derivative-price sensitivity to the underlying price. Gamma = ∂Delta/∂S = ∂²V/∂S² measures how delta changes as the underlying price changes; it is a second-order sensitivity.

排除干扰项 / ELIMINATE DISTRACTORS

A Rho 衡量利率敏感度；B Vega 衡量波动率敏感度。

A, rho, measures interest-rate sensitivity. B, vega, measures volatility sensitivity.

速记 / RECALL: Delta 的变化看 Gamma。

A change in delta is measured by gamma.

Q088 重点例题 / KEY EXAMPLE

题库 PDF 第 200 页（章节第 17 页）| 答案 PDF 第 88 页

A bank determines that its value at risk (VaR) is £5 million at 5% for one day. The bank is expecting a minimum loss of £5 million once every:

某银行的一天期 VaR 为 500 万英镑，尾部概率为 5%。按长期平均频率，至少损失 500 万英镑的情况约每多少个营业日发生一次？

A	5 business days	5 个营业日。
B	13 business days.	13 个营业日。
C	20 business days	20 个营业日。

答案 / ANSWER: C

5% 尾部概率意味着，在模型假设成立时，约 5% 的交易日损失达到或超过 500 万英镑。平均频率的倒数为 1 / 0.05 = 20 个营业日，故选 C。

A 5% tail probability means that, under the model assumptions, losses reach or exceed £5 million on about 5% of trading days. The reciprocal frequency is 1 / 0.05 = 20 business days, so C is correct.

排除干扰项 / ELIMINATE DISTRACTORS

A 把百分数 5 当成天数；B 的 13 更接近一年 250-260 个交易日内预期出现的次数，不能当作间隔天数。

A treats the number 5 as a count of days. B is approximately the expected annual number of exceedances across 250-260 trading days, not the interval in days.

速记 / RECALL: 尾部概率 5% → 长期平均约 20 天一次。

A 5% tail probability implies about one exceedance per 20 days on average.

概念辨析 / NOTE: “每 20 天一次”是模型下的平均频率，不是固定周期，也不保证任何特定 20 天内恰好出现一次。VaR 不是最大可能损失。

“Once every 20 days” is a model-based average frequency, not a fixed schedule or a guarantee of exactly one exceedance in any particular 20-day period. VaR is not the maximum possible loss.

Q080 重点例题 / KEY EXAMPLE

题库 PDF 第 198 页（章节第 15 页）| 答案 PDF 第 80 页

Which of the following metrics is most appropriate to estimate a bond's average extreme loss?

下列哪个指标最适合估计债券的平均极端损失？

A	VaR of loss	损失的风险价值（VaR）。
B	Standard deviation of loss	损失的标准差。
C	Expected loss given default	违约条件下的预期损失。

答案 / **ANSWER: C**

在给定选项及信用风险语境下，违约条件下的预期损失回答："如果债务人违约，平均会损失多少？"它反映违约这一严重信用事件发生后的平均损失程度，因此选 C。

Among the supplied choices and in the credit-risk context, expected loss given default answers: "If the debtor defaults, how much do we lose on average?" It measures average loss severity conditional on the severe credit event of default, so C is the intended answer.

排除干扰项 / ELIMINATE DISTRACTORS

A 的 VaR 是尾部损失阈值，不能给出超过阈值后的平均损失；B 的标准差描述整体离散程度，并非尾部平均损失。

A, VaR, is a tail-loss threshold and does not report the average loss beyond that threshold. B measures overall dispersion, not average tail loss.

速记 / RECALL: VaR 看阈值；CVaR 看超过阈值后的平均；LGD 看违约后的损失。

VaR: threshold. CVaR: average beyond the threshold. LGD: loss conditional on default.

概念辨析 / NOTE: 不要把 LGD 与 CVaR / expected shortfall 当作同义词。CVaR 的条件是损失超过 VaR；LGD 的条件是发生违约。一般尾部平均损失通常用 CVaR，本题因选项及债券信用风险语境选择 C。

LGD and CVaR / expected shortfall are not synonyms. CVaR conditions on losses exceeding VaR; LGD conditions on default. CVaR is the usual generic average tail-loss measure, while C is the intended choice here in the bond-credit context.

公式速查

Formula reference

符号：R 收益率；Rf 无风险利率；p 组合；m 市场；σ 标准差；σ^2 方差；ρ 相关系数；β 系统性风险；α 超额绩效；A 风险厌恶系数；n 期数/资产数（依题意）。

Notation: R return, Rf risk-free rate, p portfolio, m market, σ standard deviation, σ^2 variance, ρ correlation, β systematic risk, α abnormal performance, A risk aversion and n periods/assets as applicable.

收益与权重 / Return and weights | Q6, Q66

$$E(Rp)=\Sigma\ wiE(Ri);\ \Sigma wi=1$$

双资产方差 / Two-asset variance | Q24, Q66, Q90

$$\sigma p^2=w1^2\sigma1^2+w2^2\sigma2^2+2w1w2\rho12\sigma1\sigma2$$

协方差 / Covariance | Q53

$$Cov12=\rho12\sigma1\sigma2$$

等权组合极限 / Equal-weight limit | Q98

$$\sigma p^2=\text{average variance}/n + (n-1)/n \times \text{average covariance}$$

效用 / Utility | Q7*, Q8, Q41

$$U=E(R)-0.5A\sigma^2$$

无风险与风险组合 / Complete portfolio | Q6, Q100

$$E(Rc)=Rf+y[E(Rp)-Rf];\ \sigma c=|y|\sigma p$$

Beta | Q40, Q52

$$\beta i=Cov(i,m)/\sigma m^2=\rho im\sigma i/\sigma m$$

反求相关 / Recover correlation | Q87

$$\rho im=\beta i\sigma m/\sigma i$$

反求协方差 / Recover covariance | Q93

$$Cov(i,m)=\beta i\sigma m^2$$

残差风险 / Residual risk | Q79

$$\sigma e^2=\sigma i^2-\beta i^2\sigma m^2$$

CAPM / SML | Q50, Q52, Q54, Q81, Q94

$$E(Ri)=Rf+\beta iMRP;\ MRP=E(Rm)-Rf$$

SCL 超额收益回归 / Excess-return regression | Q43, Q73

$$Ri-Rf=\alpha i+\beta i(Rm-Rf)+\varepsilon i$$

Sharpe | Q70, Q76, Q78

$$(Rp-Rf)/\sigma p$$

Treynor | Q95

$$(Rp-Rf)/\beta p$$

Jensen alpha | Q27, Q30

$$\alpha p=Rp-[Rf+\beta p(Rm-Rf)]$$

M^2 收益水平 / Return level | Q9

$$M^2=Rf+Sharpe_p\times\sigma m$$

M^2 alpha | Q9

$$M^2\ alpha=M^2-Rm$$

退休目标 / Retirement target | Q97

FVnominal=FVtoday(1+inflation)^n; r=(FVnominal/PV)^(1/n)-1

实际与名义 / Real and nominal | Q97

1+rnominal=(1+rreal)(1+inflation)

尾部平均频率 / Average tail frequency | Q88

Average observations per exceedance=1/p

统一把收益率写成小数代入；如果采用百分数单位，必须全式一致。方差单位是收益率单位的平方。等权组合极限中的平均协方差指不同资产之间的非对角项。

Use decimal returns consistently. If using percentage-point units, keep the entire calculation consistent. Variance has squared return units. Average covariance in the equal-weight limit refers to off-diagonal covariances between distinct assets.

100 题答案总表

Complete answer key

题号为可点击链接；括号内为本讲义解析起始页。Q7 的 B* 仅抄录答案文档标示，不计分。
Click a question number to jump to its solution. Parentheses show guide pages. Q7 B* is the source-key label only and is excluded from scoring.

Q001 C (p.64)	Q002 B (p.78)	Q003 C (p.79)	Q004 C (p.53)	Q005 B (p.20)
Q006 B (p.20)	Q007 B* (p.8)	Q008 B (p.6)	Q009 A (p.44)	Q010 A (p.72)
Q011 A (p.73)	Q012 A (p.9)	Q013 B (p.75)	Q014 B (p.57)	Q015 A (p.10)
Q016 A (p.57)	Q017 B (p.73)	Q018 C (p.81)	Q019 C (p.58)	Q020 A (p.11)
Q021 A (p.74)	Q022 C (p.71)	Q023 B (p.66)	Q024 B (p.12)	Q025 B (p.37)
Q026 C (p.58)	Q027 B (p.43)	Q028 B (p.72)	Q029 B (p.38)	Q030 C (p.41)
Q031 A (p.7)	Q032 B (p.67)	Q033 B (p.74)	Q034 A (p.65)	Q035 B (p.79)
Q036 C (p.68)	Q037 B (p.49)	Q038 B (p.53)	Q039 B (p.31)	Q040 B (p.26)
Q041 C (p.6)	Q042 A (p.66)	Q043 C (p.36)	Q044 A (p.27)	Q045 B (p.51)
Q046 C (p.49)	Q047 A (p.19)	Q048 C (p.18)	Q049 C (p.50)	Q050 A (p.34)
Q051 C (p.50)	Q052 B (p.32)	Q053 A (p.13)	Q054 A (p.33)	Q055 A (p.7)
Q056 B (p.54)	Q057 A (p.69)	Q058 A (p.59)	Q059 B (p.51)	Q060 B (p.19)
Q061 C (p.59)	Q062 A (p.54)	Q063 A (p.65)	Q064 C (p.37)	Q065 C (p.67)
Q066 B (p.14)	Q067 C (p.9)	Q068 A (p.21)	Q069 A (p.60)	Q070 A (p.42)
Q071 B (p.81)	Q072 C (p.82)	Q073 B (p.36)	Q074 B (p.55)	Q075 B (p.31)
Q076 B (p.42)	Q077 C (p.55)	Q078 A (p.41)	Q079 A (p.30)	Q080 C (p.83)
Q081 C (p.35)	Q082 C (p.68)	Q083 A (p.60)	Q084 C (p.27)	Q085 C (p.52)
Q086 C (p.61)	Q087 B (p.28)	Q088 C (p.82)	Q089 B (p.80)	Q090 C (p.15)
Q091 A (p.52)	Q092 A (p.21)	Q093 A (p.29)	Q094 B (p.33)	Q095 B (p.45)
Q096 C (p.10)	Q097 C (p.56)	Q098 C (p.11)	Q099 C (p.80)	Q100 A (p.22)

评分：正确率 = 答对题数 ÷ 99（全套练习且排除 Q7）；猜对题另行标记，不视为已经掌握。
For a complete attempt, accuracy = correct answers / 99, excluding Q7. Track guessed-correct answers separately.

题号到页码索引

Question-to-page index

题库页 / Source =《1000题(4).pdf》的 PDF 页码；答案页 = 原题号。讲义页为本文件页码。
Source page refers to the supplied question PDF. The original answer page equals the question number.

题号 Q	单元 Unit	题库页 Source	讲义页 Guide	题号 Q	单元 Unit	题库页 Source	讲义页 Guide
Q001	6	184	64	Q026	5	188	58
Q002	8	184	78	Q027	4	188	43
Q003	8	184	79	Q028	7	189	72
Q004	5	184	53	Q029	3	189	38
Q005	2	184	20	Q030	4	189	41
Q006	2	185	20	Q031	1	189	7
Q007	1	185	8	Q032	6	189	67
Q008	1	185	6	Q033	7	189	74
Q009	4	185	44	Q034	6	190	65
Q010	7	185	72	Q035	8	190	79
Q011	7	186	73	Q036	6	190	68
Q012	1	186	9	Q037	5	190	49
Q013	7	186	75	Q038	5	190	53
Q014	5	186	57	Q039	3	190	31
Q015	1	186	10	Q040	3	191	26
Q016	5	186	57	Q041	1	191	6
Q017	7	187	73	Q042	6	191	66
Q018	8	187	81	Q043	3	191	36
Q019	5	187	58	Q044	3	191	27
Q020	1	187	11	Q045	5	191	51
Q021	7	187	74	Q046	5	192	49
Q022	7	187	71	Q047	2	192	19
Q023	6	188	66	Q048	2	192	18
Q024	1	188	12	Q049	5	192	50
Q025	3	188	37	Q050	3	192	34

题号索引（续）

Question index, continued

题号 Q	单元 Unit	题库页 Source	讲义页 Guide	题号 Q	单元 Unit	题库页 Source	讲义页 Guide
Q051	5	192	50	Q076	4	197	42
Q052	3	193	32	Q077	5	198	55
Q053	1	193	13	Q078	4	198	41
Q054	3	193	33	Q079	3	198	30
Q055	1	193	7	Q080	8	198	83
Q056	5	194	54	Q081	3	199	35
Q057	6	194	69	Q082	6	199	68
Q058	5	194	59	Q083	5	199	60
Q059	5	194	51	Q084	3	199	27
Q060	2	194	19	Q085	5	199	52
Q061	5	195	59	Q086	5	200	61
Q062	5	195	54	Q087	3	200	28
Q063	6	195	65	Q088	8	200	82
Q064	3	195	37	Q089	8	200	80
Q065	6	195	67	Q090	1	200	15
Q066	1	196	14	Q091	5	201	52
Q067	1	196	9	Q092	2	201	21
Q068	2	196	21	Q093	3	201	29
Q069	5	196	60	Q094	3	201	33
Q070	4	196	42	Q095	4	201	45
Q071	8	197	81	Q096	1	202	10
Q072	8	197	82	Q097	5	202	56
Q073	3	197	36	Q098	1	202	11
Q074	5	197	55	Q099	8	202	80
Q075	3	197	31	Q100	2	202	22

原文校勘与口径说明

Source issues and conventions

以下说明把“题库答案是什么”和“概念应如何准确理解”分开呈现。两处原文排版/内容问题已明确处理；其他条目是术语、假设或题目语境的辨析。

These notes distinguish the supplied answer from the precise interpretation of the concept. Two source defects are explicitly handled; other entries clarify terminology, assumptions or question context.

Q7 | 原文严重错配 / Corrupted source

题干与周转率表属于现金转换周期，选项和答案却讨论效用。无法恢复原本完整题干。保留原文，B* 仅为答案页标示，排除计分。

The cash-conversion-cycle stem/table does not match the utility choices/solution. The intended complete stem cannot be recovered. The source is preserved, B* only records the answer key, and the item is excluded from scoring.

Q66 | 表头错位 / Misaligned table header

原题表头错位。依答案第 66 页的计算，重排为证券、预期收益率、标准差；数据保持为 17%/24% 与 6%/12%。

The source header is misaligned. Answer page 66 supports the corrected columns: security, expected return and standard deviation, retaining 17%/24% and 6%/12%.

Q5、Q20 | 无风险资产 / Risk-free asset

原答案按“零相关”作答；严格数学上相关系数未定义，协方差为零。

The source uses zero correlation. Strictly, the correlation is undefined and covariance is zero.

Q9 | M^2 与 M^2 alpha

11.6% 是风险调整收益水平；减去市场收益后 3.6% 才是本题 M^2 alpha。

11.6% is the adjusted return level; subtracting market return gives the requested 3.6% M^2 alpha.

Q15 | 分散化比率 / Diversification ratio

按本题口径为组合标准差除以平均单项标准差；不能与倒数定义混用。

Use portfolio volatility divided by average standalone volatility; do not mix it with the reciprocal convention.

Q53 | 5.5 $\%^2$

换成小数单位是 0.00055；标准差为 2.708%。

In decimal-return units this is 0.00055, implying standard deviation of 2.708%.

Q76 | Treynor 与 SML

CAL 斜率为 Sharpe；SML 斜率为市场风险溢价。仅在使用预期收益且 CAPM 公允定价、β 非零时，Treynor 等于市场风险溢价。

The CAL slope is Sharpe; the SML slope is the market premium. Treynor equals the premium when based on expected returns under CAPM fair pricing with nonzero beta.

Q80 | LGD 与 CVaR

LGD 的条件是已经违约；CVaR 的条件是处于损失尾部，两者不能直接互换。

LGD conditions on default; CVaR conditions on tail loss. They are not interchangeable.

Q88 | 20 天 / 20 days

表示平均频率，不是固定的亏损周期。

An average frequency, not a fixed loss schedule.

其他说明见逐题解析：Q23/Q65 产品惯例；Q31 峰度；Q48 框架；Q50 假设；Q64 收益与残差。

Other local notes: Q23/Q65 product conventions; Q31 kurtosis; Q48 framework; Q50 assumption; Q64 return versus residual.

最后一轮：只复盘真正没掌握的题

Final pass: focus on the remaining gaps

每题只写一个最主要错因：概念混淆、公式选择、代入/单位、读题遗漏、记忆分类。猜对也记为待复习。重做时遮住解析，先说规则，再计算或选项排除。

Assign one primary error cause: concept confusion, formula selection, substitution/units, missed wording or classification recall. Guessed-correct items also need review. On reattempts, cover the solution, state the rule, then calculate or eliminate choices.

题号 / Q	首次答案 / First	错因 / Error cause	一句规则 / One rule	复做 / Reattempt

考前优先排查的五种错误

Five errors to catch before the exam

方差和标准差：问 variance 就停在 σ^2；问 standard deviation 才开平方。

Variance versus standard deviation: stop at σ^2 for variance; take the square root for standard deviation.

市场收益和风险溢价：已给 market risk premium 就不要再减 Rf。

Market return versus premium: do not subtract Rf again from a stated market risk premium.

风险口径：Sharpe/M^2 看 σ；Treynor/Jensen 看 β。

Risk basis: Sharpe/M^2 use σ; Treynor/Jensen use β.

IPS 位置：评估业绩看 Evaluation；更新 IPS 看 Procedures。

IPS location: performance evaluation belongs to Evaluation; updating the IPS belongs to Procedures.

相关性方向：类内正相关；类间低相关；ρ=0 不等于零风险。

Correlation: positive within classes, low across classes; ρ=0 does not mean zero risk.

核心术语对照

Core bilingual glossary

English	中文
Risk aversion	风险厌恶
Risk tolerance	风险承受度 / 风险容忍度
Risk capacity / Ability	承担风险的客观能力
Willingness to take risk	承担风险的主观意愿
Variance / Standard deviation	方差 / 标准差
Covariance / Correlation	协方差 / 相关系数
Diversification	分散化
Systematic risk	系统性风险
Idiosyncratic / Nonsystematic risk	特有 / 非系统性风险
Efficient frontier	有效前沿
Global minimum-variance portfolio	全局最小方差组合
Optimal risky portfolio	最优风险组合
Complete portfolio	完整组合：风险组合加无风险资产
Capital allocation line (CAL)	资本配置线
Capital market line (CML)	资本市场线
Security market line (SML)	证券市场线
Security characteristic line (SCL)	证券特征线
Market risk premium	市场风险溢价
Investment policy statement (IPS)	投资政策说明书
Strategic / Tactical asset allocation	战略 / 战术资产配置
Rebalancing	再平衡
Defined benefit / Defined contribution	设定受益 / 设定提存
Cognitive error / Emotional bias	认知错误 / 情绪偏差
Value at risk (VaR)	在险价值
Expected shortfall / CVaR	预期尾部损失 / 条件在险价值
Loss given default (LGD)	违约损失（率或金额依定义）

资料来源与编制说明

Sources and preparation notes

主要来源 / PRIMARY SOURCES

S1：《1000题(4).pdf》，用户上传，整个文件 202 页。提取组合管理第 184-202 页（章节第 1-19 页），共 Q1-Q100。

S1: User-supplied question PDF, 202 pages total. Portfolio Management is extracted from PDF pages 184-202, chapter pages 1-19, containing Q1-Q100.

S2：《组合管理(1).pdf》，用户上传，共 100 页扫描答案。逐页识别并核对；第 n 页对应 Qn。所有答案字母以此为核对基础；需限定或校勘的内容已明确标注。

S2: User-supplied answer PDF, 100 scanned pages. Each page was recognized and checked; page n corresponds to Qn. Answer letters were matched to this source, with qualifications and defects explicitly marked.

官方术语与主题核对 / OFFICIAL CROSS-CHECKS

R1 | CFA Institute (2026): Portfolio Risk and Return: Part I

R2 | CFA Institute (2026): Portfolio Risk and Return: Part II

R3 | CFA Institute (2026): Portfolio Management: An Overview

R4 | CFA Institute (2026): Basics of Portfolio Planning and Construction

R5 | CFA Institute (2026): The Behavioral Biases of Individuals

R6 | CFA Institute (2026): Introduction to Risk Management

R7 | CFA Institute (2026): Credit Risk

上述官方页面用于交叉核对阅读主题和术语；不用于证明本题库出自 CFA Institute。完整题干、数值和答案来源仍为 S1/S2。讲解、中文翻译、归类、图表与计算复核为本讲义新增整理。

These official pages cross-check reading themes and terminology; they do not establish CFA Institute authorship of the source questions. Question wording, inputs and answer letters come from S1/S2. Explanations, translations, organization, charts and arithmetic checks were prepared for this guide.

核对日期 / Accessed and checked: 2026-11-02

完整性记录：100 个唯一原题号；8 个学习单元；原题与答案页一一对应；Q7 作为原文损坏项保留；Q66 表头重排并留痕；来源未核实的题目不标作“官方题”。

Completeness record: 100 unique source numbers; eight learning units; one-to-one question/answer page mapping; Q7 retained as corrupted; Q66 header repaired transparently; unverified items are not labeled official questions.